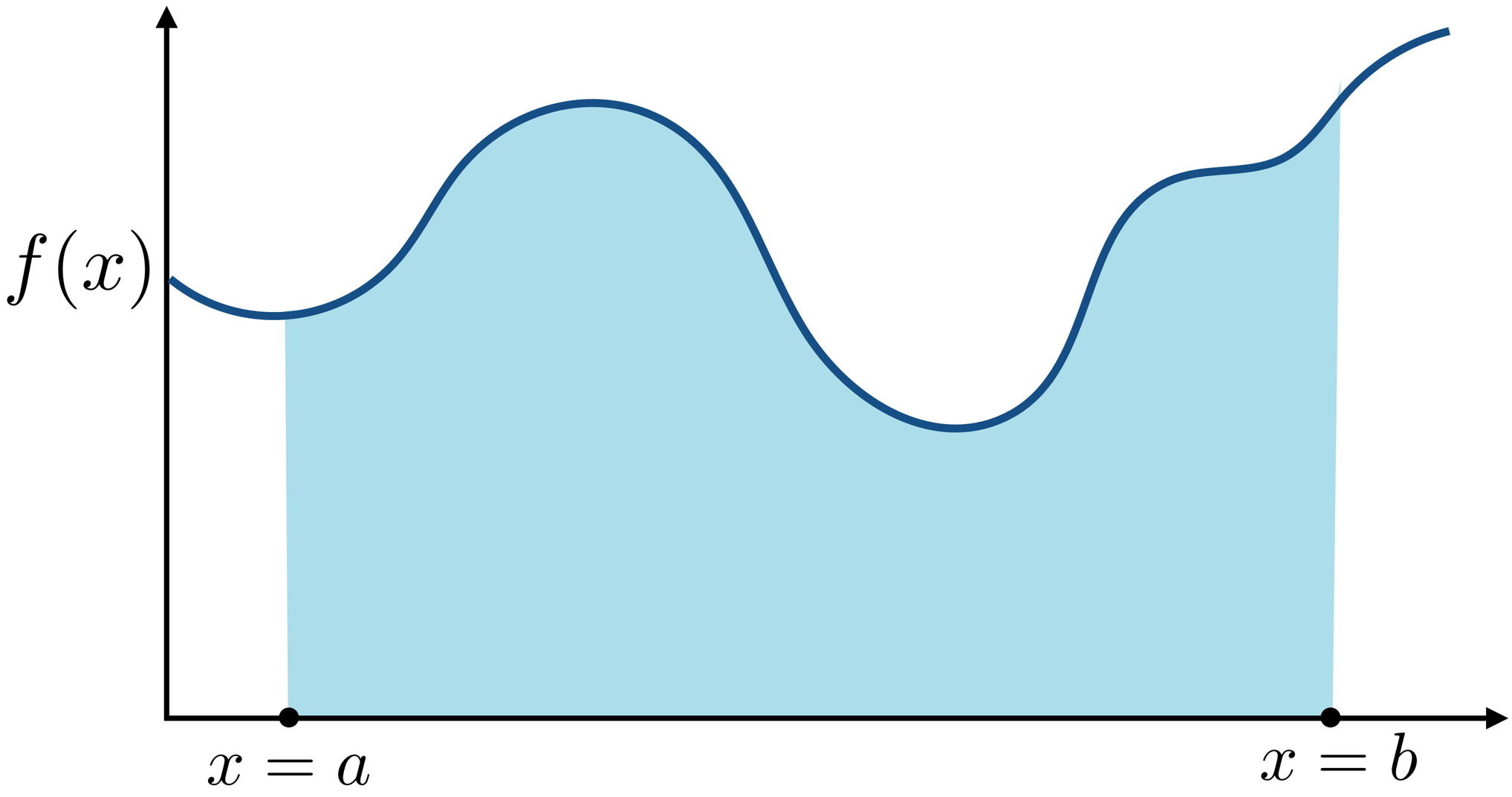
Lecture 12:

Monte Carlo Integration

Computer Graphics and Imaging
UC Berkeley CS184/284A

Definite Integral as "Area under the Curve"

$$\int_a^b f(x)dx$$

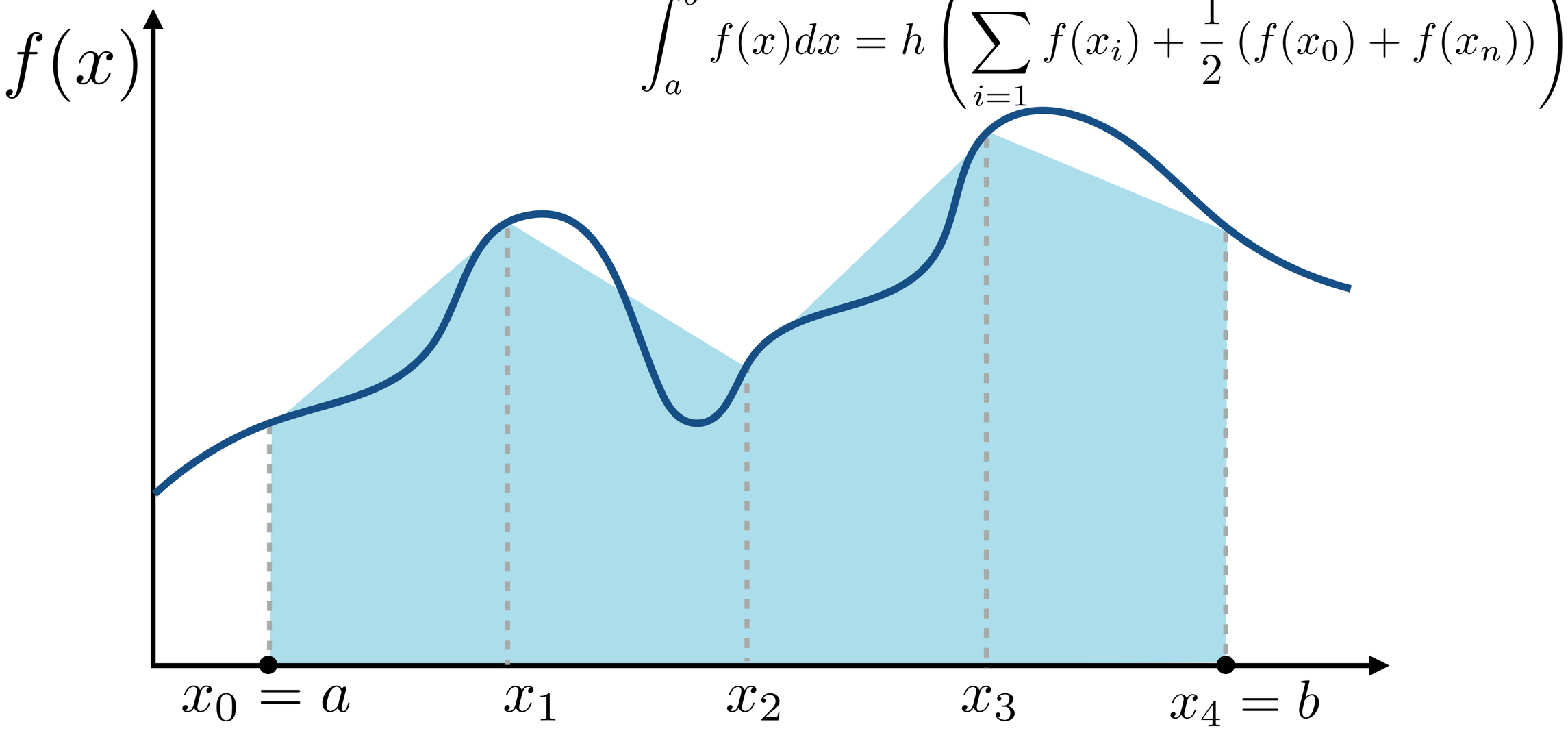


Numerical Integration: e.g. Trapezoidal Rule

Approximate integral of f(x) by assuming function is piecewise linear

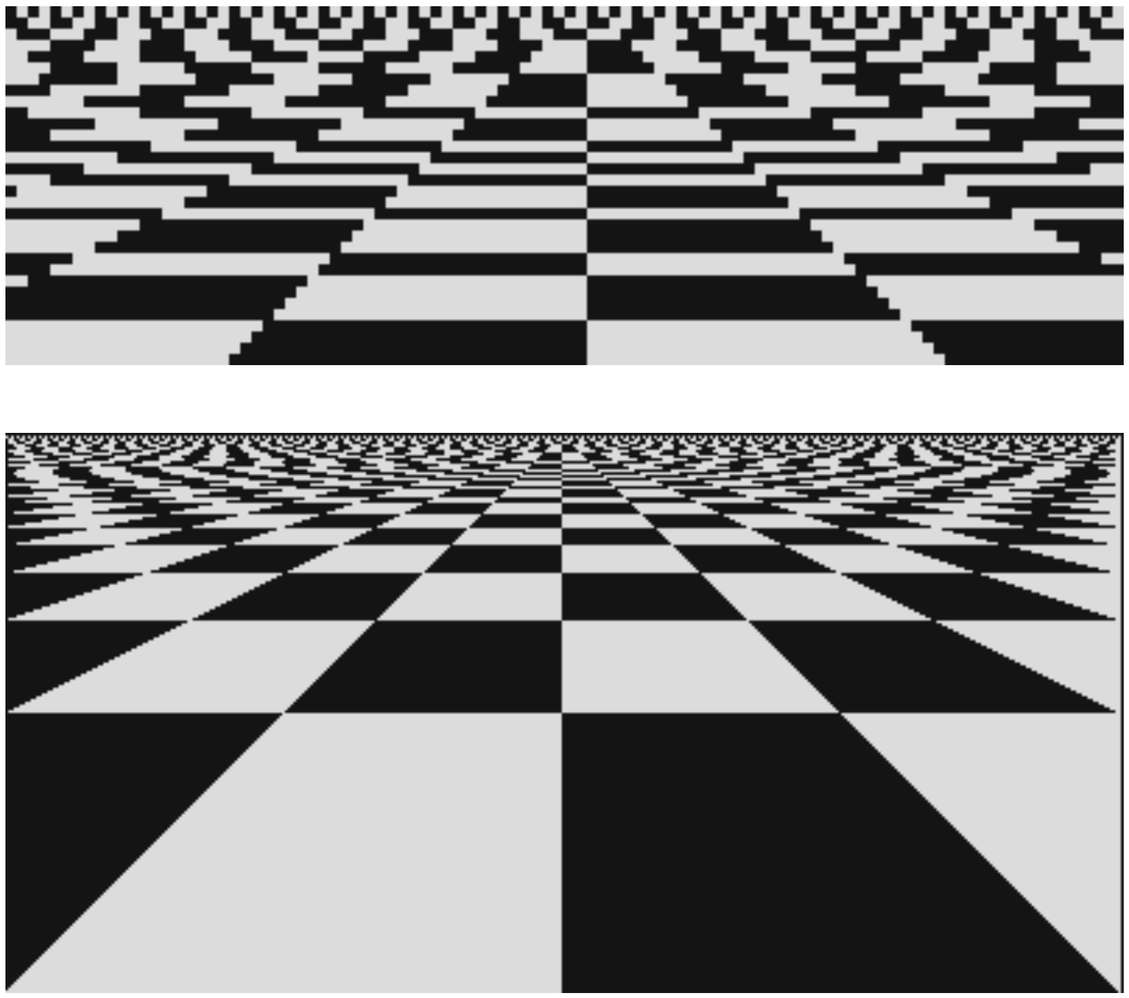
For equal length segments: $h = \frac{b-a}{n-1}$

$$\int_a^b f(x)dx = h\left(\sum_{i=1}^{n-1} f(x_i) + \frac{1}{2}\left(f(x_0) + f(x_n)\right)\right)$$

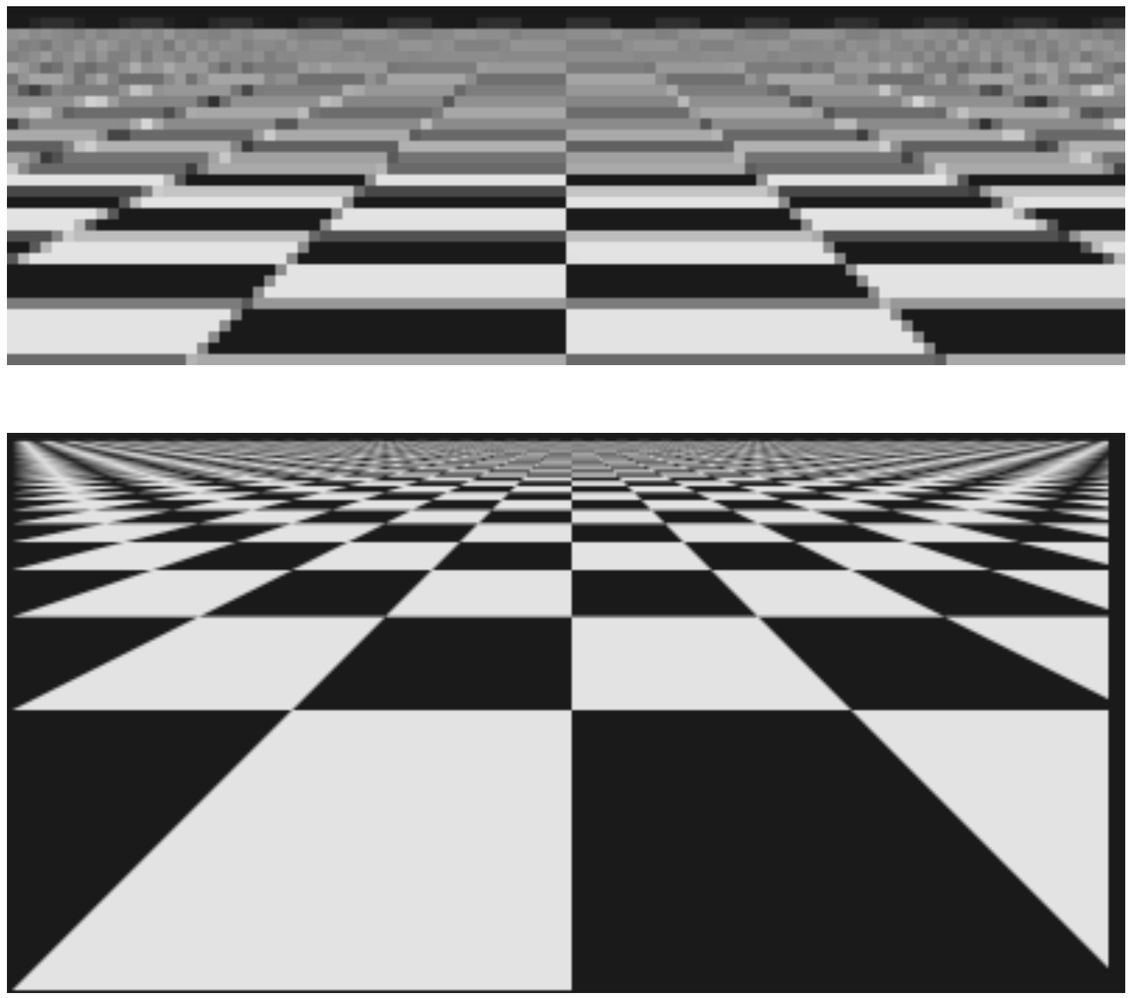


Multi-Dimensional Integrals (Rendering Examples)

2D Integral: Recall Antialiasing By Area Sampling



Point Sampling

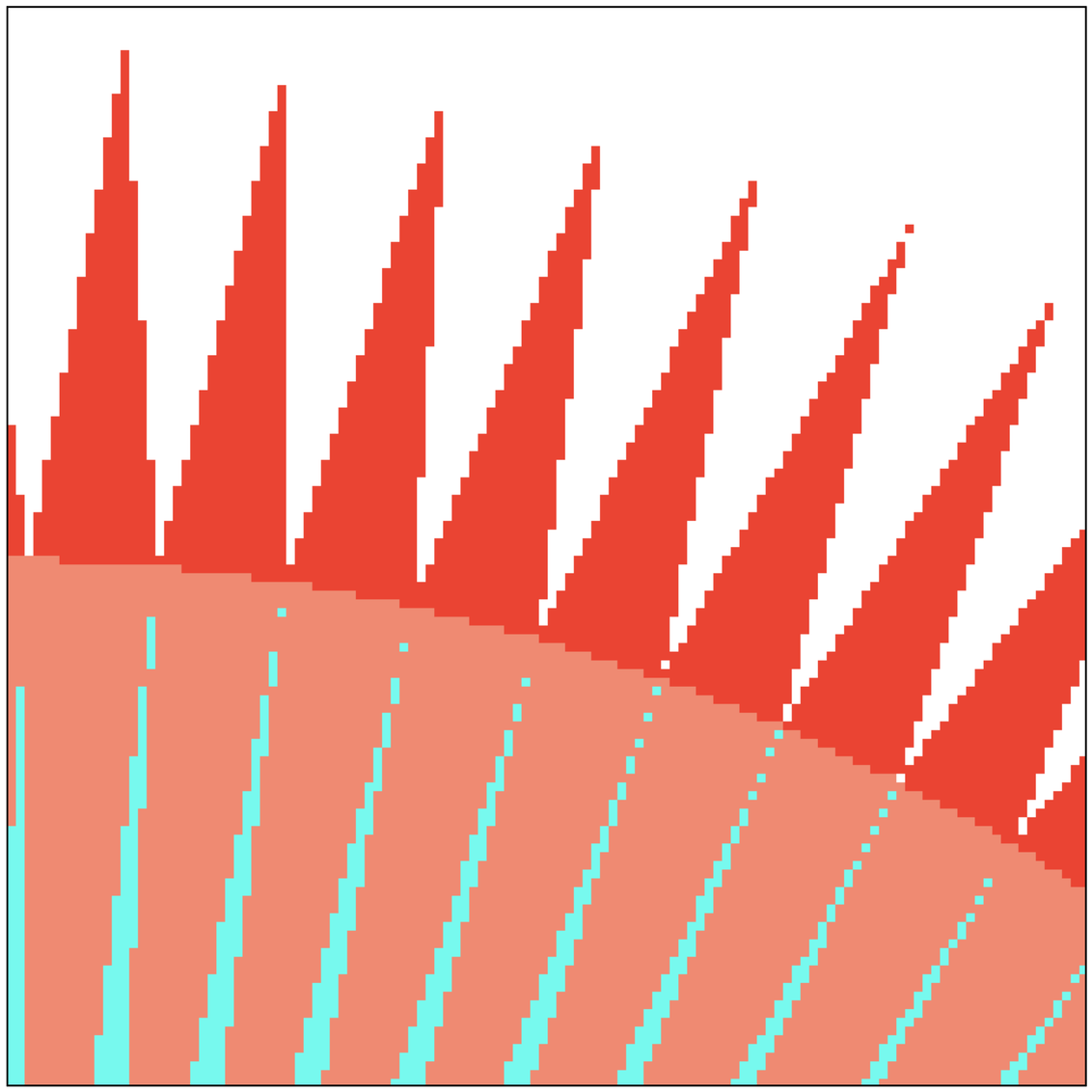


Area Sampling

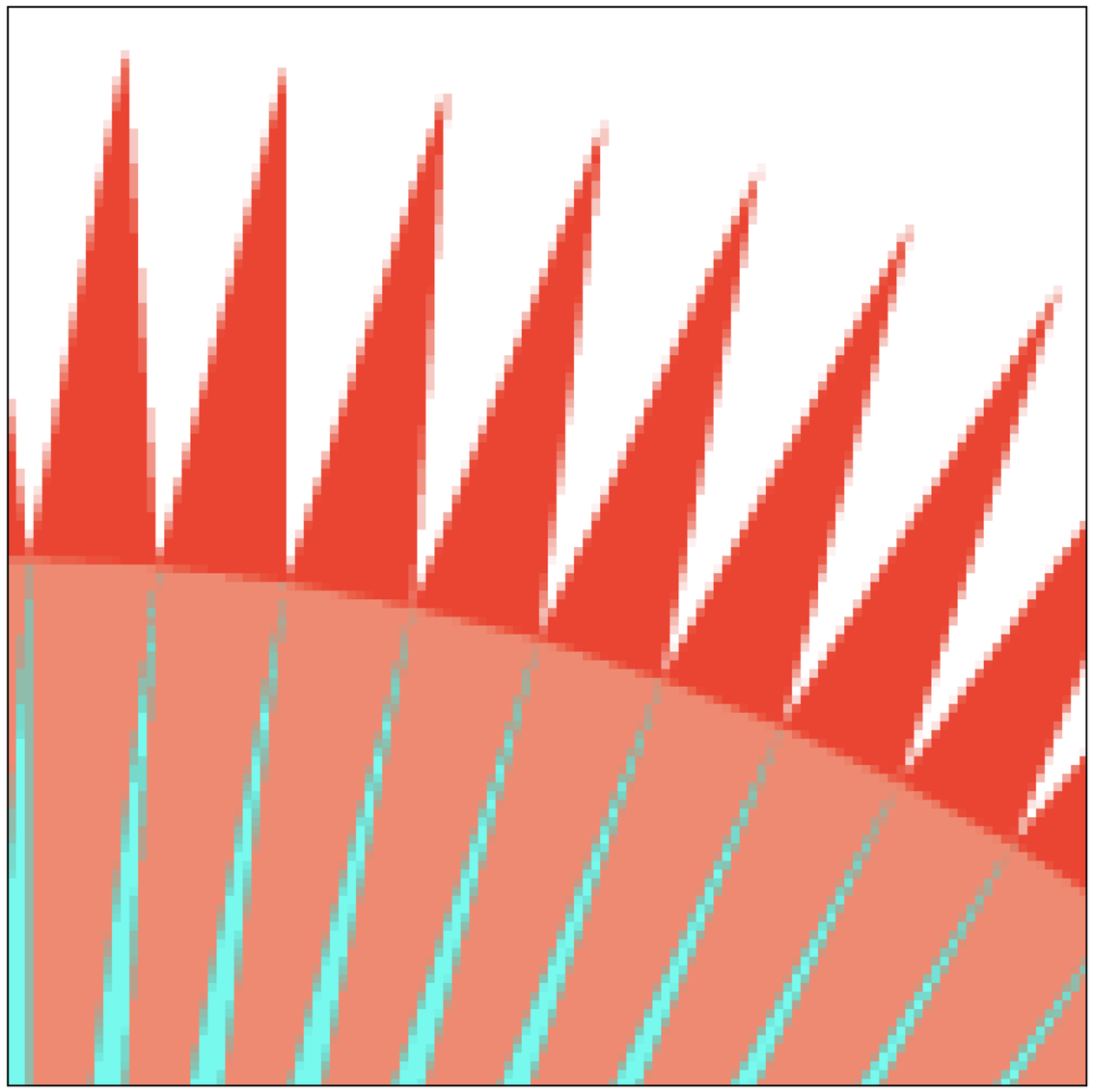
Checkerboard sequence by Tom Duff

Integrate over 2D area of pixel

2D Integral: Recall Antialiasing By Area Sampling



Point sampling



Area sampling

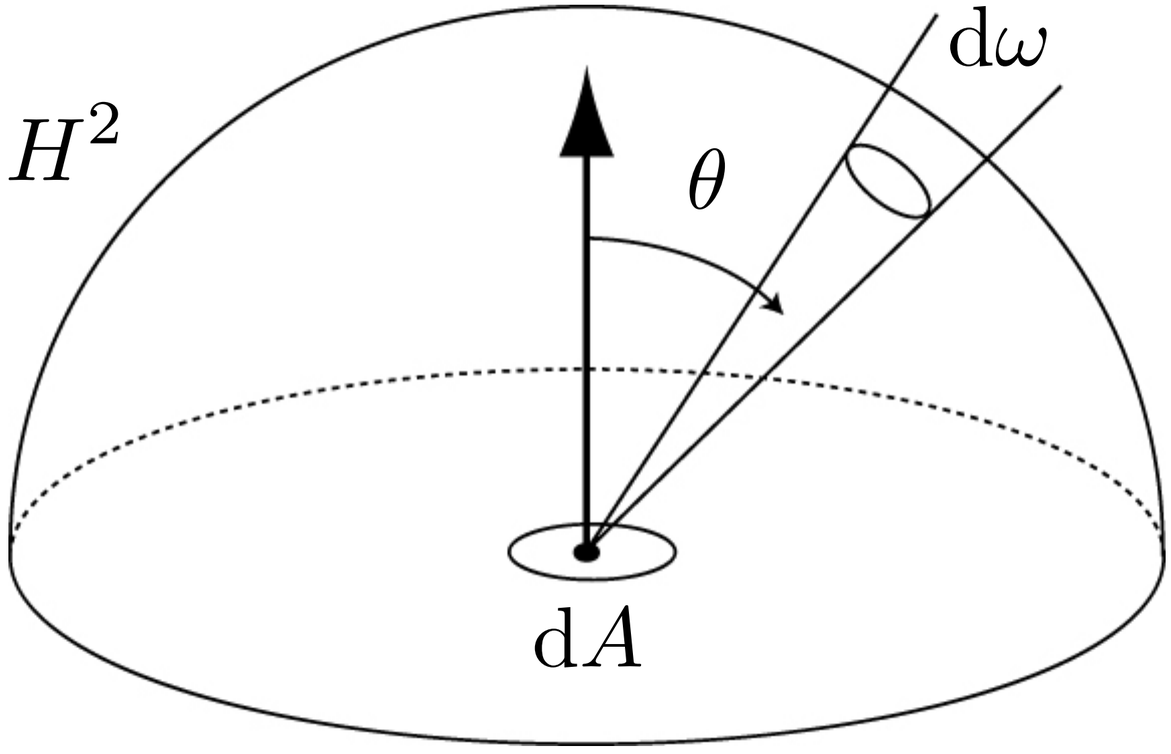
Integrate over 2D area of pixel

2D Integral: Irradiance from the Environment

Computing flux per unit area on surface, due to incoming light from all directions.

$$E(\mathrm{p}) = \int_{H^2} L_i(\mathrm{p}, \omega) \cos\theta \, \mathrm{d}\omega$$

Contribution to irradiance from light arriving from direction ω



Hemisphere: H^2



Light meter

3D Integral: Motion Blur



Cook et al. “1984”

Integrate over area of pixel and over exposure time.

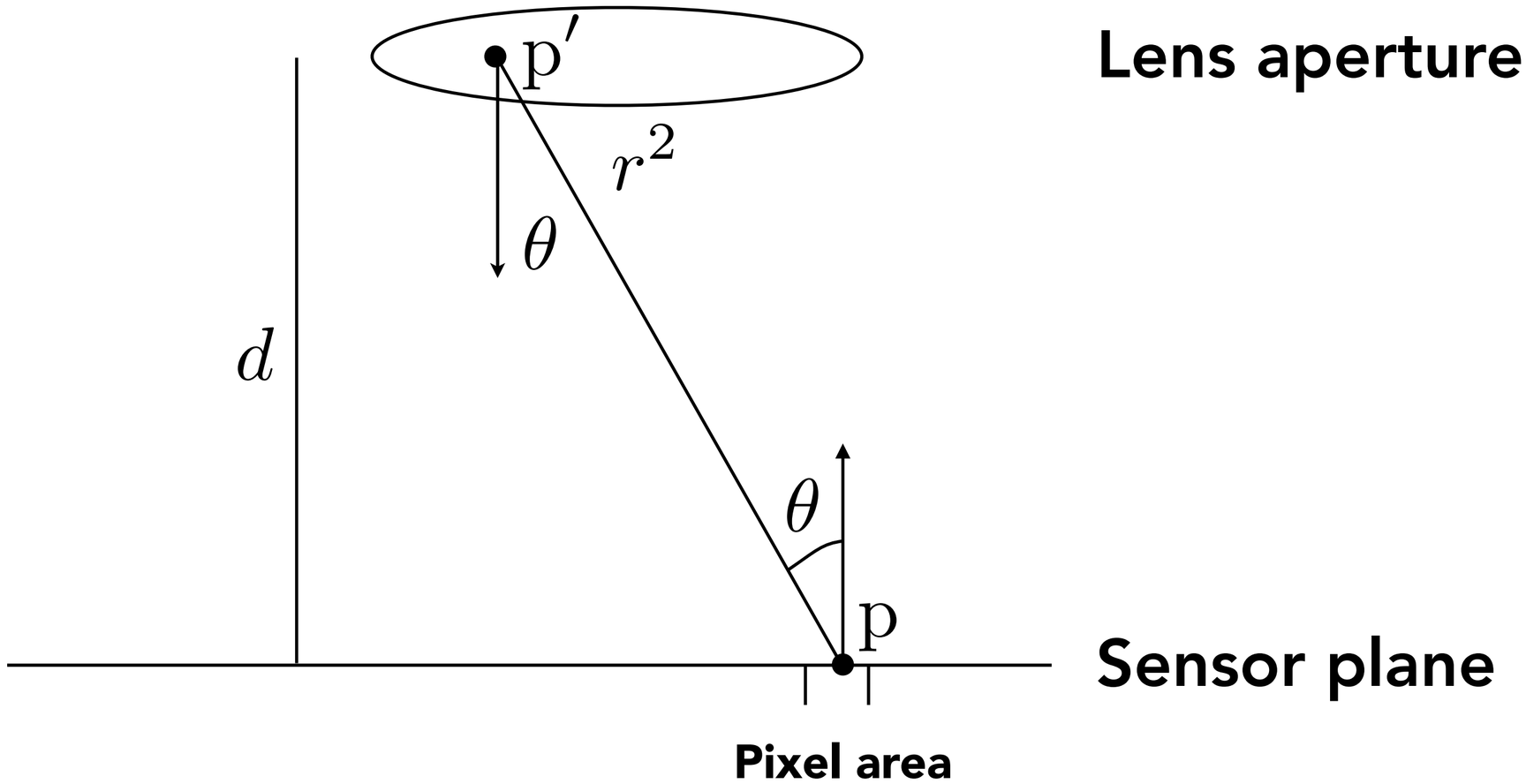
5D Integral: Real Camera Pixel Exposure



Credit: lycheng99, http://flic.kr/p/x4DEZh

Integrate over 2D lens pupil, 2D pixel, and over exposure time

5D Integral: Real Camera Pixel Exposure



$$Q_{\text{pixel}} = \frac{1}{d^2} \int_{t_0}^{t^1} \int_{A_{\text{lens}}} \int_{A_{\text{pixel}}} L(\text{p}' \to \text{p}, t) \cos^4 \theta \, \text{d}p \, \text{d}p' \, \text{d}t$$

The Curse of Dimensionality

High-Dimensional Integration

High-Dimensional Integration

Complete set of samples:

$$N = \underbrace{n \times n \times \cdots \times n}_{d} = n^d$$

- "Curse of dimensionality"

High-Dimensional Integration

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- **"Curse of dimensionality"**

Numerical integration error: $\text{Error} \sim \frac{1}{n} = \frac{1}{N^{1/d}}$

High-Dimensional Integration

Complete set of samples: $N = \underbrace{n \times n \times \cdots \times n}_{d} = n^d$

- **"Curse of dimensionality"**

Numerical integration error: $\text{Error} \sim \frac{1}{n} = \frac{1}{N^{1/d}}$

Random sampling error: $\text{Error} = \text{Variance}^{1/2} \sim \frac{1}{\sqrt{N}}$

High-Dimensional Integration

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In high dimensions, Monte Carlo integration requires fewer samples than quadrature-based numerical integration

High-Dimensional Integration

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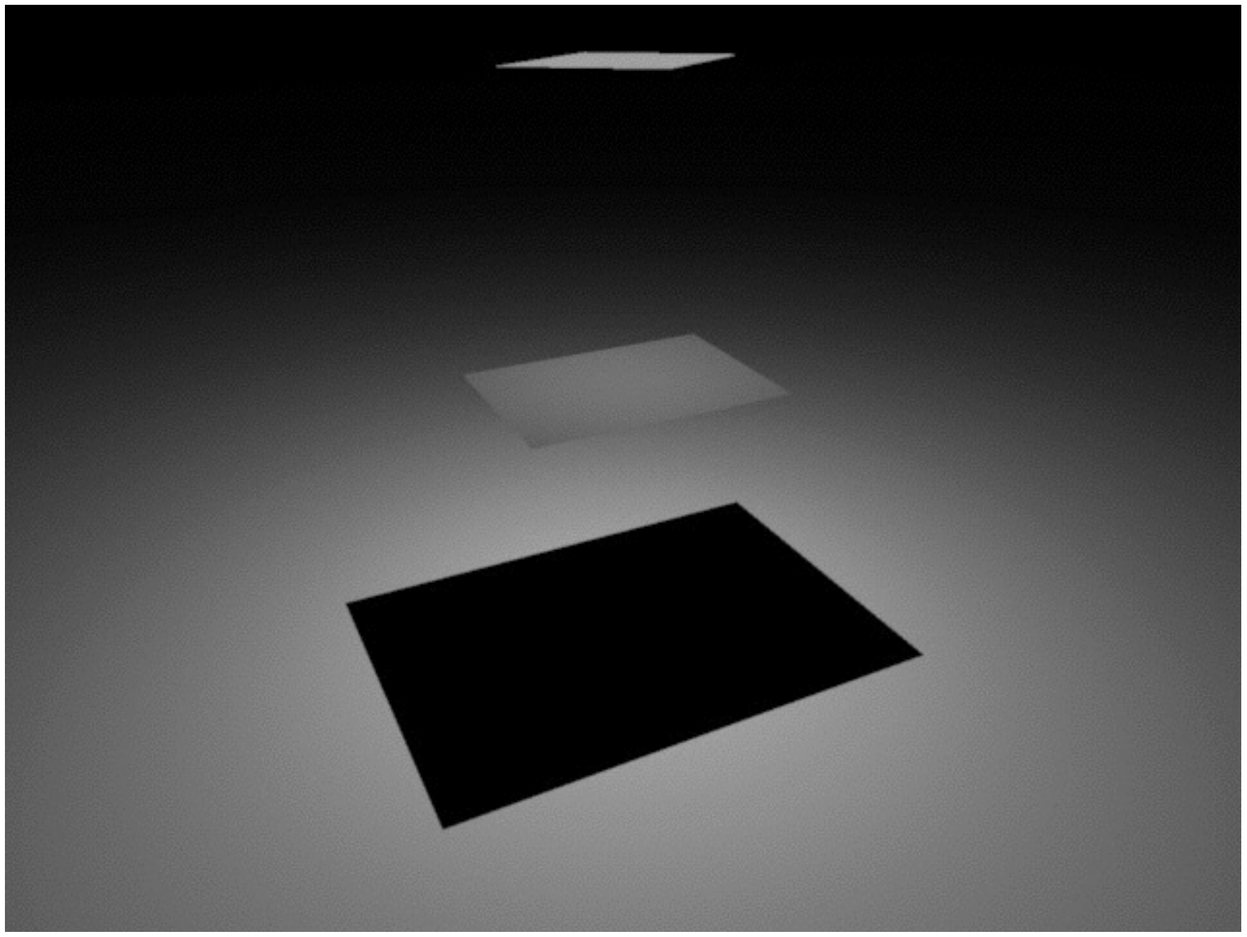
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In high dimensions, Monte Carlo integration requires fewer samples than quadrature-based numerical integration

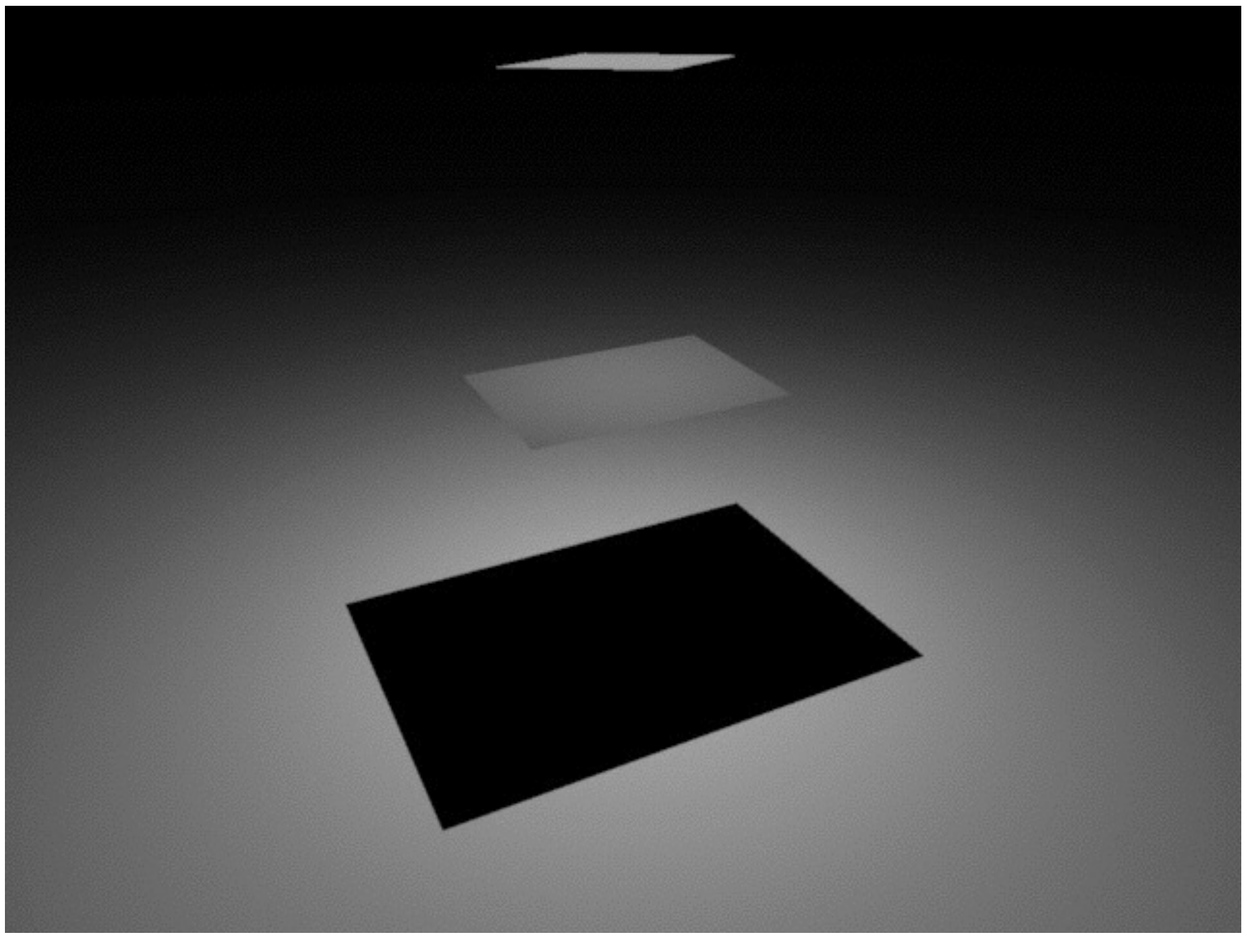
Global illumination = infinite-dimensional integrals

Example: Discrete vs Monte Carlo - Shadows

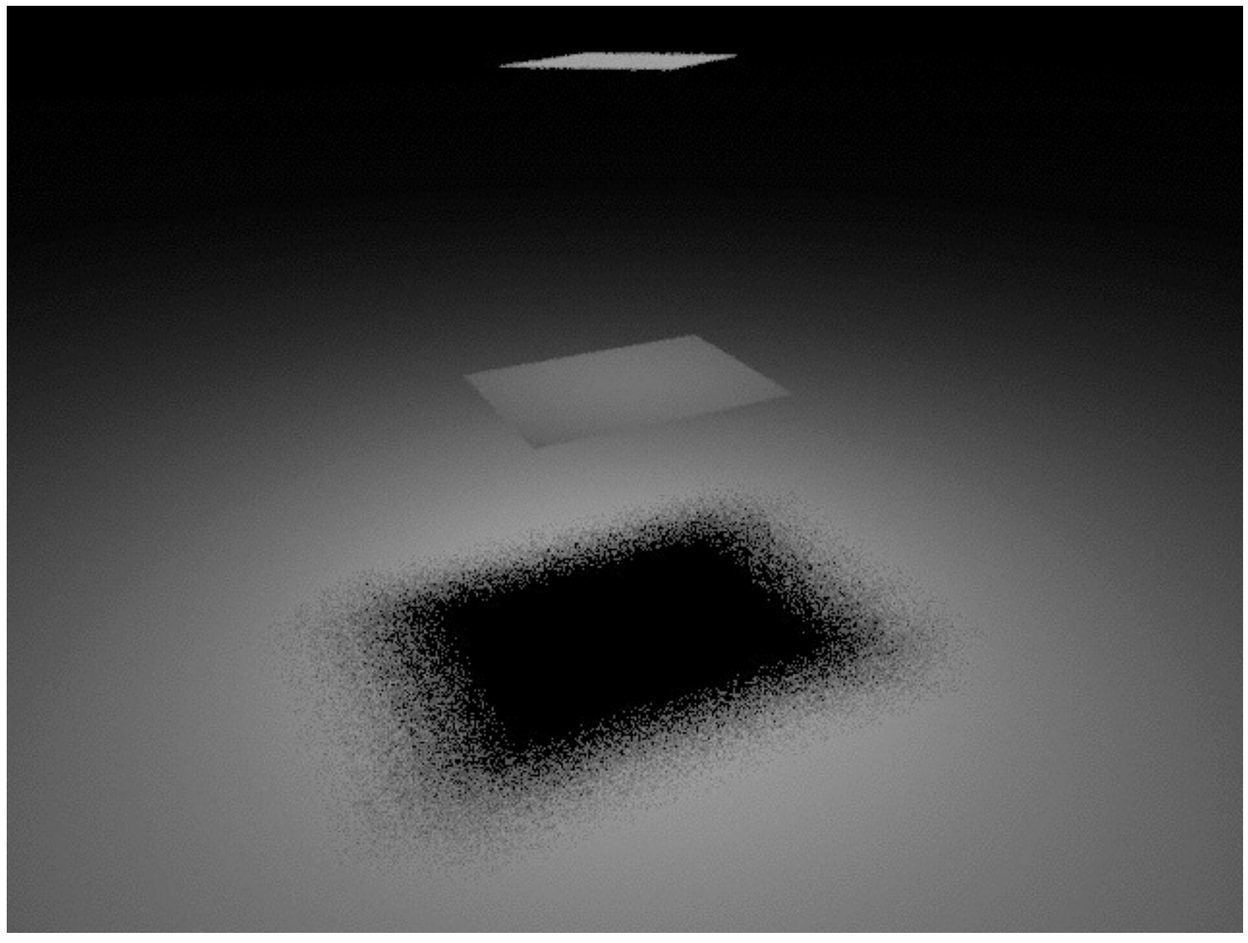


1 sample per pixel
Sample center of light

Example: Discrete vs Monte Carlo - Shadows

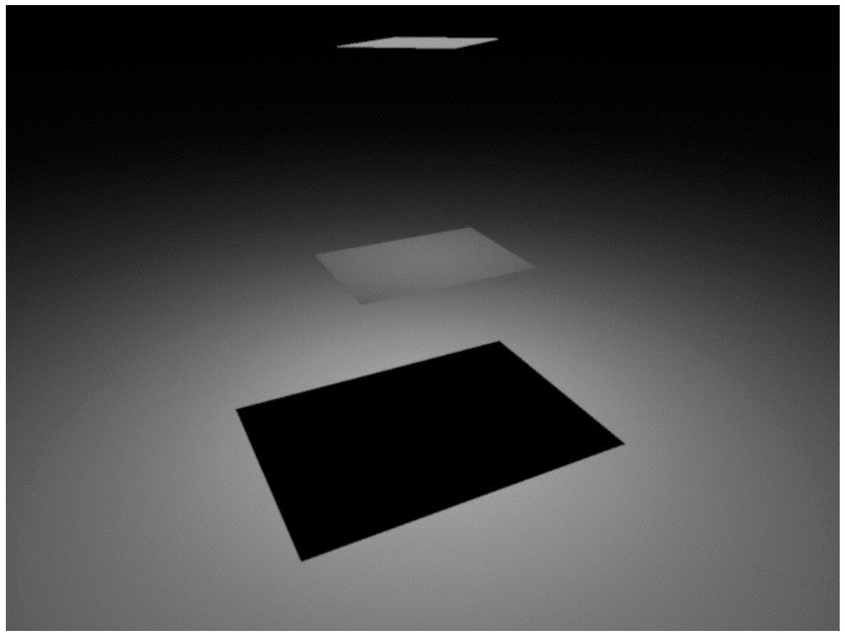


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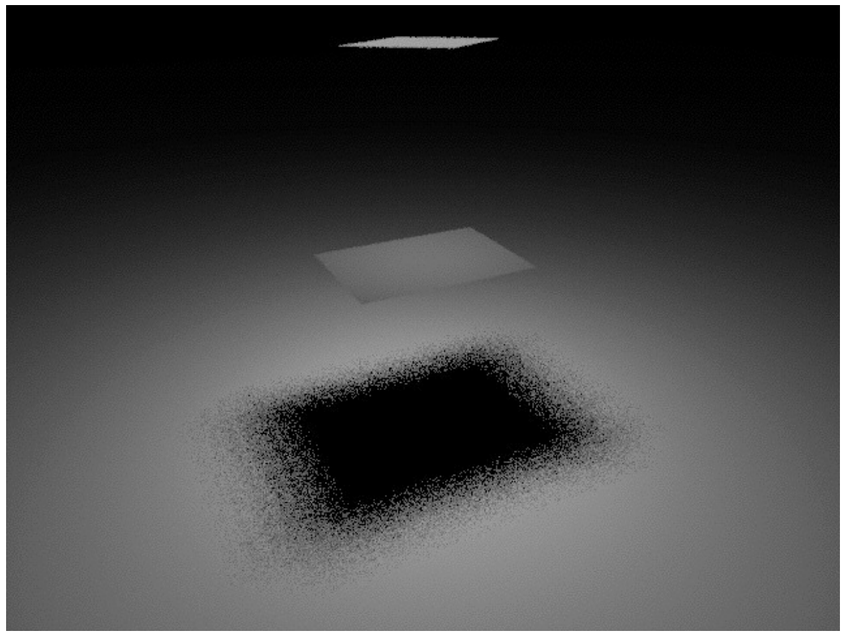


1 sample per pixel
Sample random point on light

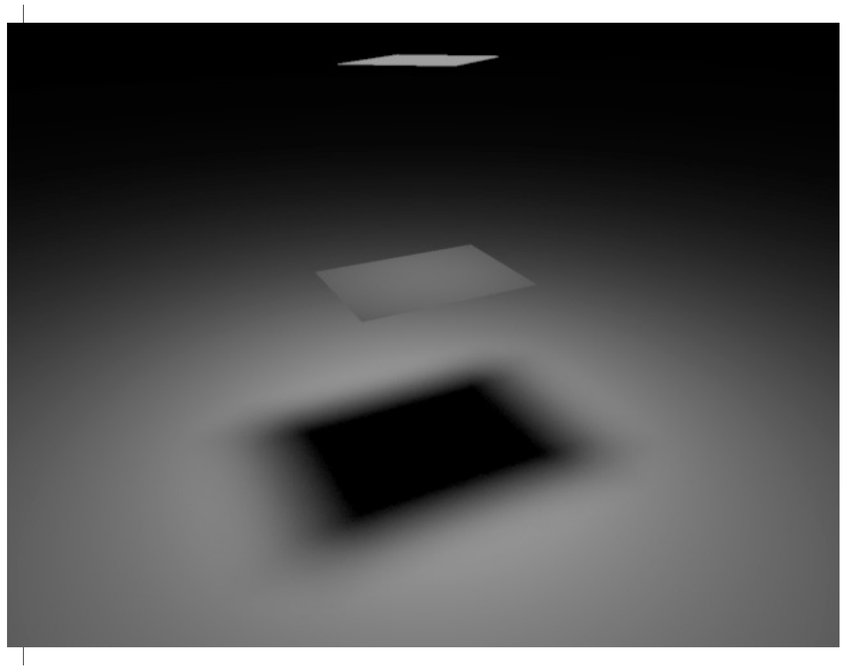
Example: Discrete vs Monte Carlo - Shadows



Sample center of light



Sample random point on light



True answer

Overview: Monte Carlo Integration

Overview: Monte Carlo Integration

Idea: estimate integral based on random sampling of function

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Idea: estimate integral based on random sampling of function

Advantages:

- General and relatively simple method
- Requires only function evaluation at any point
- Works for very general functions, including discontinuities
- Efficient for high-dimensional integrals — avoids “curse of dimensionality”

Overview: Monte Carlo Integration

Idea: estimate integral based on random sampling of function

Advantages:

- General and relatively simple method
- Requires only function evaluation at any point
- Works for very general functions, including discontinuities
- Efficient for high-dimensional integrals — avoids “curse of dimensionality”

Disadvantages:

- Noise. Integral estimate is random, only correct “on average”
- Can be slow to converge — need a lot of samples

Probability Review

Random Variables

Random Variables

X random variable. Represents a distribution of potential values

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$X \sim p(x)$ probability density function (PDF). Describes relative probability of a random process choosing value x

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X random variable. Represents a distribution of potential values

$X \sim p(x)$ probability density function (PDF). Describes relative probability of a random process choosing value x

Example: uniform PDF: all values over a domain are equally likely

e.g. A six-sided die

X **takes on values 1, 2, 3, 4, 5, 6**

$$p(1) = p(2) = p(3) = p(4) = p(5) = p(6)$$

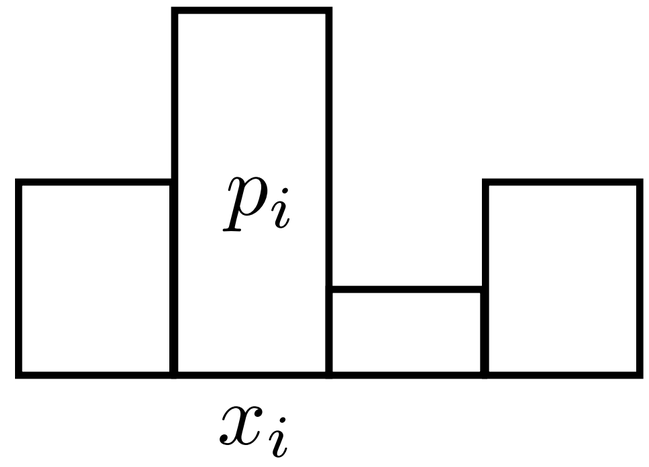


Probability Distribution Function (PDF)

Probability Distribution Function (PDF)

n **discrete values** x_i

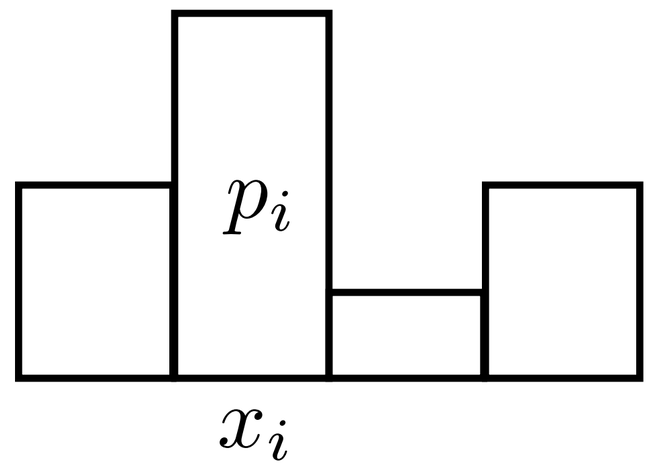
With probability p_i



Probability Distribution Function (PDF)

n **discrete values** x_i

With probability p_i



Requirements of a probability distribution:

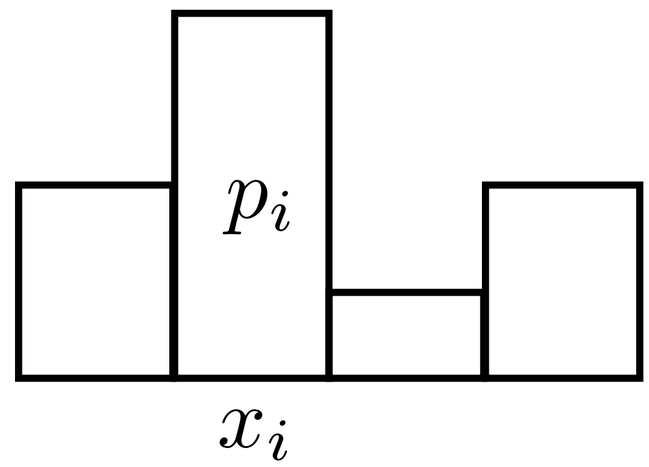
$$p_i \geq 0$$

$$\sum_{i=1}^{n} p_i = 1$$

Probability Distribution Function (PDF)

n **discrete values** x_i

With probability p_i



Requirements of a probability distribution:

$$p_i \geq 0$$

$$\sum_{i=1}^{n} p_i = 1$$



Six-sided die example: $p_i = \frac{1}{6}$

Think: p_i **is the probability that a random measurement of** X **will yield the value** x_i

X **takes on the value** x_i **with probability** p_i

Expected Value of a Random Variable

Expected Value of a Random Variable

The average value that one obtains if repeatedly drawing samples from the random distribution.

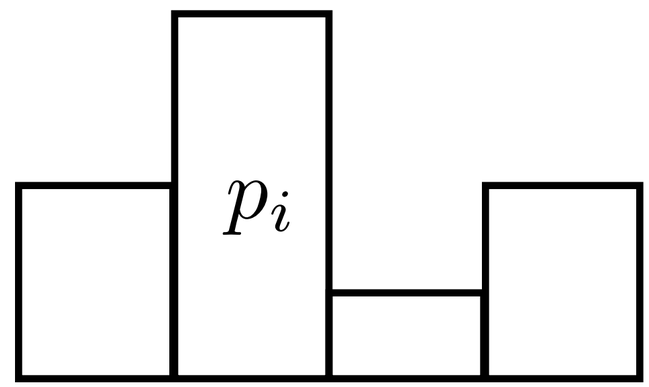
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X **drawn from distribution with**

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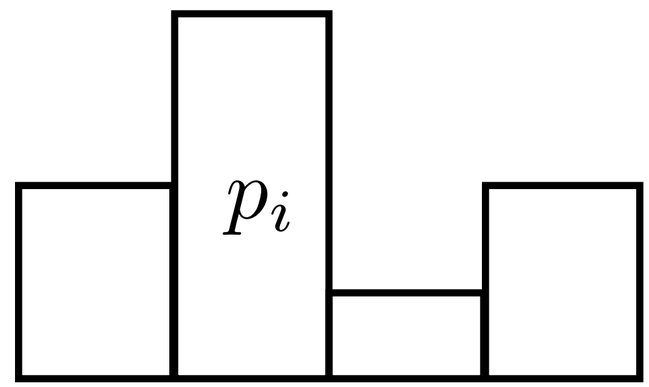
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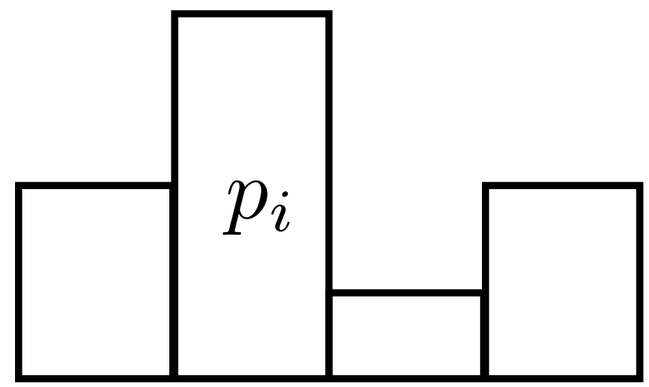


Expected value of X: $E[X] = \sum_{i=1}^{n} x_i p_i$ x_i

Expected Value of a Random Variable

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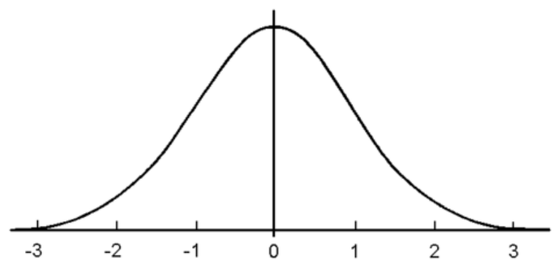
Die example: $E[X] = \sum_{i=1}^{n} \frac{i}{6}$



$$= (1 + 2 + 3 + 4 + 5 + 6)/6 = 3.5$$

Continuous Probability Distribution Function

$$X \sim p(x)$$



A random variable X that can take any of a continuous set of values, where the relative probability of a particular value is given by a continuous probability density function p(x).

Conditions on p(x): $p(x) \geq 0 \text{ and } \int p(x)\,dx = 1$

Expected value of X: $E[X] = \int x\,p(x)\,dx$

Function of a Random Variable

A function Y of a random variable X is also a random variable:

$$X \sim p(x)$$
$$Y = f(X)$$

Expected value of a function of a random variable:

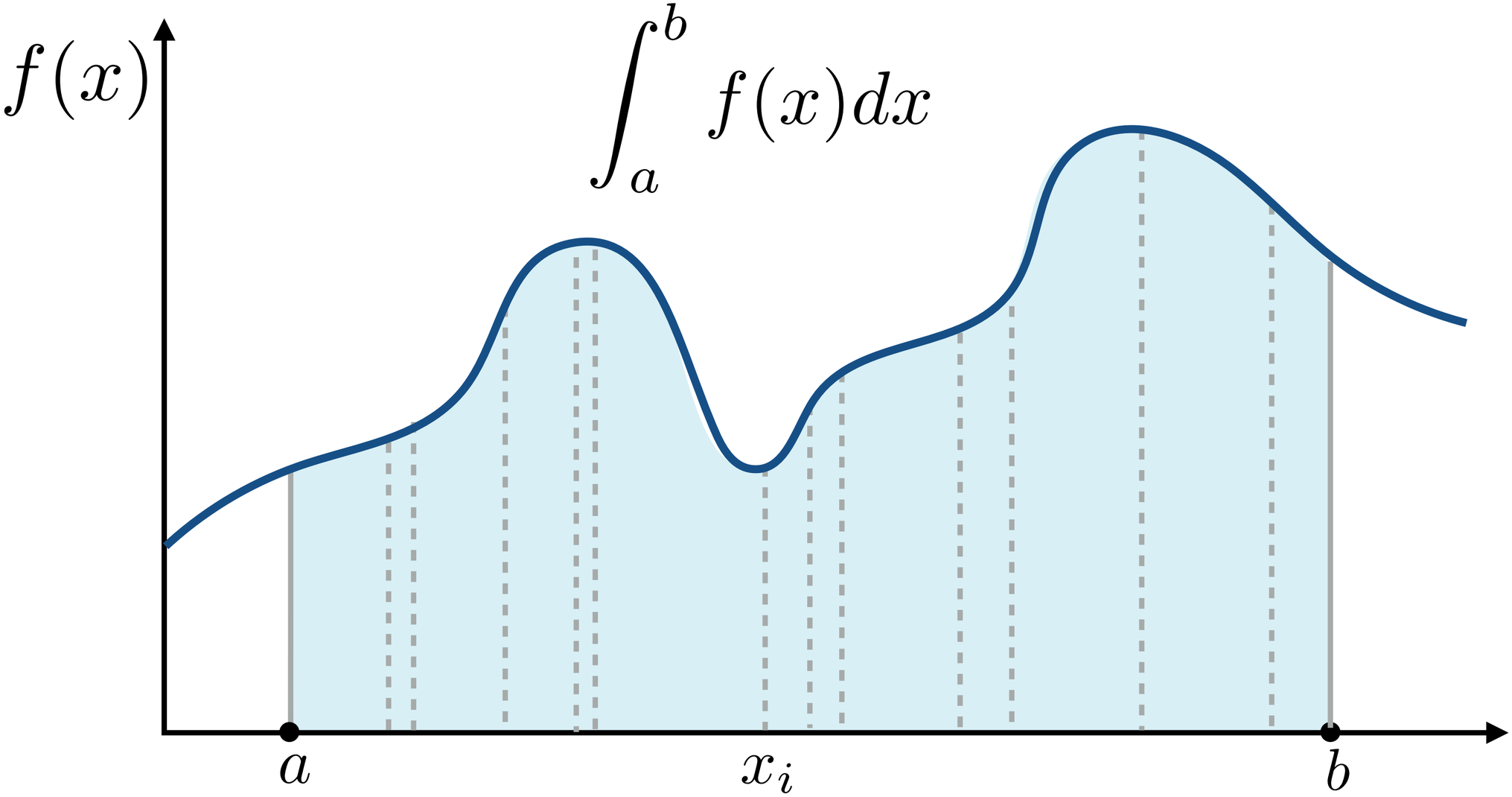
$$E[Y] = E[f(X)] = \int f(x)\, p(x)\, dx$$

Monte Carlo Integration

Monte Carlo Integration

Simple idea: estimate the integral of a function by averaging random samples of the function's value.

$$\int_a^b f(x)dx$$



Monte Carlo Integration

Let us define the Monte Carlo estimator for the definite integral of given function $f(x)$

Definite integral

$$\int_a^b f(x)dx$$

Random variable

$$X_i \sim p(x)$$

Note: p(x) must be nonzero for all x where f(x) is nonzero

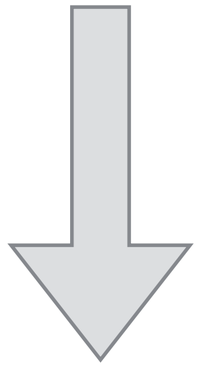
Monte Carlo estimator

$$F_N = \frac{1}{N}\sum_{i=1}^{N}\frac{f(X_i)}{p(X_i)}$$

Example: Basic Monte Carlo Estimator

The basic Monte Carlo estimator is a simple special case where we sample with a uniform random variable

Uniform random variable $X_i \sim p(x) = \frac{1}{b-a}$



Basic Monte Carlo estimator $F_N = \frac{b-a}{N} \sum_{i=1}^{N} f(X_i)$

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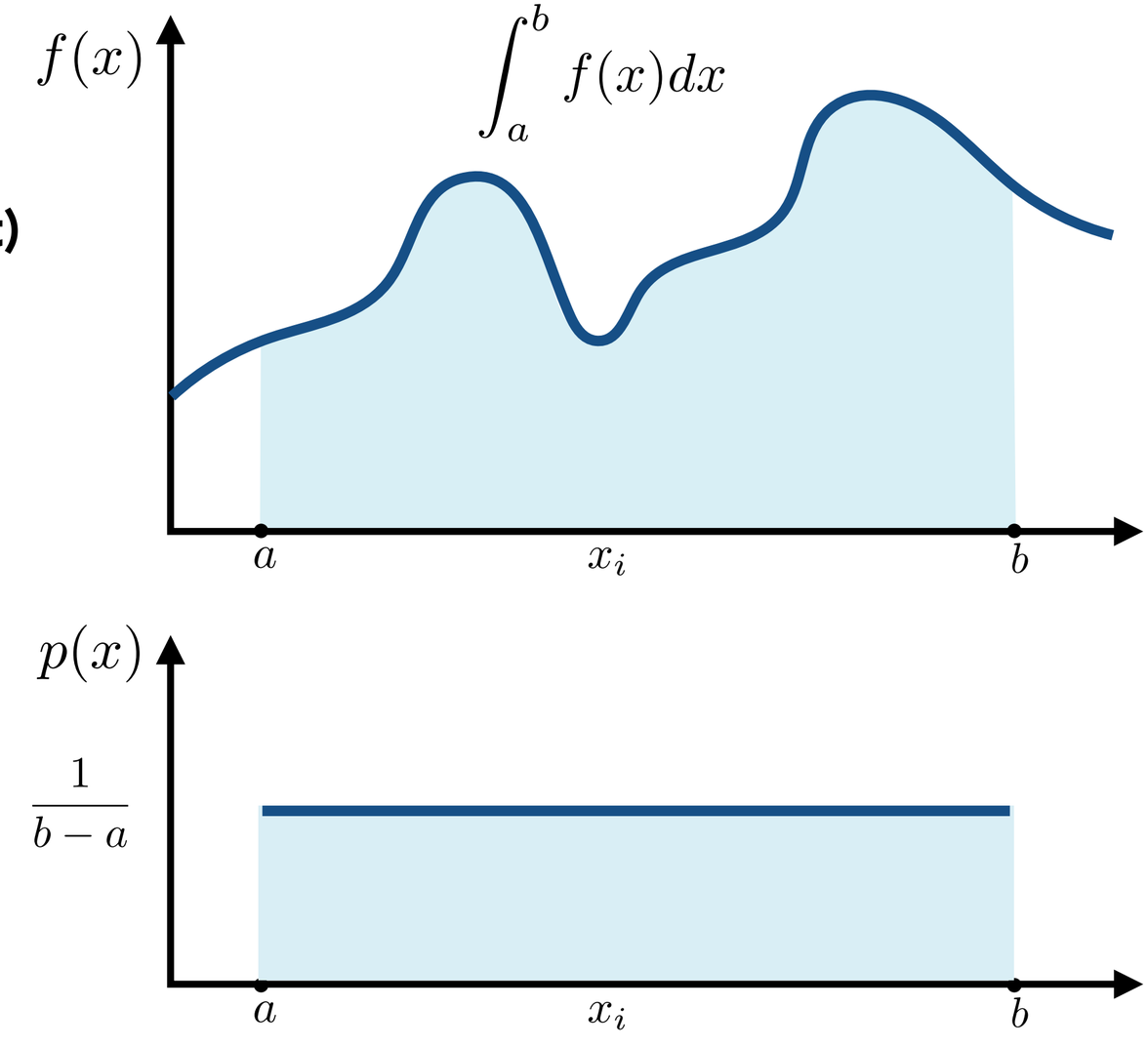
Uniform random variable

$$X_i \sim p(x) = C \text{ (constant)}$$

$$\int_a^b p(x)\,dx = 1$$

$$\implies \int_a^b C\,dx = 1$$

$$\implies C = \frac{1}{b-a}$$



Example: Basic Monte Carlo Estimator

The basic Monte Carlo estimator is a simple special case where we sample with a uniform random variable

Basic Monte Carlo estimator (derivation)

$$F_N = \frac{1}{N}\sum_{i=1}^{N}\frac{f(X_i)}{p(X_i)} \quad \textbf{(MC Estimator)}$$

$$= \frac{1}{N}\sum_{i=1}^{N}\frac{f(X_i)}{1/(b-a)}$$

$$= \frac{b-a}{N}\sum_{i=1}^{N} f(X_i)$$

Example: Basic Monte Carlo Estimator

Let us define the Monte Carlo estimator for the definite integral of given function $f(x)$

Definite integral $$\int_a^b f(x)dx$$

Uniform random variable $$X_i \sim p(x) = \frac{1}{b-a}$$

Basic Monte Carlo estimator $$F_N = \frac{b-a}{N}\sum_{i=1}^{N} f(X_i)$$

Unbiased Estimator

Definition: A randomized integral estimator is *unbiased* if its expected value is the desired integral.

Fact: the general and basic Monte Carlo estimators are unbiased (proof on next slide)

Why do we want unbiased estimators?

Proof That Monte Carlo Estimator Is Unbiased

$$E[F_N] = E\left[\frac{1}{N}\sum_{i=1}^{N}\frac{f(X_i)}{p(X_i)}\right]$$

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Properties of expected values:

$$E\left[\sum_i Y_i\right] = \sum_i E[Y_i]$$

$$E[aY] = aE[Y]$$

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\begin{aligned}
E[F_N] &= E\left[\frac{1}{N}\sum_{i=1}^{N}\frac{f(X_i)}{p(X_i)}\right] \\
&= \frac{1}{N}\sum_{i=1}^{N} E\left[\frac{f(X_i)}{p(X_i)}\right] \\
&= \frac{1}{N}\sum_{i=1}^{N}\int_a^b \frac{f(x)}{p(x)}\, p(x)\mathrm{d}x
\end{aligned}
$$

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\end{aligned}
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Properties of expected values:

$$E\left[\sum_i Y_i\right] = \sum_i E[Y_i]$$

$$E[aY] = aE[Y]$$

The expected value of the Monte Carlo estimator is the desired integral.

Variance of a Random Variable

Definition

$$V[Y] = E[(Y - E[Y])^2]$$
$$= E[Y^2] - E[Y]^2$$

Because $E[Y]E[Y] = (E[Y])^2$

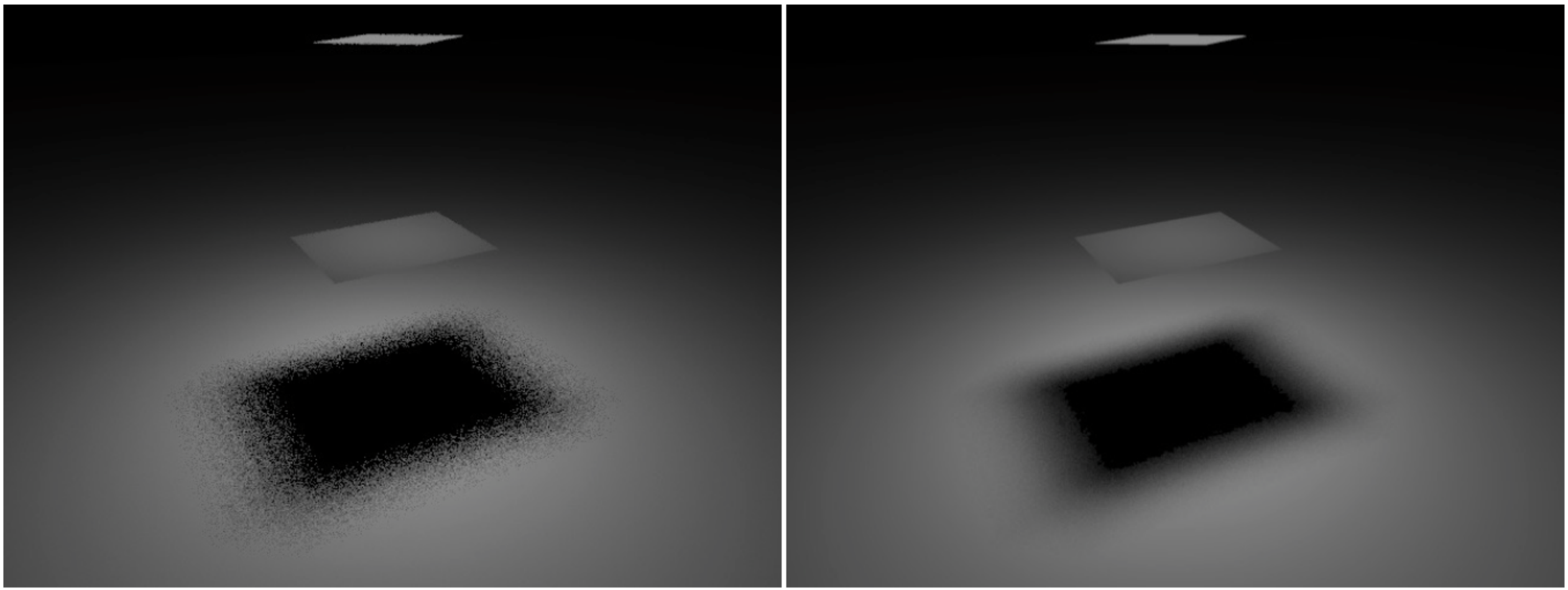
Properties of variance

$$V\left[\sum_{i=1}^{N} Y_i\right] = \sum_{i=1}^{N} V[Y_i] \qquad V[aY] = a^2\, V[Y]$$

Variance decreases linearly with number of samples

$$V\left[\frac{1}{N}\sum_{i=1}^{N} Y_i\right] = \frac{1}{N^2}\sum_{i=1}^{N} V[Y_i] = \frac{1}{N^2} N\, V[Y] = \frac{1}{N} V[Y]$$

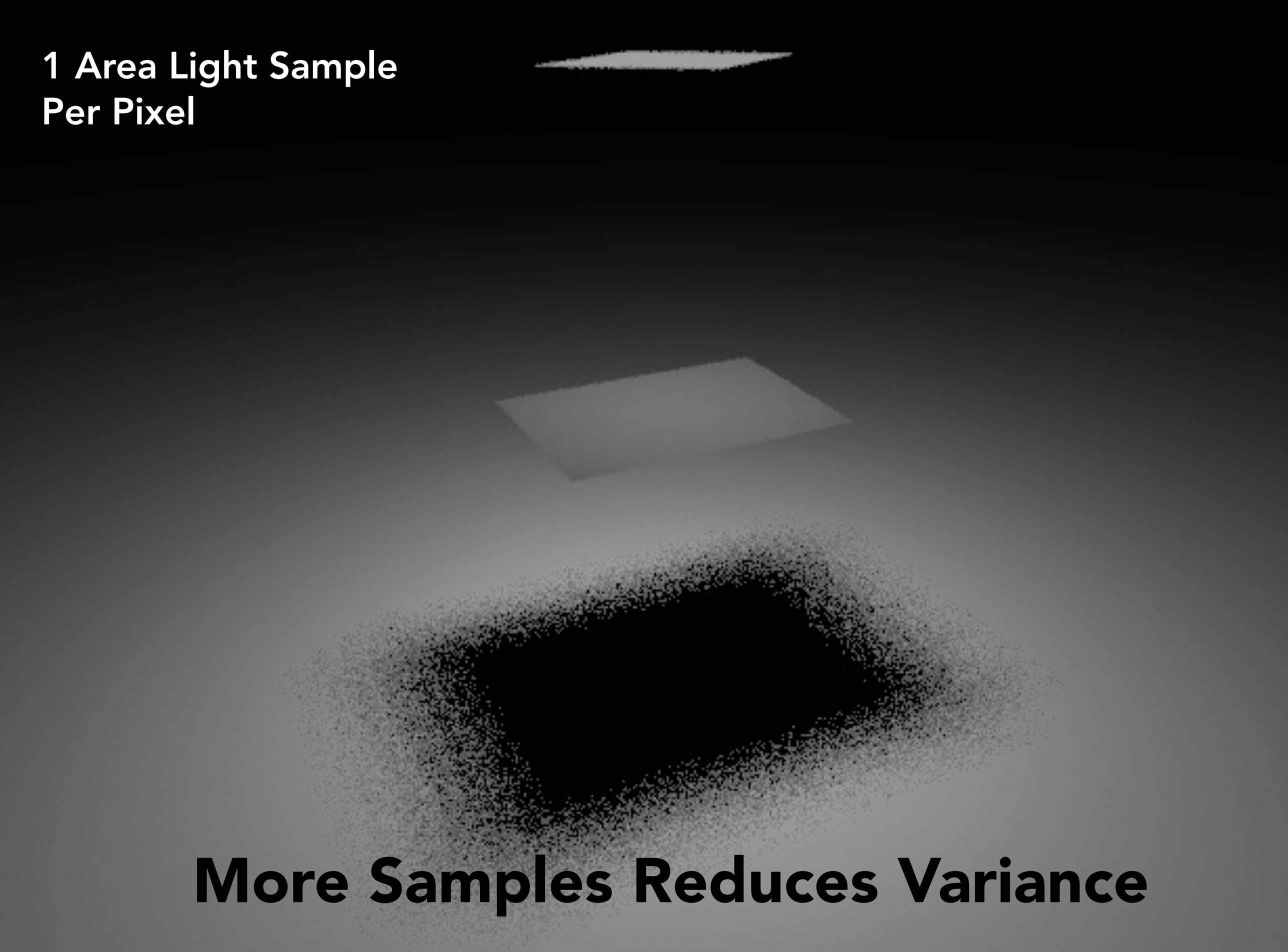
More Random Samples Reduces Variance



1 shadow ray

16 shadow rays

1 Area Light Sample
Per Pixel



More Samples Reduces Variance

16 Area Light Samples
Per Pixel

More Samples Reduces Variance

Definite Integral Can Be N-Dimensional

Example in 3D:

$$\int_{x_0}^{x_1} \int_{y_0}^{y_1} \int_{z_0}^{z_1} f(x, y, z)\, dx\, dy\, dz$$

Definite Integral Can Be N-Dimensional

Example in 3D:

$$\int_{x_0}^{x_1} \int_{y_0}^{y_1} \int_{z_0}^{z_1} f(x, y, z)\, dx\, dy\, dz$$

Uniform 3D random variable*

$$X_i \sim p(x, y, z) = \frac{1}{x_1 - x_0} \frac{1}{y_1 - y_0} \frac{1}{z_1 - z_0}$$

*** Generalizes to arbitrary N-dimensional PDFs**

Definite Integral Can Be N-Dimensional

Example in 3D:

$$\int_{x_0}^{x_1} \int_{y_0}^{y_1} \int_{z_0}^{z_1} f(x, y, z)\, dx\, dy\, dz$$

Uniform 3D random variable*

$$X_i \sim p(x, y, z) = \frac{1}{x_1 - x_0} \frac{1}{y_1 - y_0} \frac{1}{z_1 - z_0}$$

Basic 3D MC estimator*

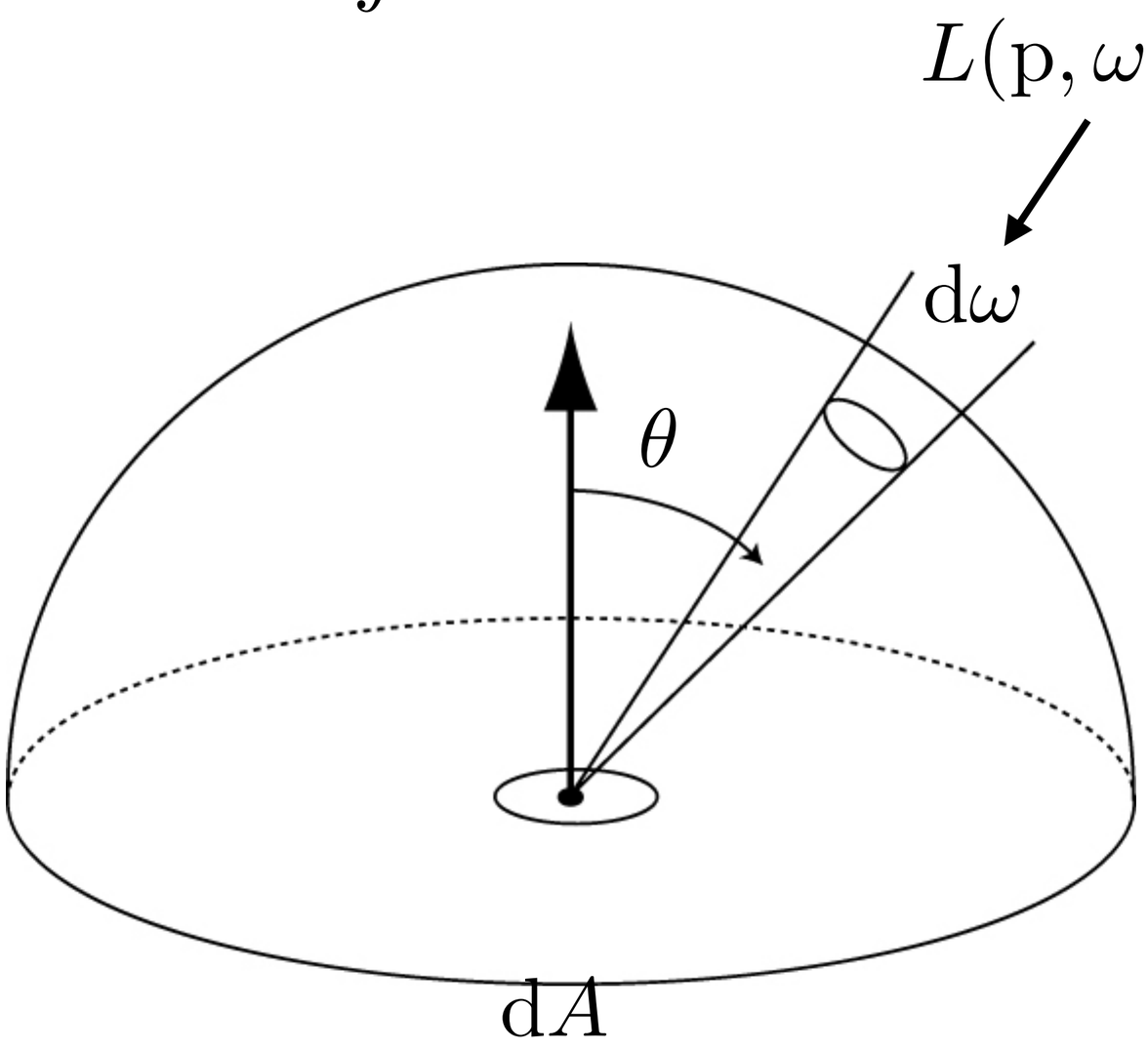
$$F_N = \frac{(x_1 - x_0)(y_1 - y_0)(z_1 - z_0)}{N} \sum_{i=1}^{N} f(X_i)$$

*** Generalizes to arbitrary N-dimensional PDFs**

Example: Monte Carlo Estimate Of Direct Lighting Integral

Direct Lighting (Irradiance) Estimate

$$E(\mathrm{p}) = \int L(\mathrm{p}, \omega)\ \cos\theta \, \mathrm{d}\omega$$



Idea: sample directions over hemisphere uniformly in solid angle

Estimator:

$$X_i \sim p(\omega) \qquad p(\omega) = \frac{1}{2\pi}$$

$$Y_i = f(X_i)$$

$$Y_i = L(\mathrm{p}, \omega_i) cos\, \theta_i$$

$$F_N = \frac{2\pi}{N} \sum_{i=1}^{N} Y_i$$

Direct Lighting (Irradiance) Estimate

Sample directions over hemisphere uniformly in solid angle

$$E(\mathrm{p}) = \int L(\mathrm{p}, \omega)\, \cos\theta \, \mathrm{d}\omega$$

Given surface point *p*

Initialize Monte Carlo estimator F_N to 0

For each of N samples:

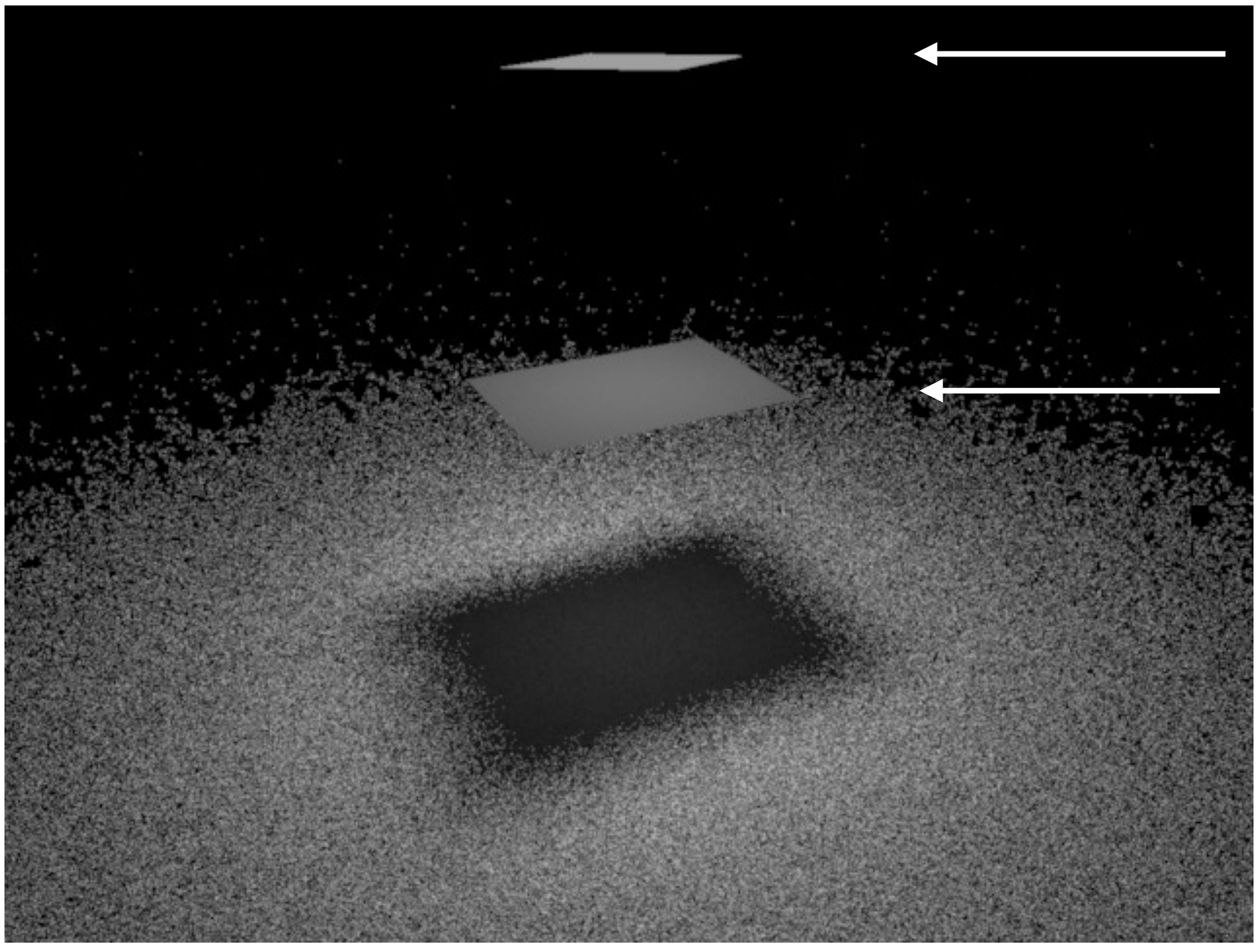
A ray tracer evaluates radiance along a ray

Generate random direction: ω_i

Compute incoming radiance L_i arriving at *p* from direction ω_i

Increment the Monte Carlo estimator: $F_N := F_N + \frac{2\pi}{N} L_i \cos\theta_i$

Direct Lighting - Solid Angle Sampling



Trace 100 rays per pixel

Hemispherical Solid Angle
Sampling 100 rays

(random directions drawn
uniformly from hemisphere)

The MC estimator uses different random directions at each pixel. Only some directions point towards the light.

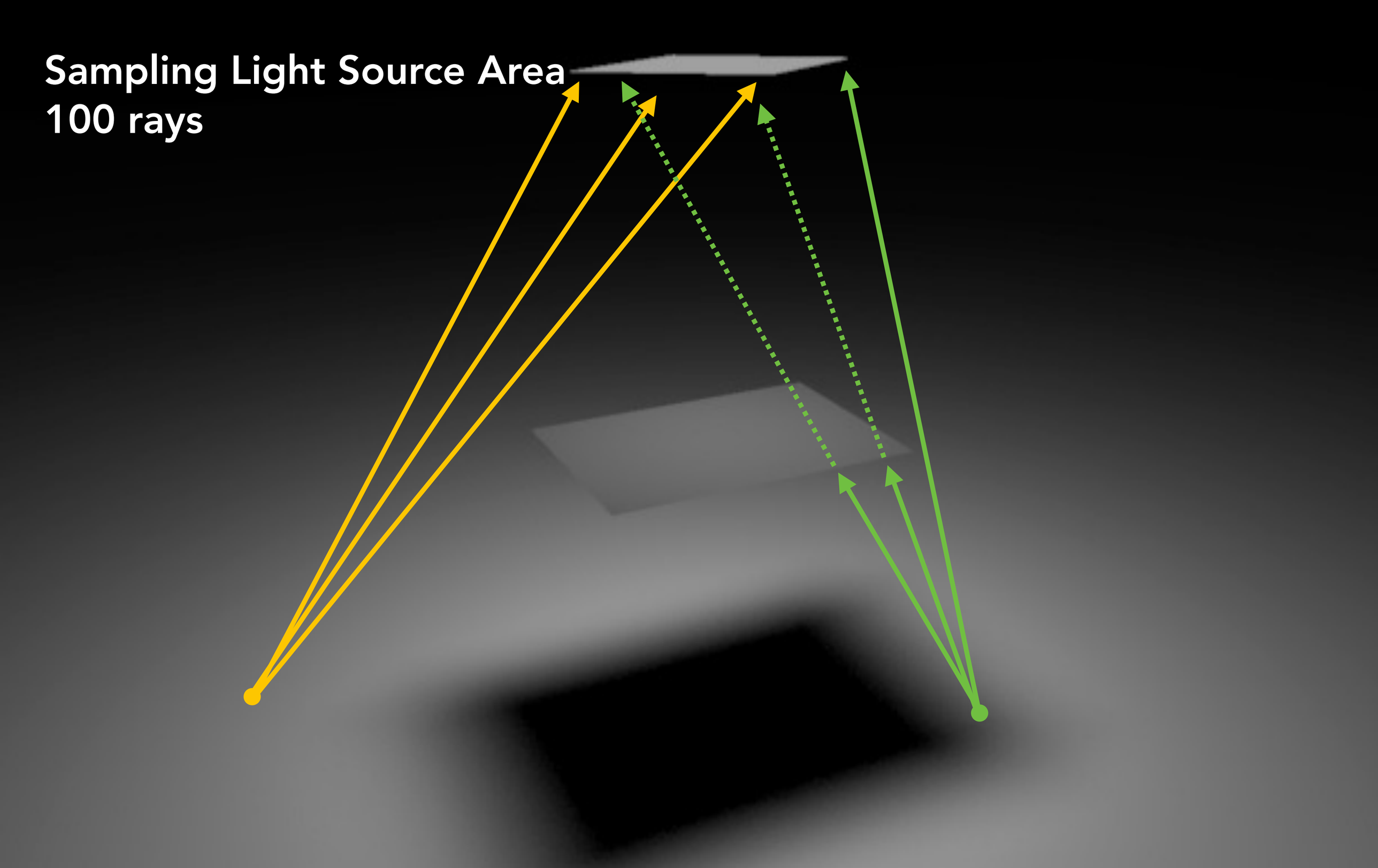
(Estimator is a random variable)

Why does this image look “noisy”?

Observation: incoming radiance is zero for most directions in this scene

Idea: integrate only over the area of the light (directions where incoming radiance could be non-zero)

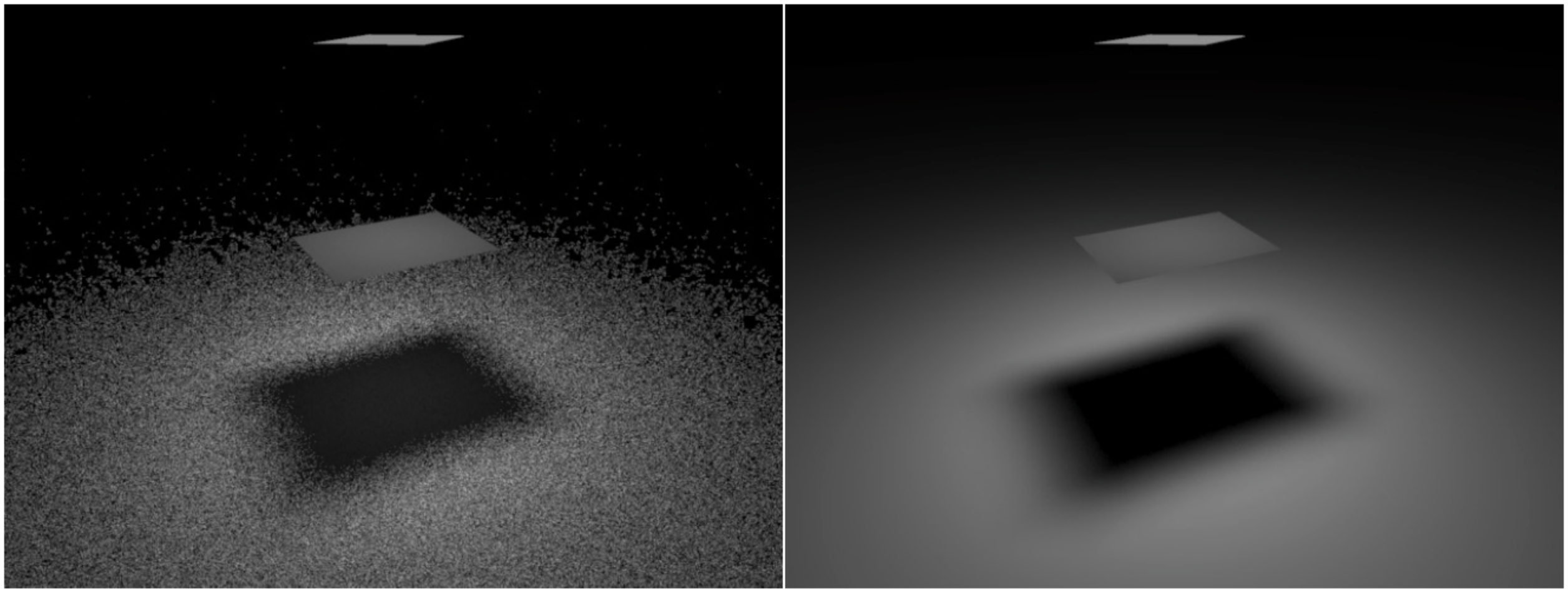
Sampling Light Source Area
100 rays



If no occlusion is present, all directions chosen in computing estimate “hit” the light source.

(Choice of direction only matters if portion of light is occluded from surface point *p*.)

Solid Angle Sampling vs Light Area Sampling



Sampling solid angle

100 random directions on hemisphere

Sampling light source area

100 random points on area of light source

Here, only important to sample directions that could hit the light

Attendance

If you are seated in class, go to this form and sign in:

- https://tinyurl.com/184class

Word of the day:

- variance

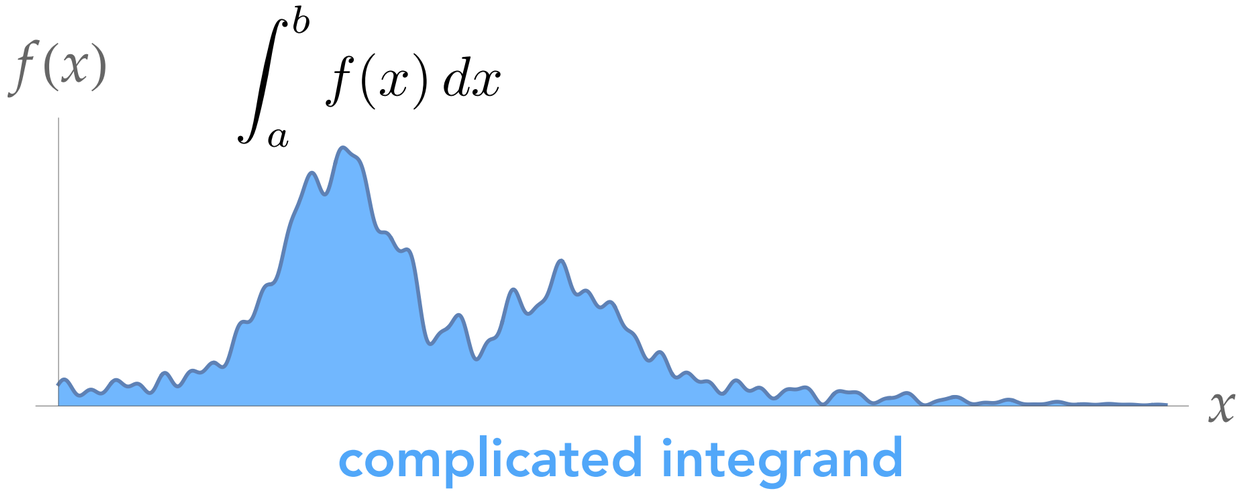
Importance Sampling

Importance Sampling

Simple idea: sample the integrand according to how much we expect it to contribute to the integral.

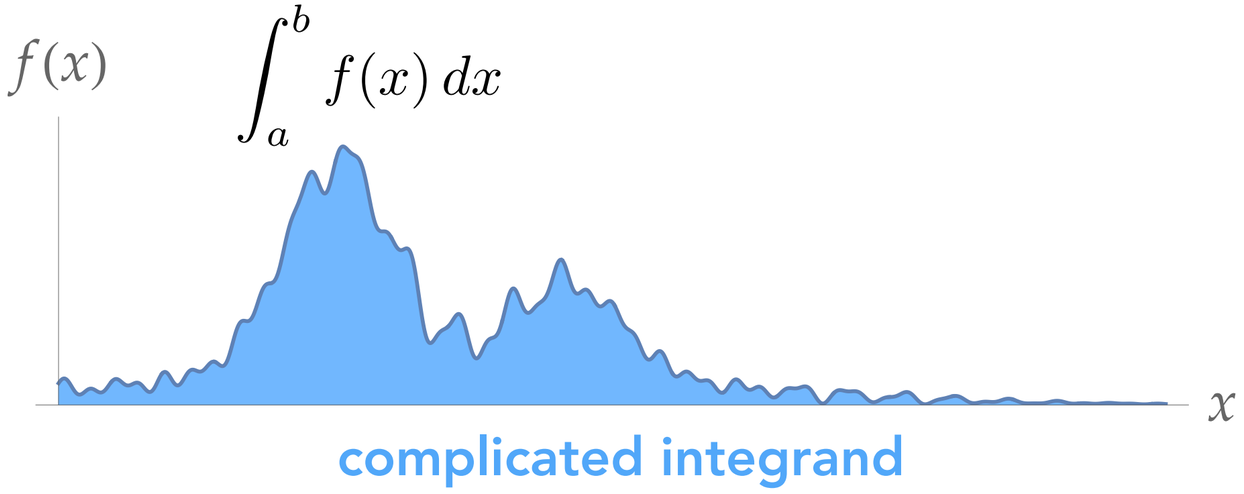
Importance Sampling

Simple idea: sample the integrand according to how much we expect it to contribute to the integral.



Importance Sampling

Simple idea: sample the integrand according to how much we expect it to contribute to the integral.



complicated integrand

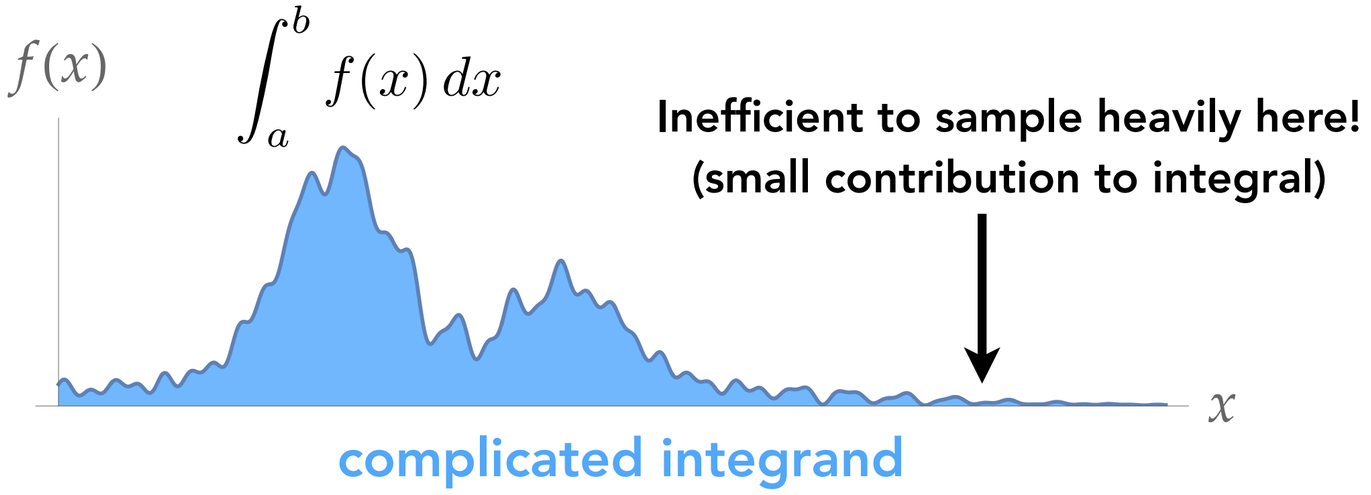
Basic Monte Carlo:

$$\frac{b-a}{N}\sum_{i=1}^{N} f(X_i)$$

(x_i are sampled *uniformly*)

Importance Sampling

Simple idea: sample the integrand according to how much we expect it to contribute to the integral.



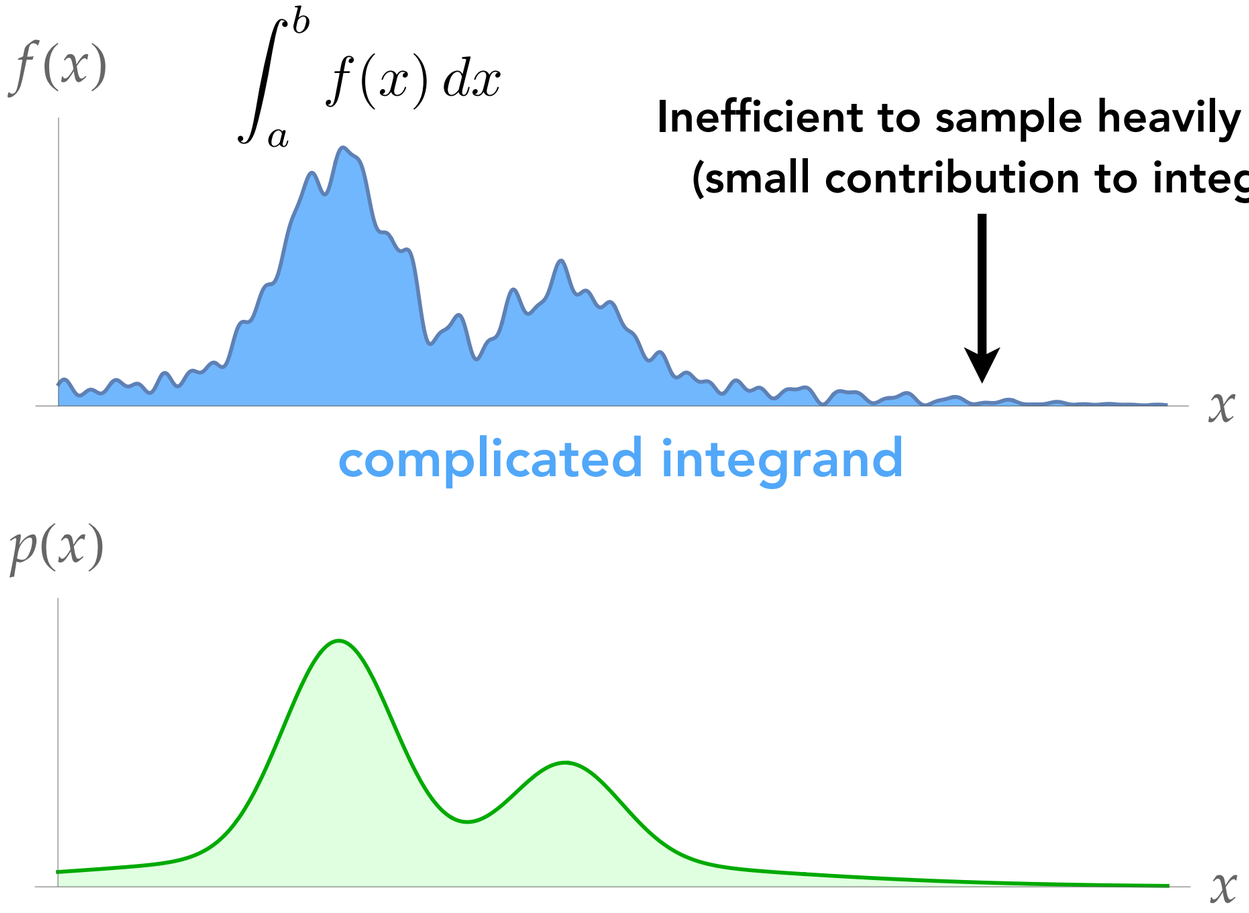
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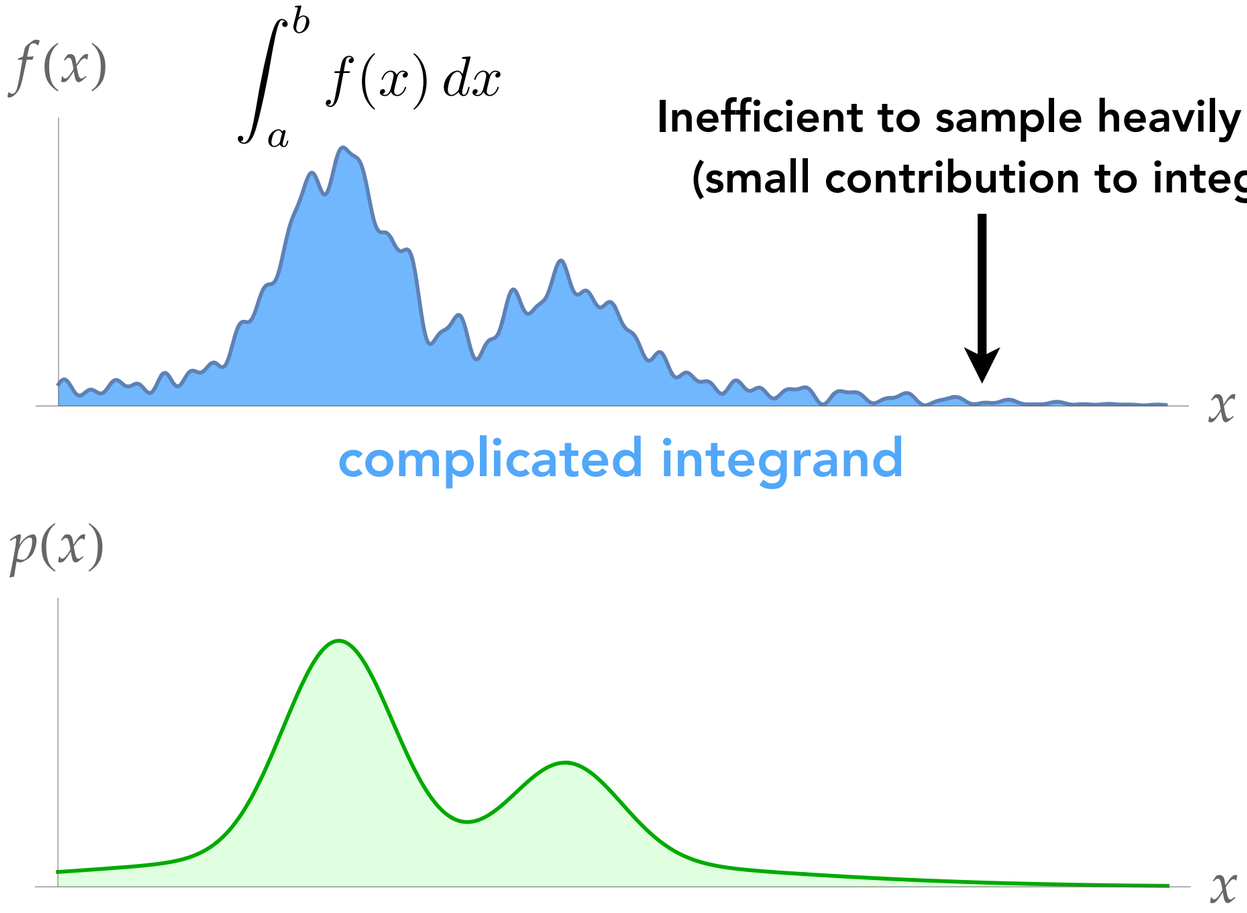
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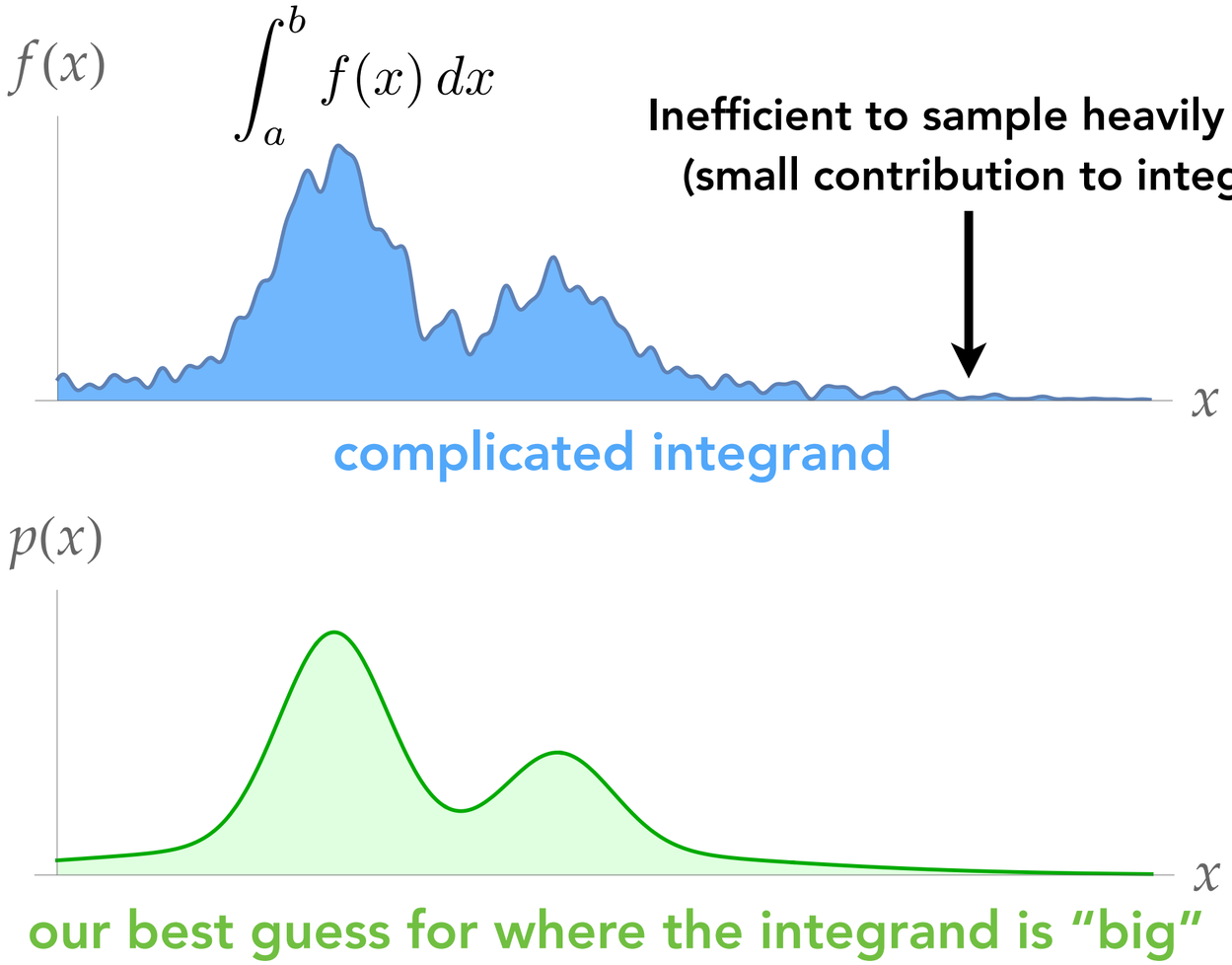
Importance-Sampled Monte Carlo:

$$\frac{1}{n}\sum_{i=1}^{n} \frac{f(x_i)}{p(x_i)}$$

(x_i are sampled proportional to *p*)

Importance Sampling

Simple idea: sample the integrand according to how much we expect it to contribute to the integral.



Basic Monte Carlo:

$$\frac{b-a}{N}\sum_{i=1}^{N} f(X_i)$$

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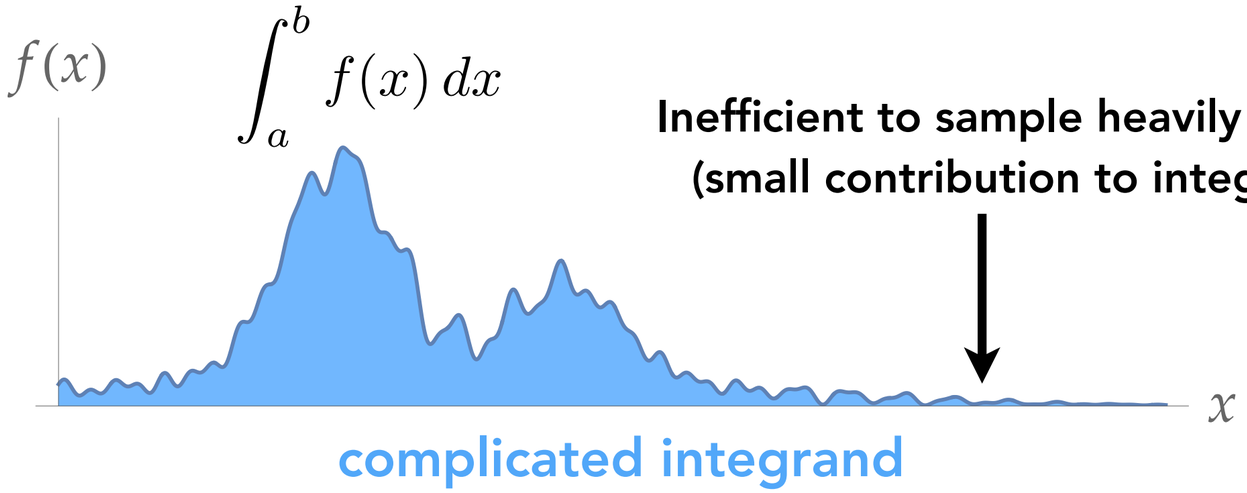
$$\frac{1}{n}\sum_{i=1}^{n} \frac{f(x_i)}{p(x_i)}$$

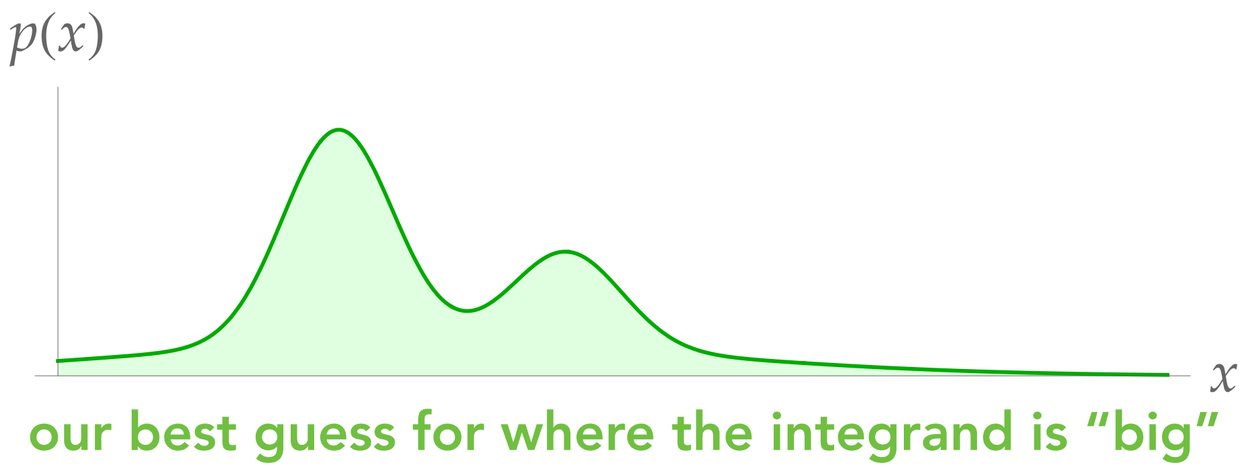
(x_i are sampled proportional to *p*)

"If I sample x less frequently, each sample should count for more."

Importance Sampling

Simple idea: sample the integrand according to how much we expect it to contribute to the integral.





Note: p(x) must be non-zero where f(x) is non-zero

Basic Monte Carlo:

$$\frac{b-a}{N}\sum_{i=1}^{N} f(X_i)$$

(x_i are sampled *uniformly*)

Importance-Sampled Monte Carlo:

$$\frac{1}{n}\sum_{i=1}^{n} \frac{f(x_i)}{p(x_i)}$$

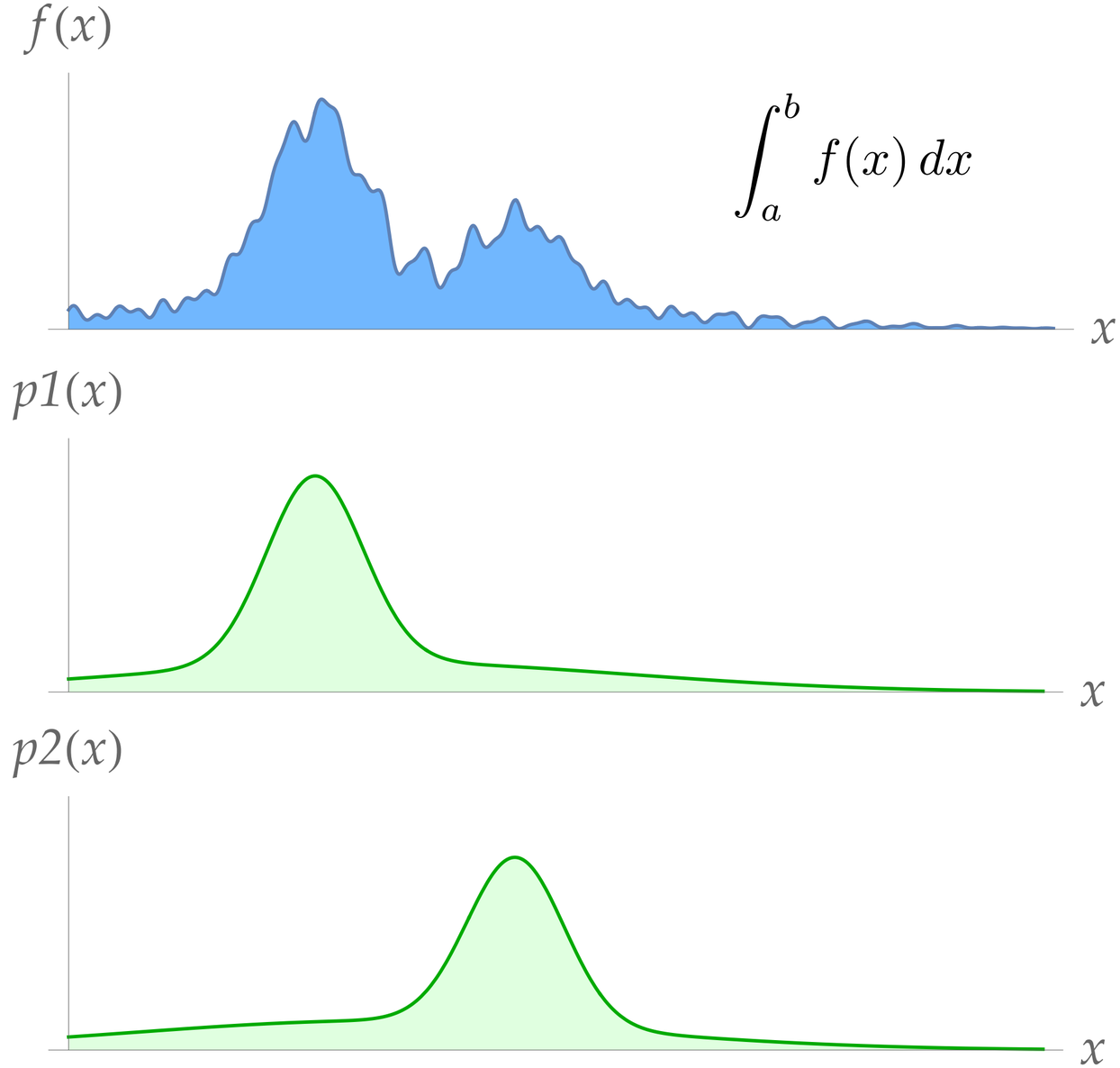
(x_i are sampled proportional to *p*)

"If I sample x less frequently, each sample should count for more."

Many Importance Sampling Strategies

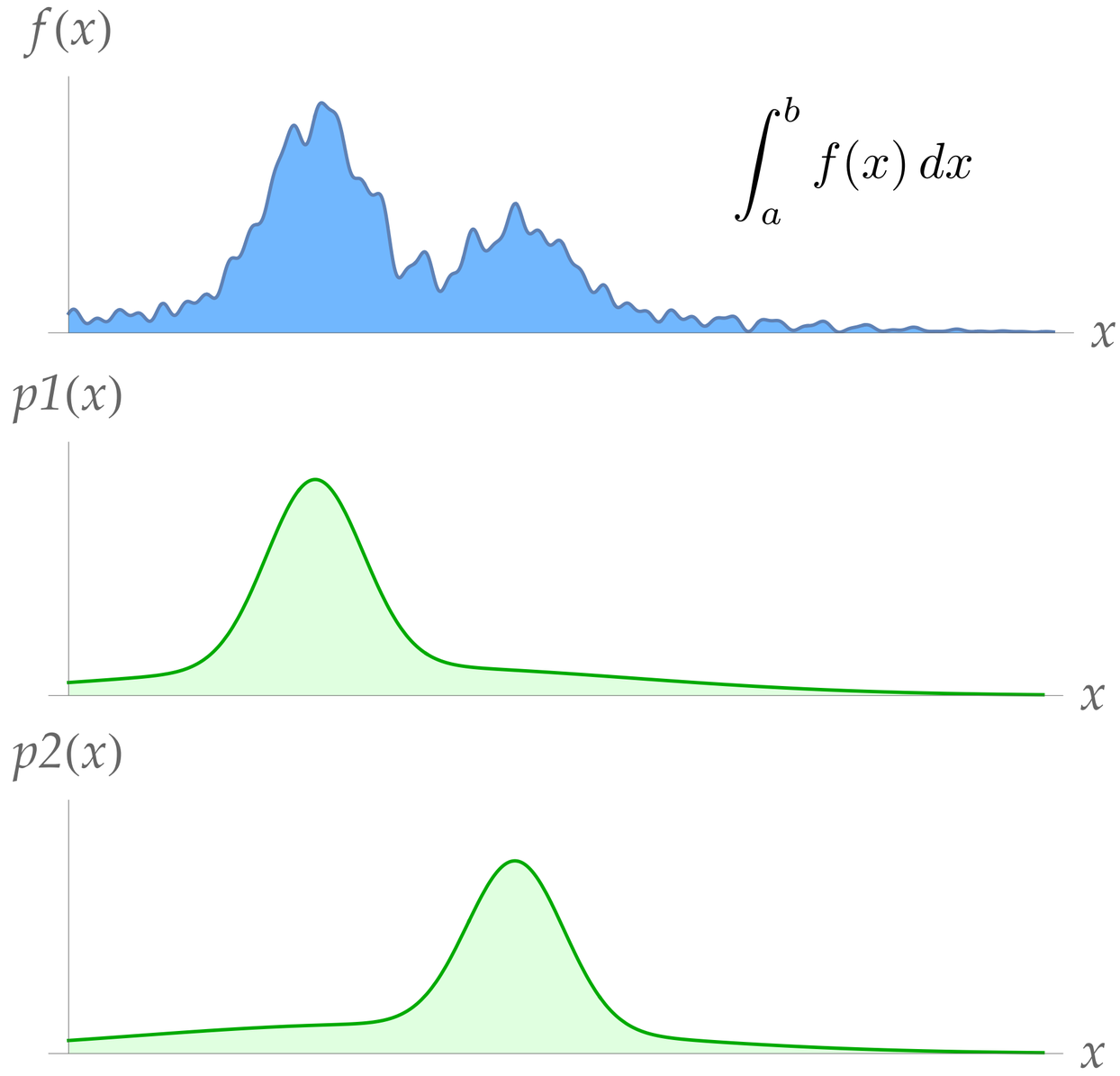
Many Importance Sampling Strategies

- Many possible importance sampling strategies (pdfs we could choose to sample from)



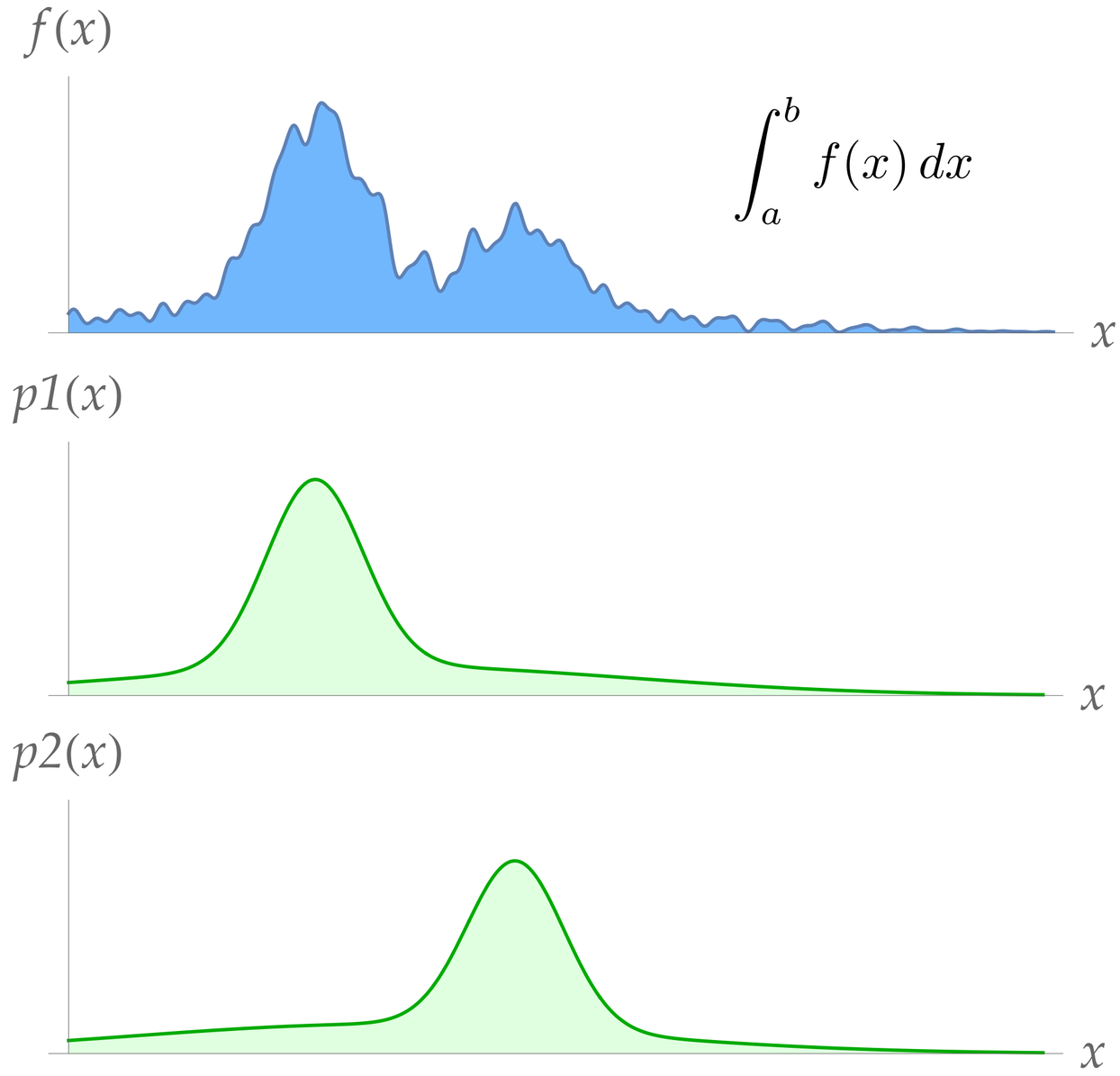
Many Importance Sampling Strategies

- Many possible importance sampling strategies (pdfs we could choose to sample from)
- A good fit to f(x) will decrease noise, but poor fit will increase noise!

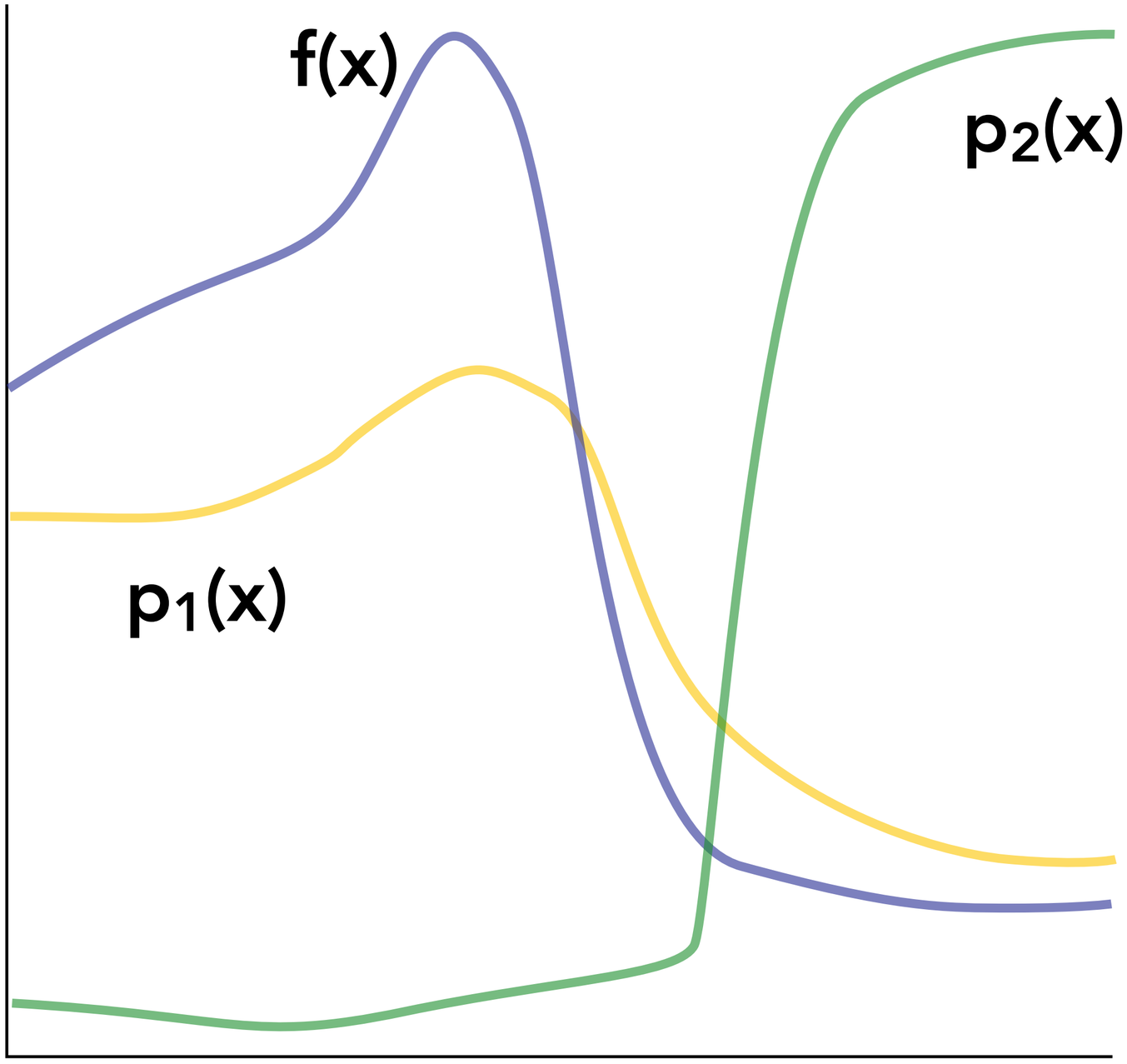


Many Importance Sampling Strategies

- Many possible importance sampling strategies (pdfs we could choose to sample from)
- A good fit to f(x) will decrease noise, but poor fit will increase noise!
- Not just about magnitude, also relates to regional sample variance

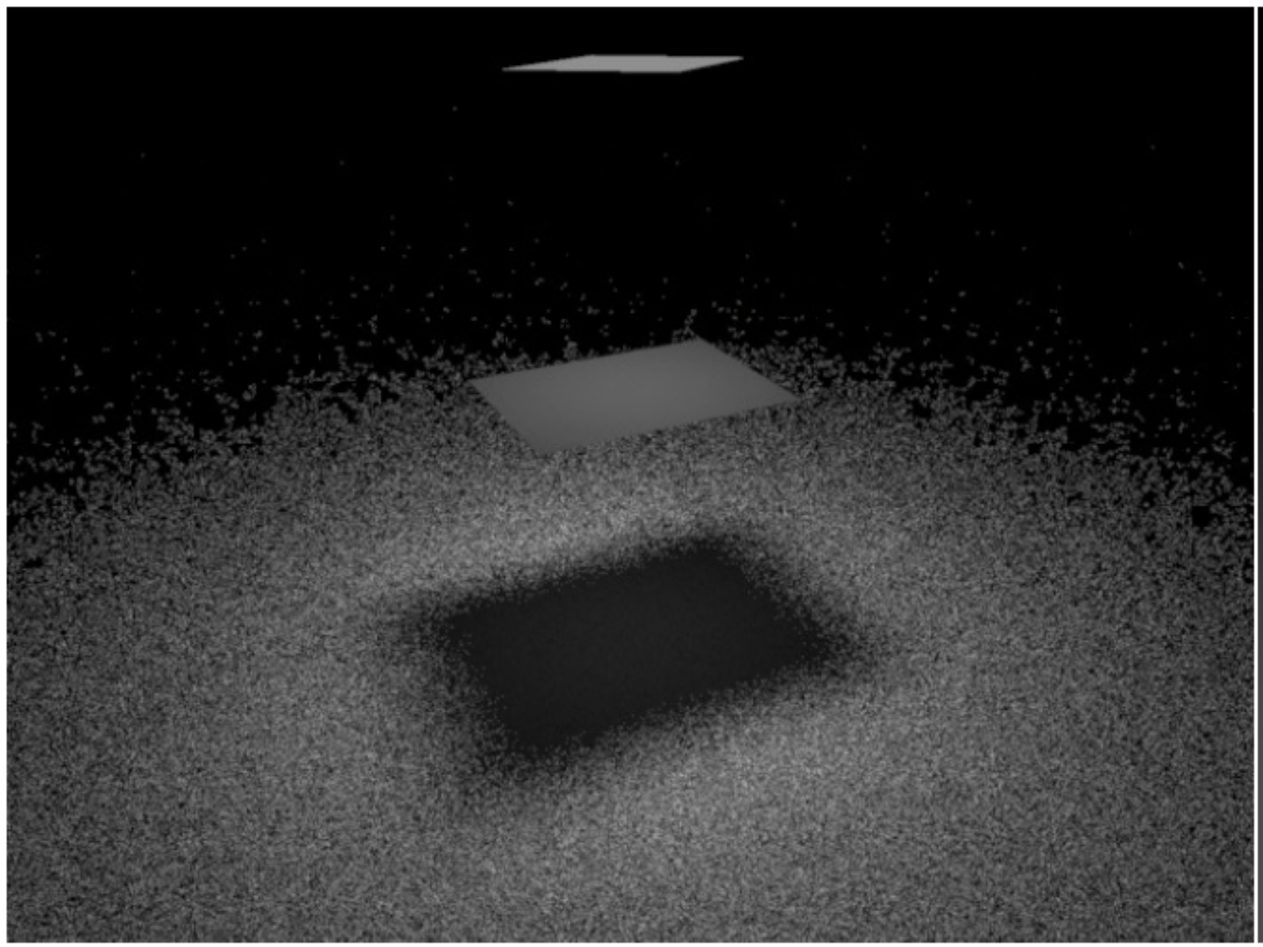


Effect of Sampling Distribution "Fit"



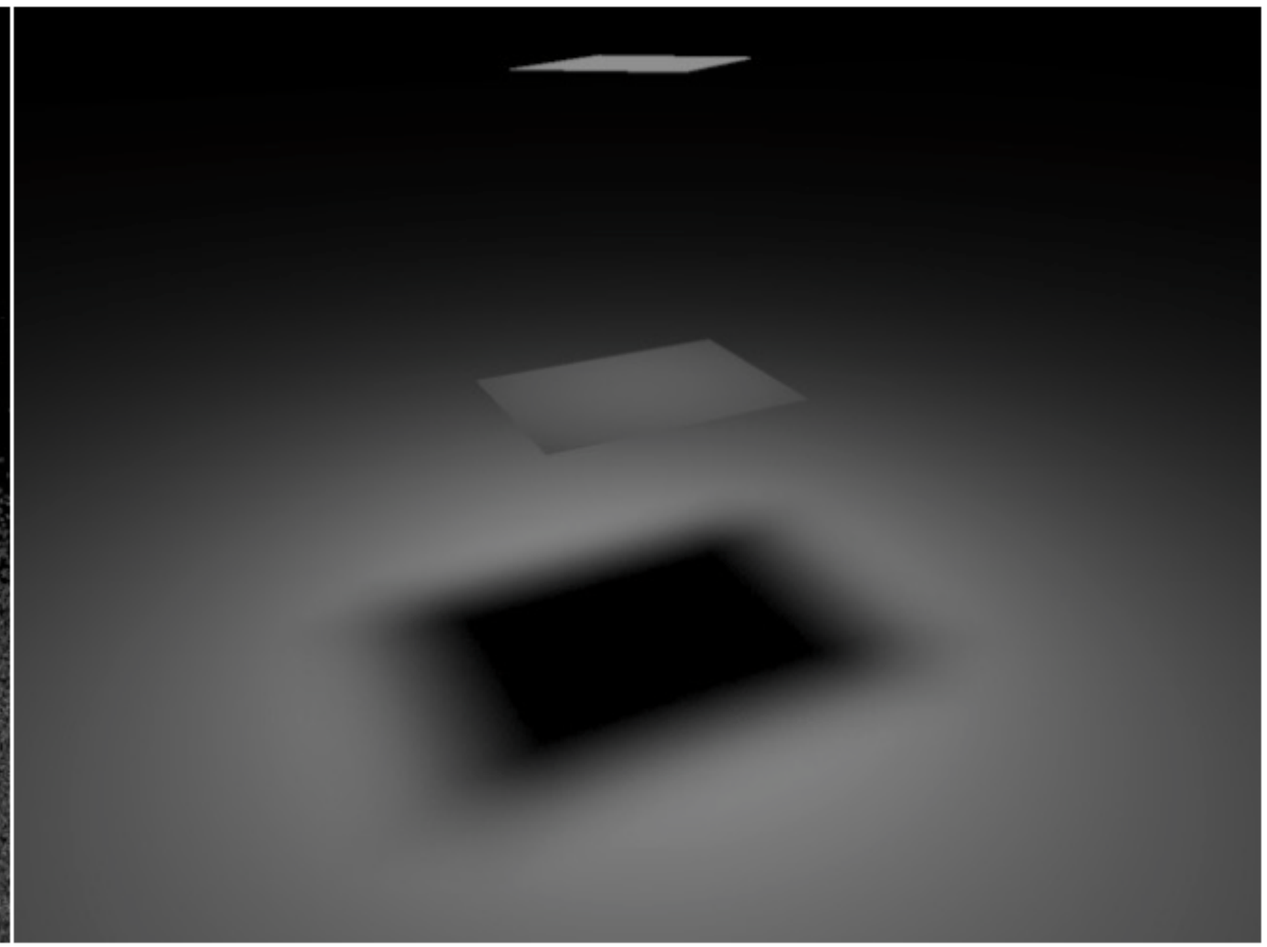
What is the behavior of $f(x)/p_1(x)$? $f(x)/p_2(x)$?
How does this impact the variance of the estimator?

Solid Angle Sampling vs Light Area Sampling



Sampling solid angle

100 random directions on hemisphere

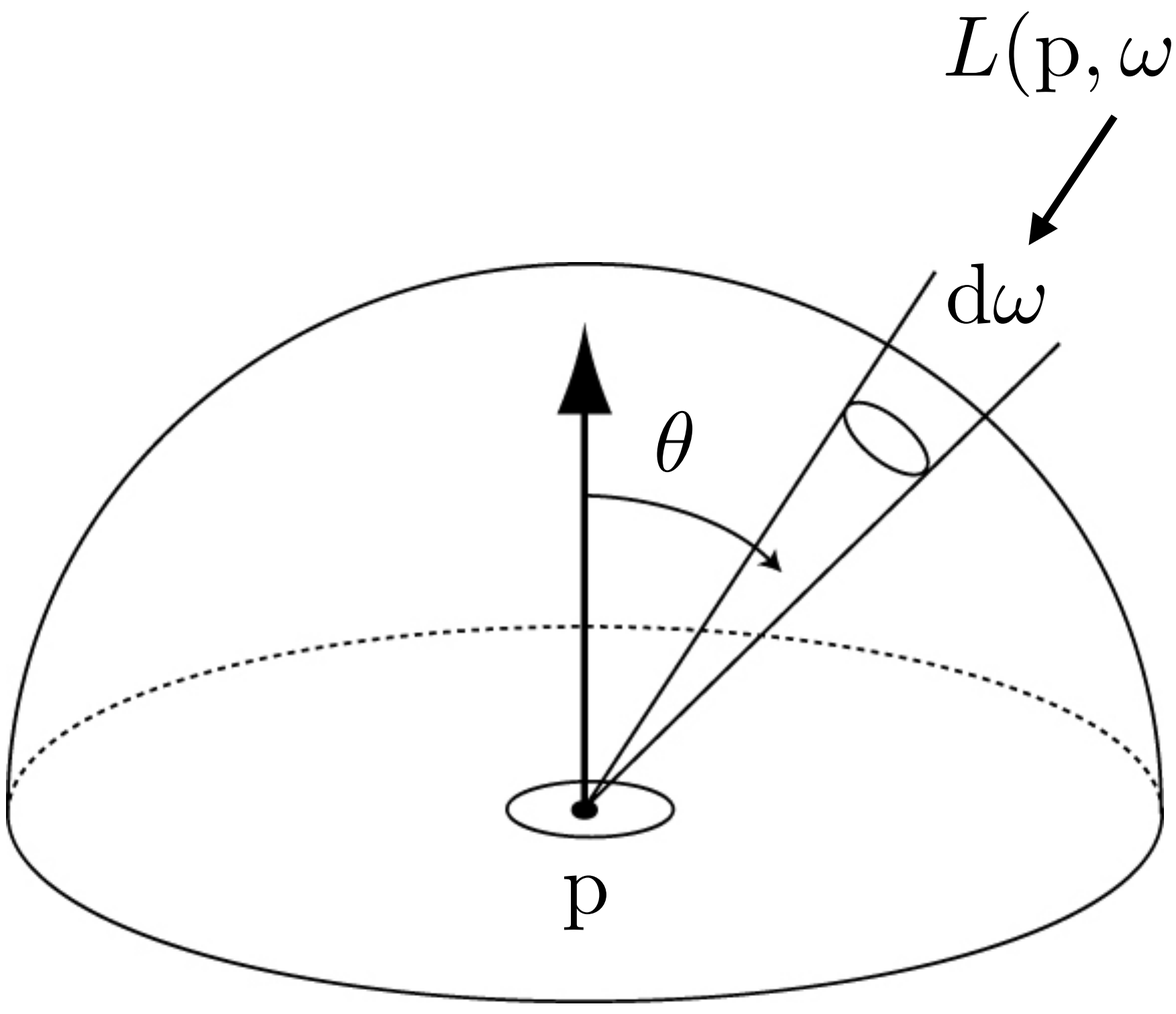


Sampling light source area

100 random points on area of light source

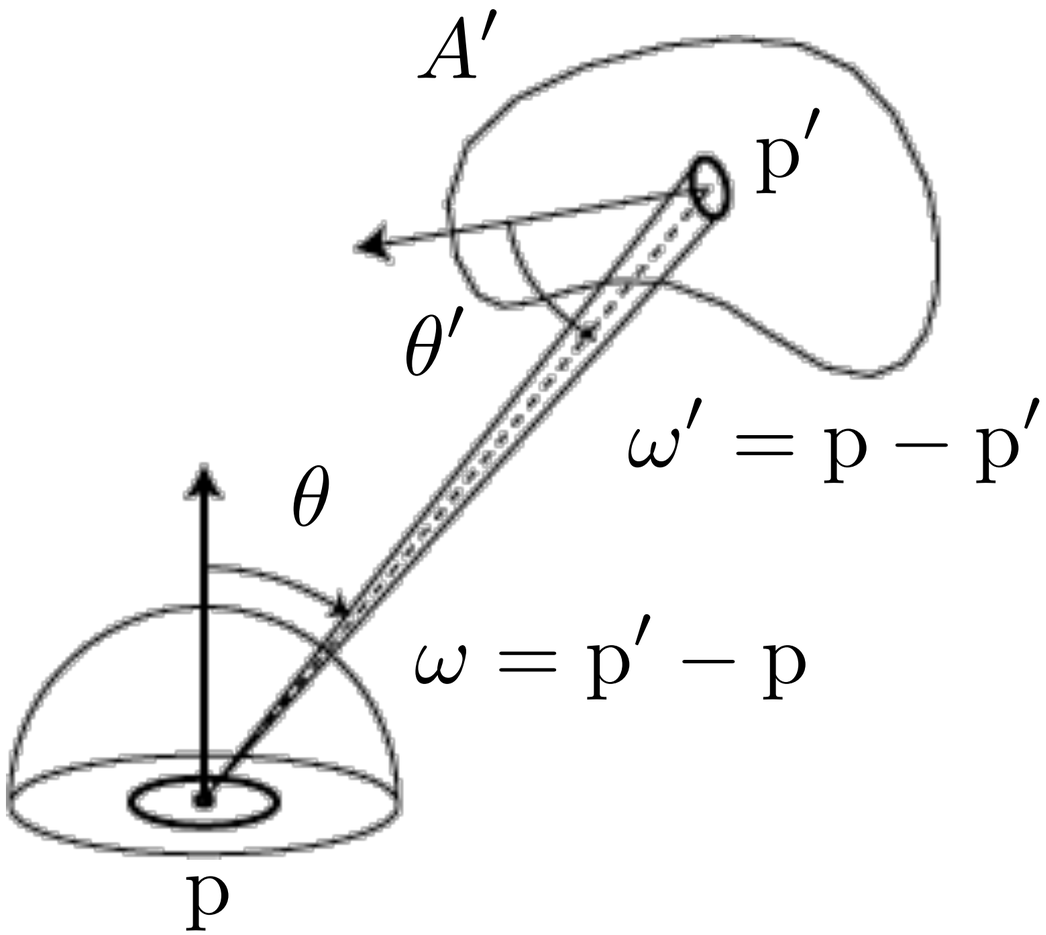
Changing Basis of Integration: Sampling Hemisphere

$$E(\mathrm{p}) = \int L(\mathrm{p}, \omega)\ \cos\theta \, \mathrm{d}\omega$$



Changing Basis of Integration: Sampling Light Source Area

$$E(\mathrm{p}) = \int_{A'} L_o(\mathrm{p}', \omega')\, V(\mathrm{p}, \mathrm{p}')\, \frac{\cos\theta \cos\theta'}{|\mathrm{p} - \mathrm{p}'|^2}\, \mathrm{d}A'$$

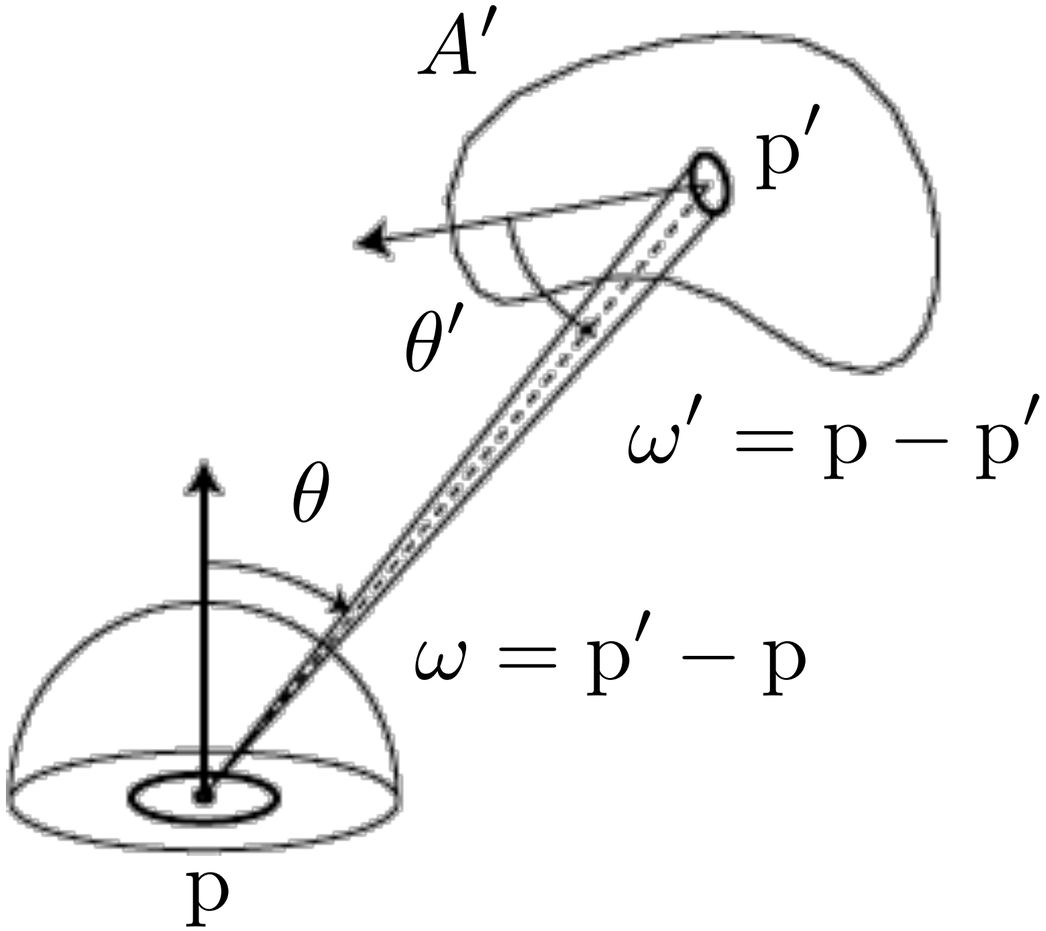


Changing Basis of Integration: Sampling Light Source Area

$$E(\mathrm{p}) = \int_{A'} L_o(\mathrm{p}', \omega')\, V(\mathrm{p}, \mathrm{p}')\, \frac{\cos\theta \cos\theta'}{|\mathrm{p} - \mathrm{p}'|^2}\, \mathrm{d}A'$$

← Change of variables to integral over area of light

$$dw = \frac{dA' cos\,\theta'}{|\mathrm{p}' - \mathrm{p}|^2}$$

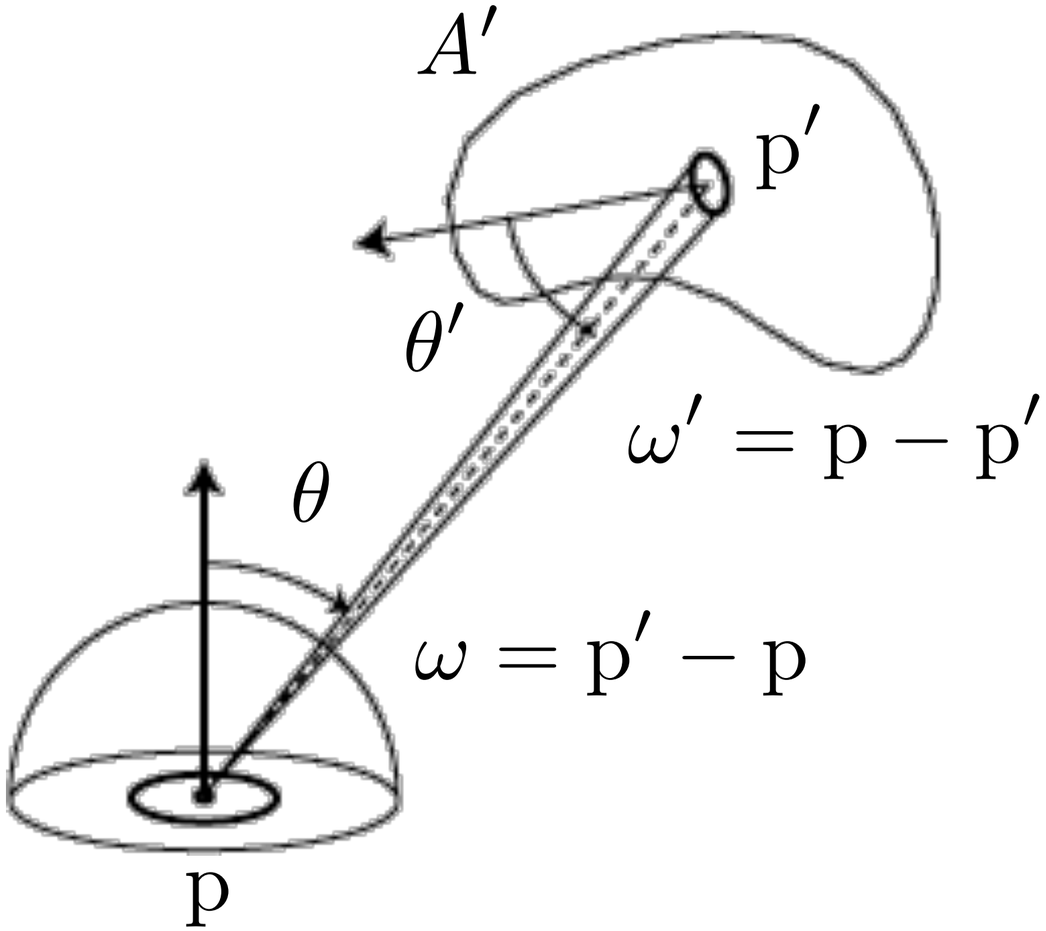


Changing Basis of Integration: Sampling Light Source Area

$$E(\mathrm{p}) = \int_{A'} L_o(\mathrm{p}', \omega')\, V(\mathrm{p}, \mathrm{p}')\, \frac{\cos\theta \cos\theta'}{|\mathrm{p} - \mathrm{p}'|^2}\, \mathrm{d}A'$$

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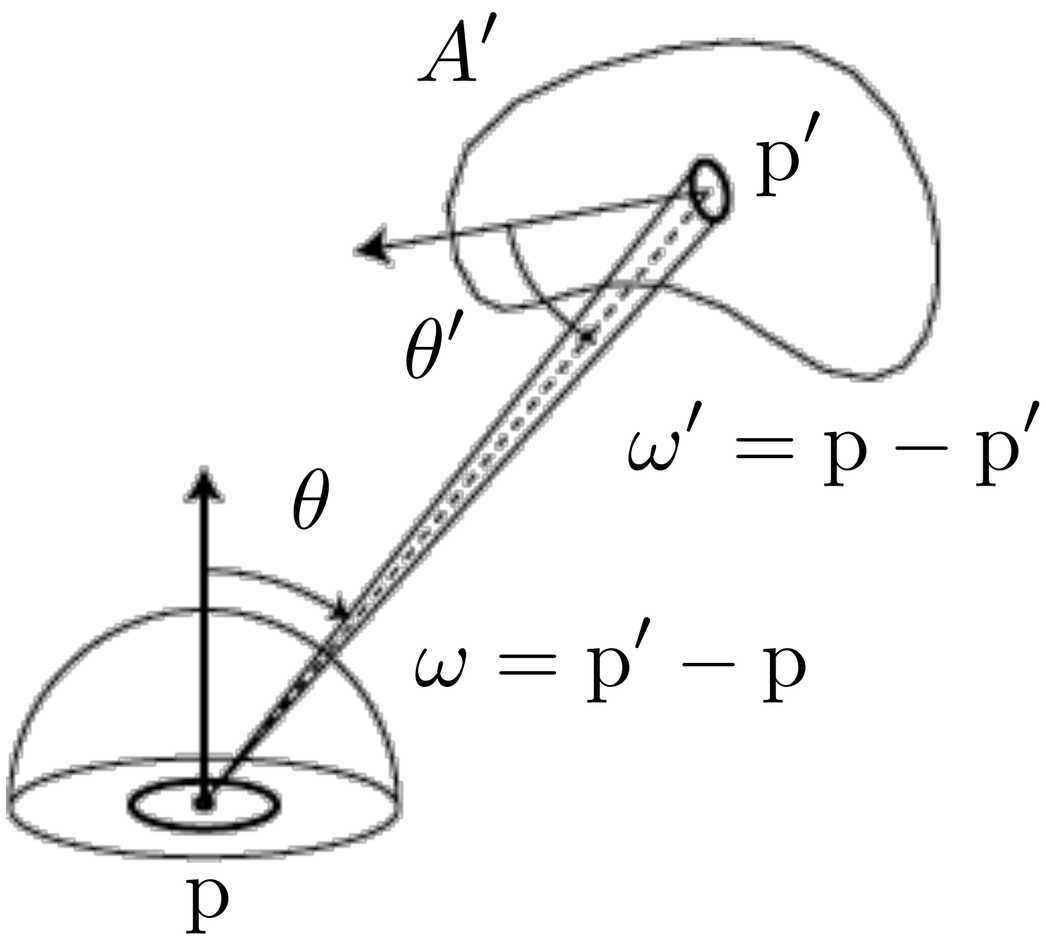
Outgoing radiance from light point p, in direction w′ towards p

Changing Basis of Integration: Sampling Light Source Area

$$E(\mathrm{p}) = \int_{A'} L_o(\mathrm{p}', \omega')\, V(\mathrm{p}, \mathrm{p}')\, \frac{\cos\theta \cos\theta'}{|\mathrm{p} - \mathrm{p}'|^2}\, \mathrm{d}A'$$

← Change of variables to integral over area of light

$$dw = \frac{dA' cos\,\theta'}{|\mathrm{p}' - \mathrm{p}|^2}$$

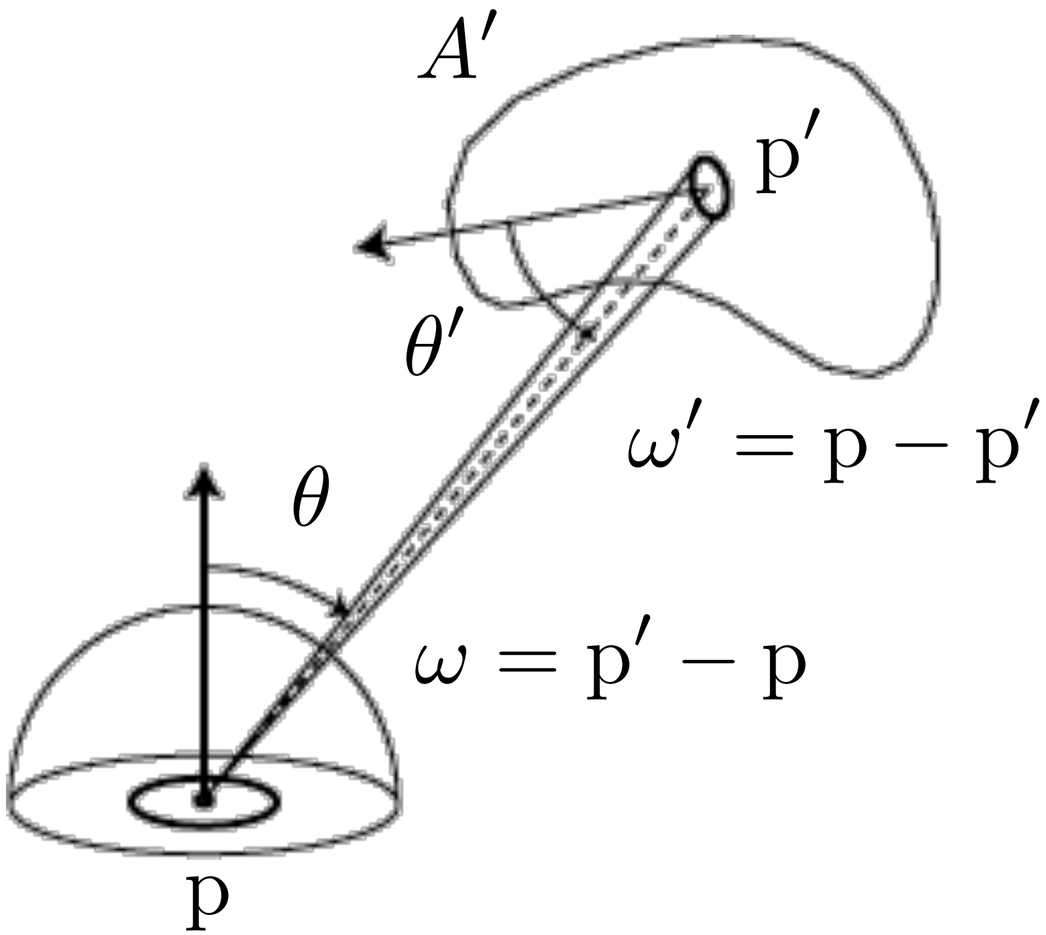


Binary visibility function:
1 if p′ is visible from p, 0 otherwise
(accounts for light occlusion)

Outgoing radiance from light point p, in direction w′ towards p

Monte Carlo Estimate by Sampling Light Source Area

$$E(\mathrm{p}) = \int_{A'} L_o(\mathrm{p}', \omega')\, V(\mathrm{p}, \mathrm{p}')\, \frac{\cos\theta \cos\theta'}{|\mathrm{p} - \mathrm{p}'|^2}\, \mathrm{d}A'$$



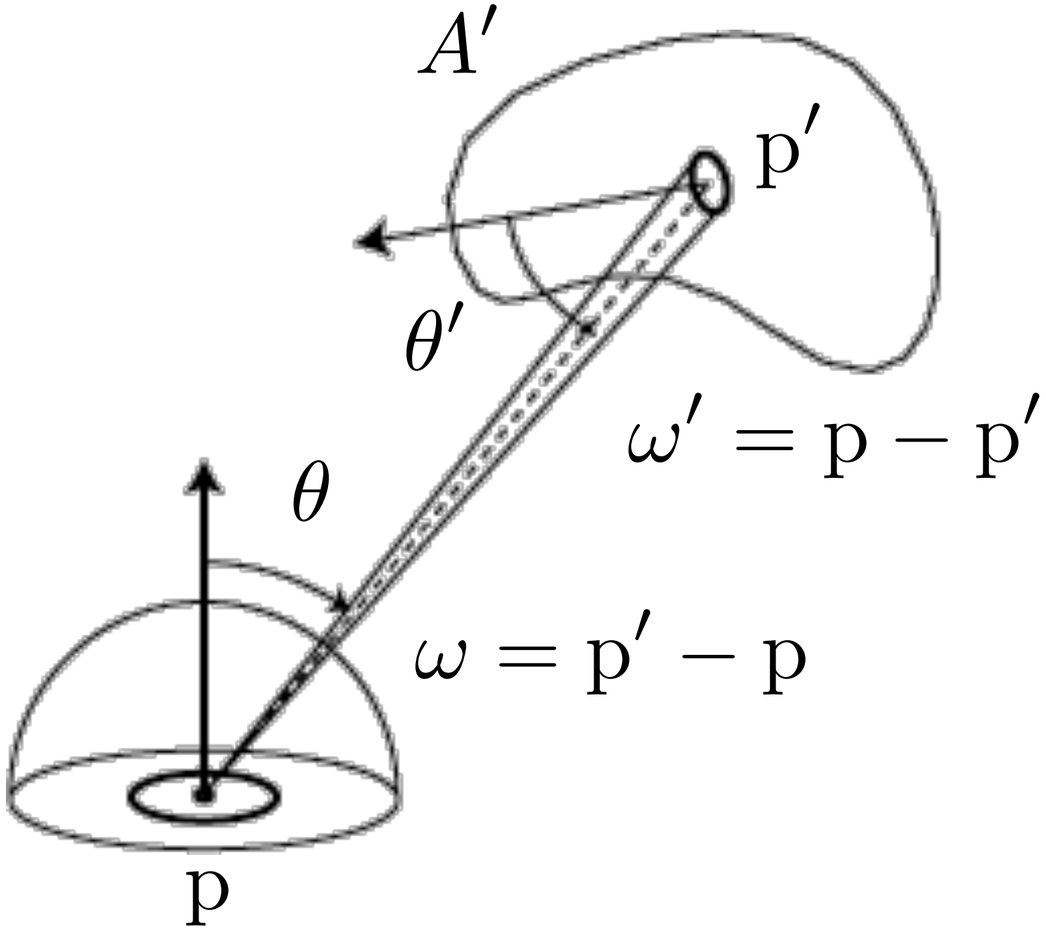
Monte Carlo Estimate by Sampling Light Source Area

$$E(\mathrm{p}) = \int_{A'} L_o(\mathrm{p}', \omega')\, V(\mathrm{p}, \mathrm{p}')\, \frac{\cos\theta \cos\theta'}{|\mathrm{p} - \mathrm{p}'|^2}\, \mathrm{d}A'$$

Randomly sample light source area A' (assume uniformly over area)

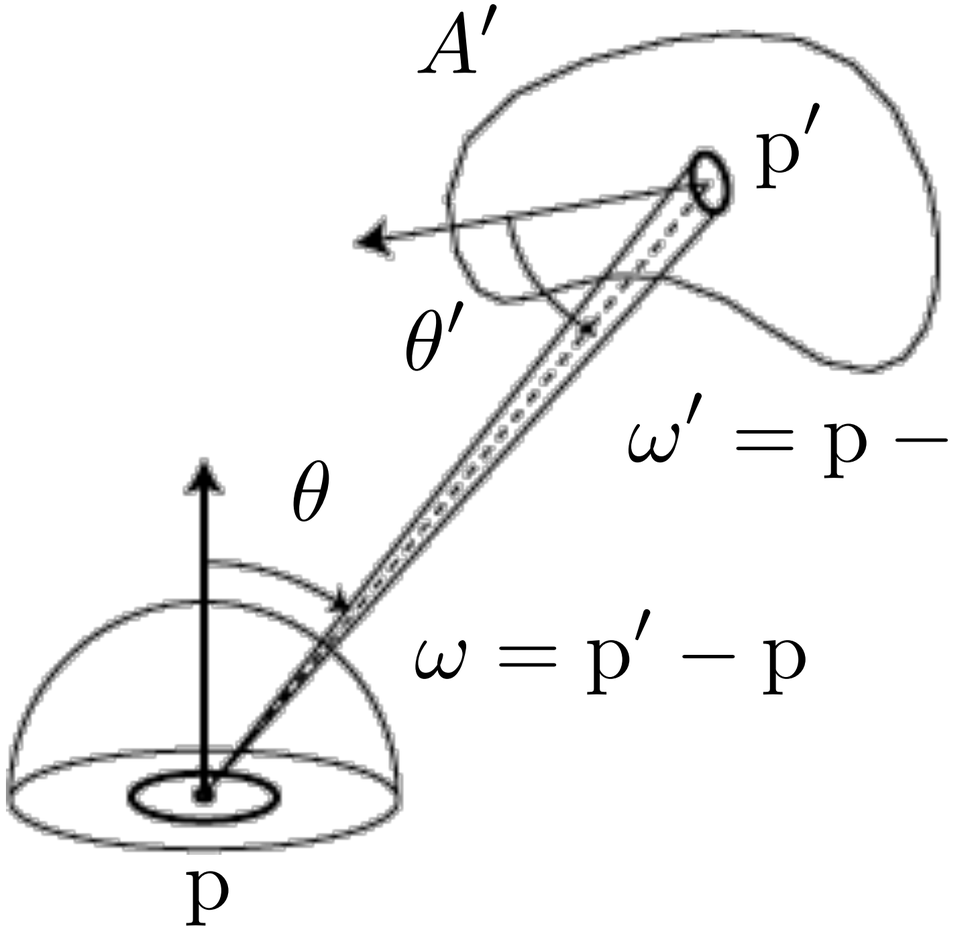
$$\int_{A'} p(\mathrm{p}')\, \mathrm{d}A' = 1$$

$$p(\mathrm{p}') = \frac{1}{A'}$$



Monte Carlo Estimate by Sampling Light Source Area

$$E(\mathrm{p}) = \int_{A'} L_o(\mathrm{p}', \omega')\, V(\mathrm{p}, \mathrm{p}')\, \frac{\cos\theta \cos\theta'}{|\mathrm{p} - \mathrm{p}'|^2}\, \mathrm{d}A'$$



Randomly sample light source area A' (assume uniformly over area)

$$\int_{A'} p(\mathrm{p}')\, \mathrm{d}A' = 1$$

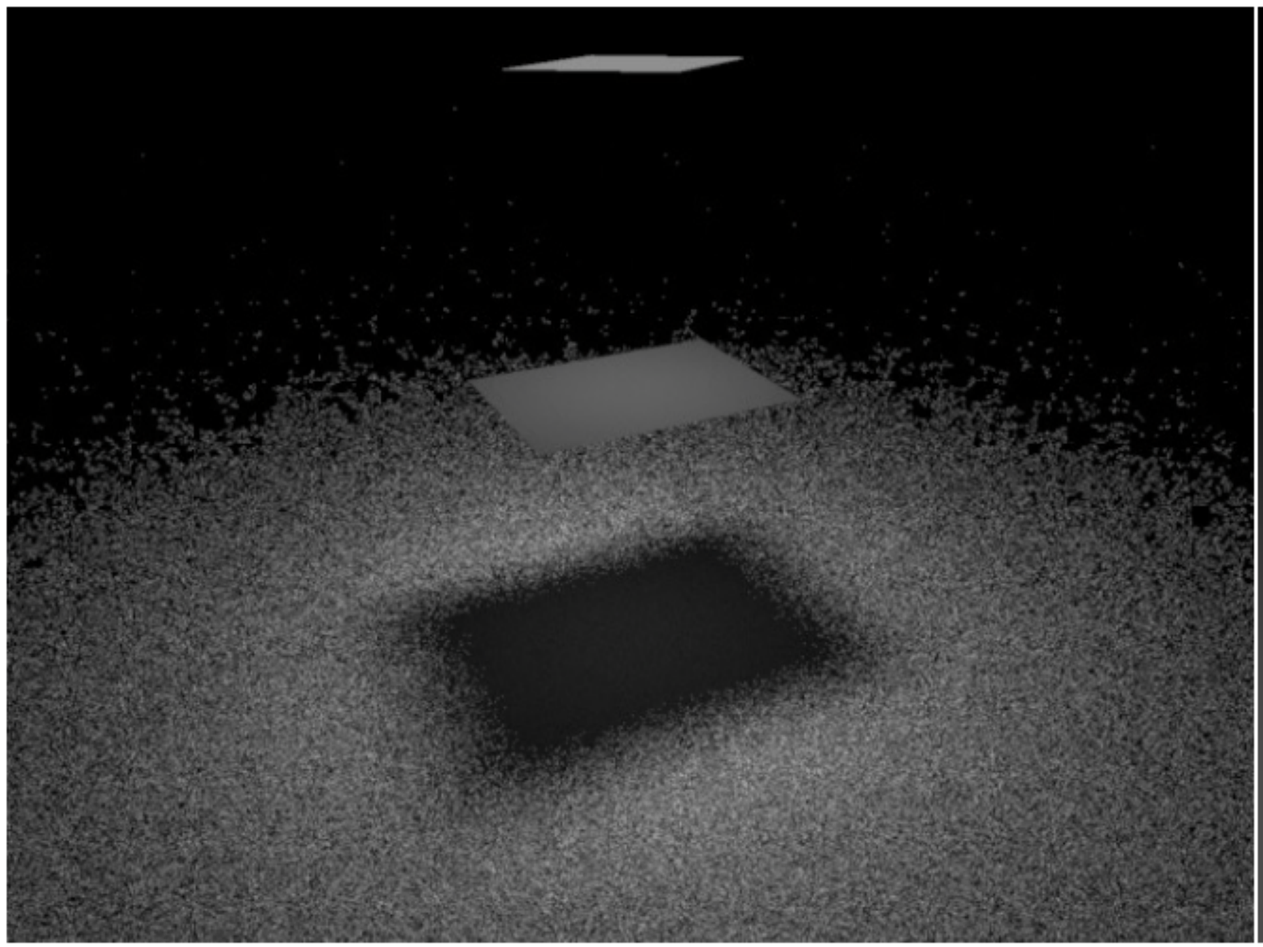
$$p(\mathrm{p}') = \frac{1}{A'}$$

Monte Carlo Estimator

$$F_N = \frac{A'}{N} \sum_{i=1}^{N} Y_i$$

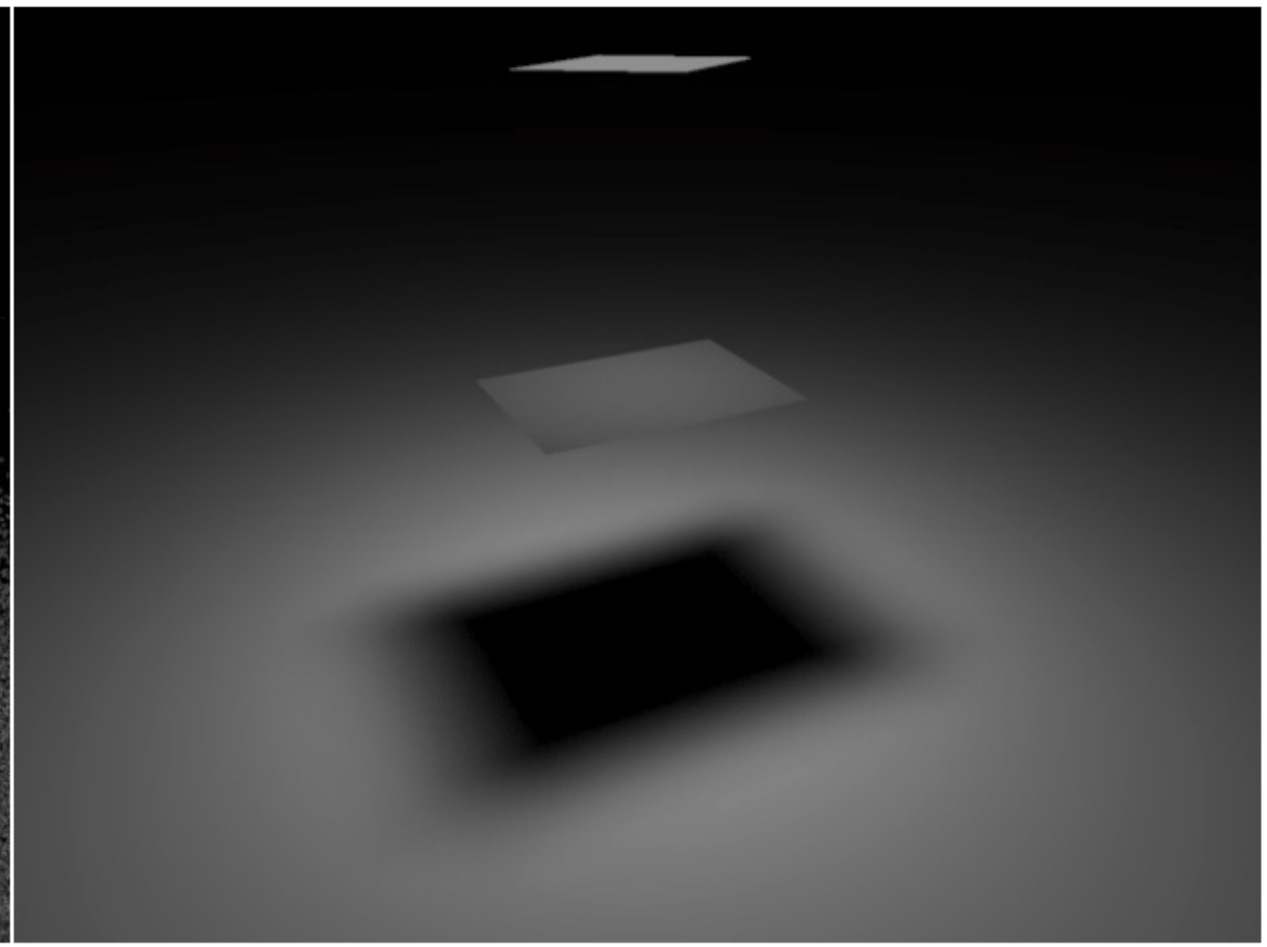
$$Y_i = L_o(\mathrm{p}'_i, \omega'_i)\, V(\mathrm{p}, \mathrm{p}'_i)\, \frac{\cos\theta_i \cos\theta'_i}{|\mathrm{p} - \mathrm{p}'_i|^2}$$

Solid Angle Sampling vs Light Area Sampling



Sampling solid angle

100 random directions on hemisphere



Sampling light source area

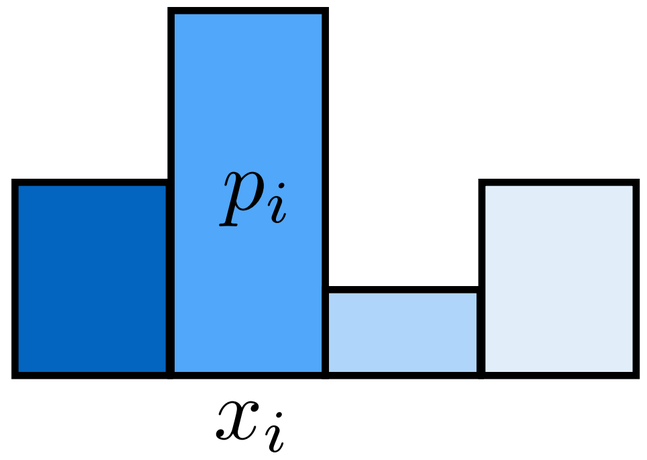
100 random points on area of light source

How to Draw Samples From a Desired Probability Distribution? One Approach: Inversion Method

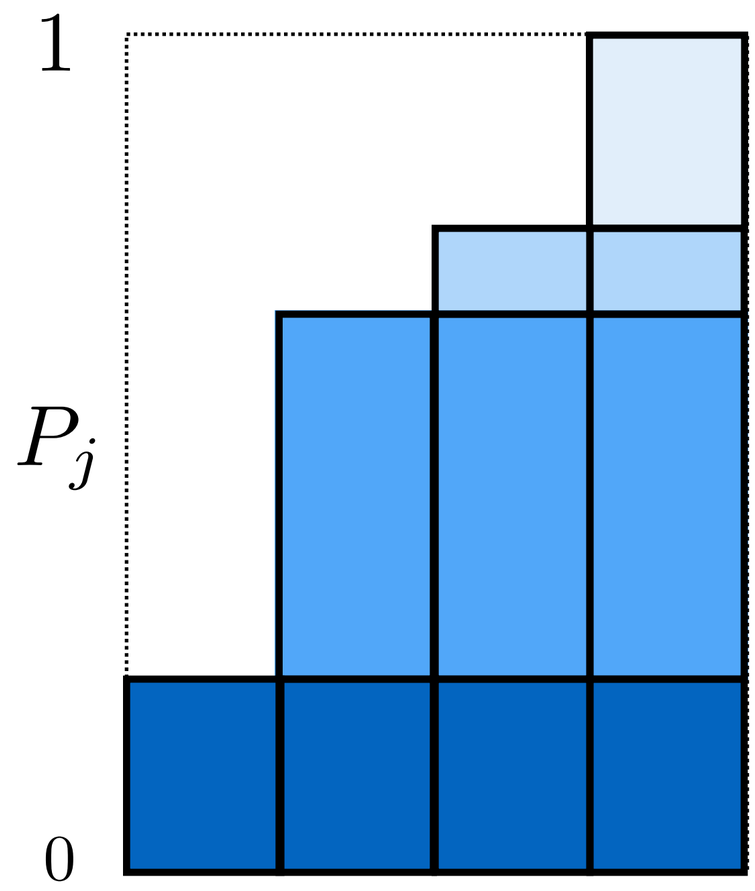
Task: Draw A Random Value From a Given PDF

Task:

Given a PDF for a discrete random variable, probability p_i for each value x_i,



Draw a random value X from this PDF.



Step 1:

Calculate cumulative PDF: $P_j = \sum_{i=1}^{j} p_i$

Note: must have

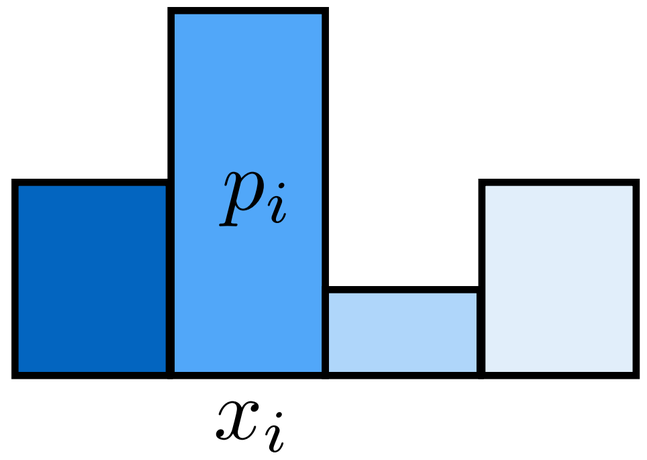
$$0 \leq P_i \leq 1$$

$$P_n = 1$$

Task: Draw A Random Value From a Given PDF

Task:

Given a PDF for a discrete random variable, probability p_i for each value x_i,

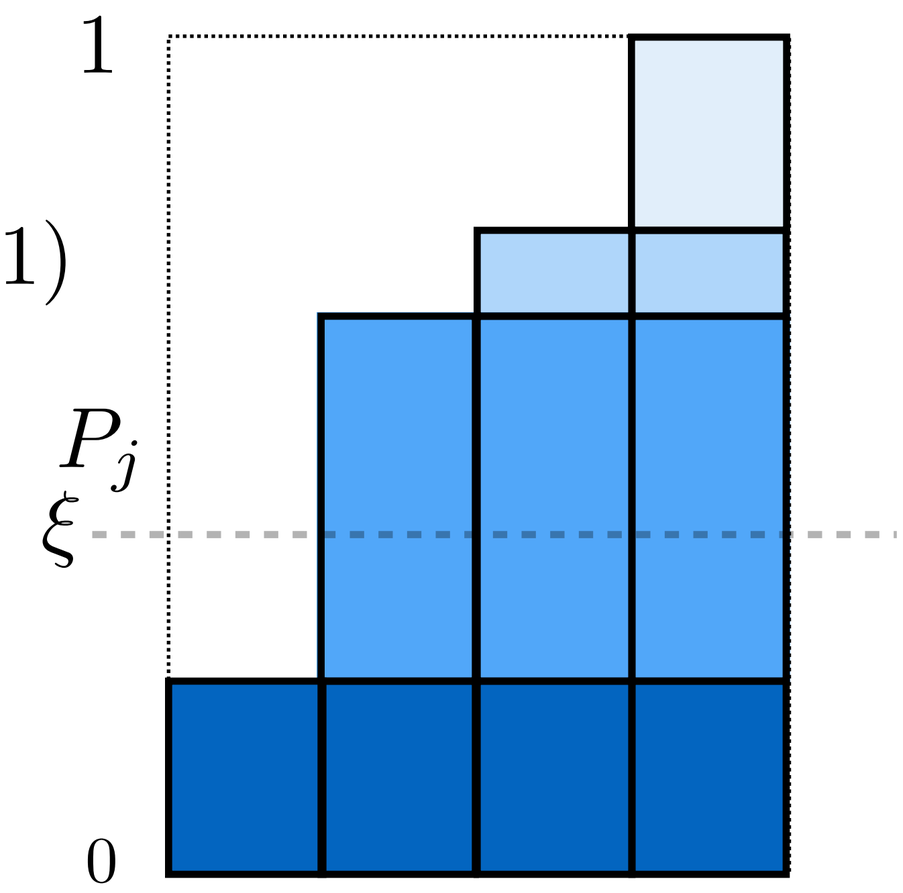


Draw a random value X from this PDF.

Step 2:

Given a uniform random variable $\xi \in [0, 1)$

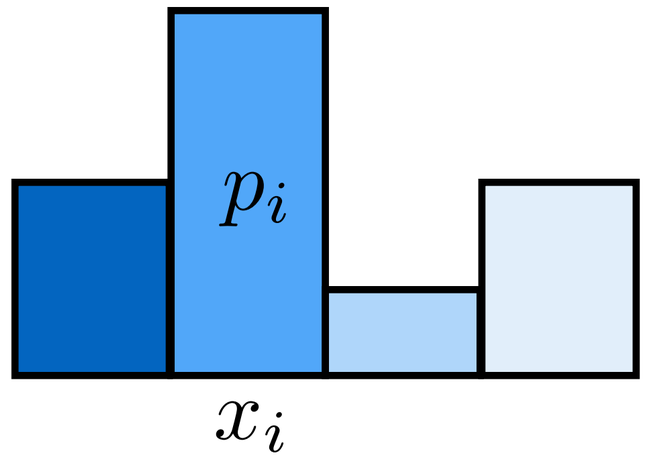
choose $X = x_i$
such that $P_{i-1} < \xi \leq P_i$



Task: Draw A Random Value From a Given PDF

Task:

Given a PDF for a discrete random variable, probability p_i for each value x_i,

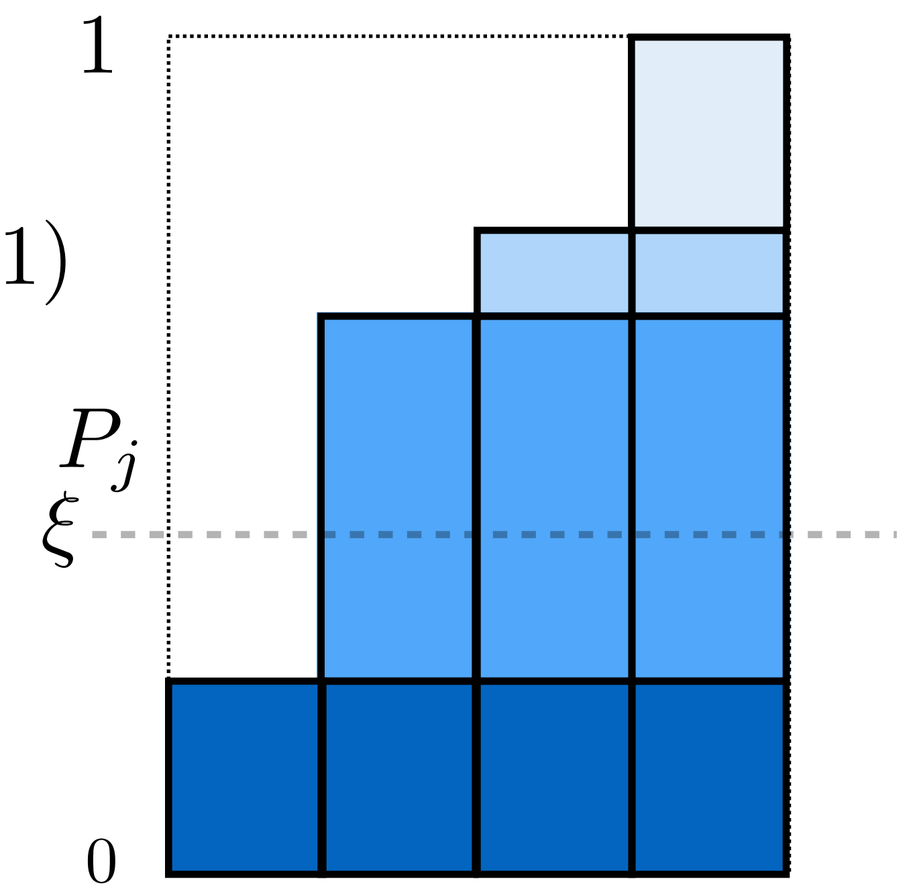


Draw a random value X from this PDF.

Step 2:

Given a uniform random variable $\xi \in [0, 1)$

choose $X = x_i$
such that $P_{i-1} < \xi \leq P_i$

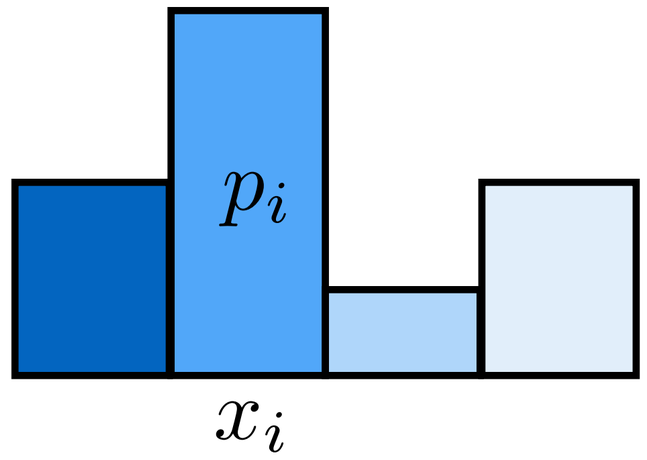


How to compute?

Task: Draw A Random Value From a Given PDF

Task:

Given a PDF for a discrete random variable, probability p_i for each value x_i,

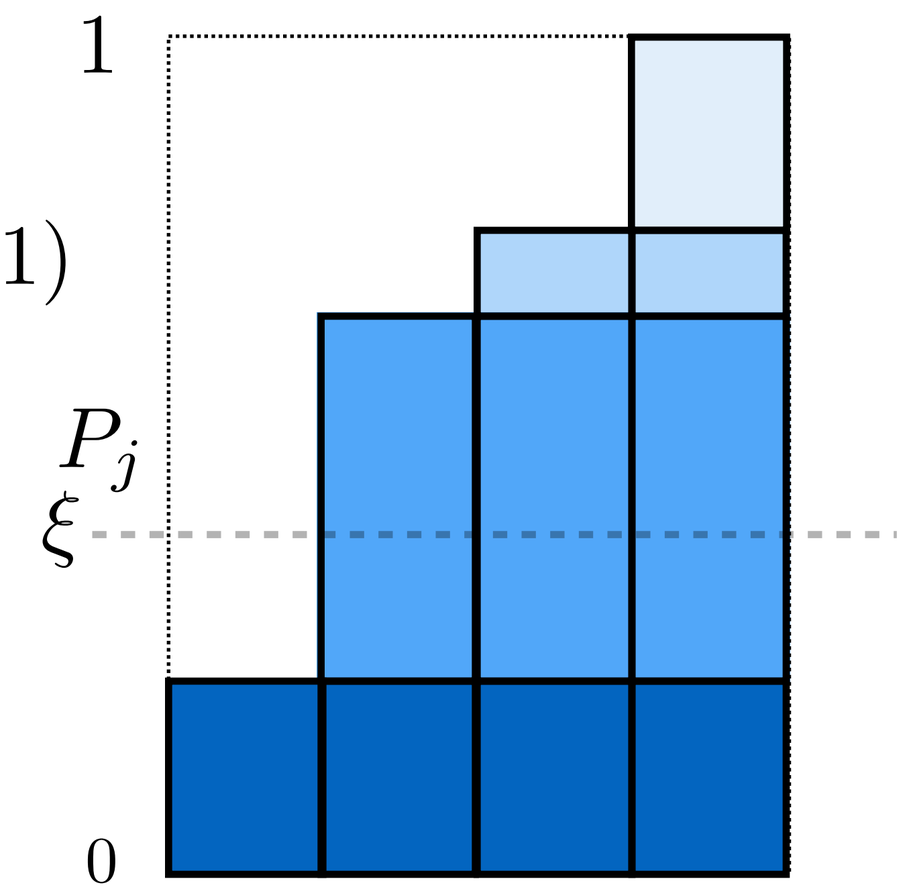


Draw a random value X from this PDF.

Step 2:

Given a uniform random variable $\xi \in [0, 1)$

choose $X = x_i$
such that $P_{i-1} < \xi \leq P_i$



How to compute? Binary search.

Cumulative Density Function (CDF) - Continuous Case

PDF $p(x)$

$$p(x) \geq 0$$

CDF $P(x)$

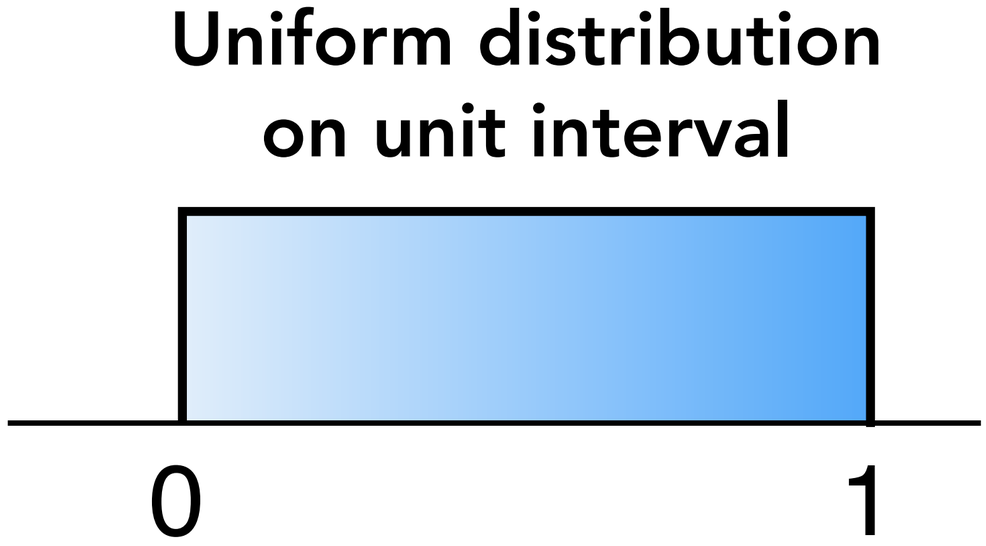
$$P(x) = \int_0^x p(x)\,\mathrm{d}x$$

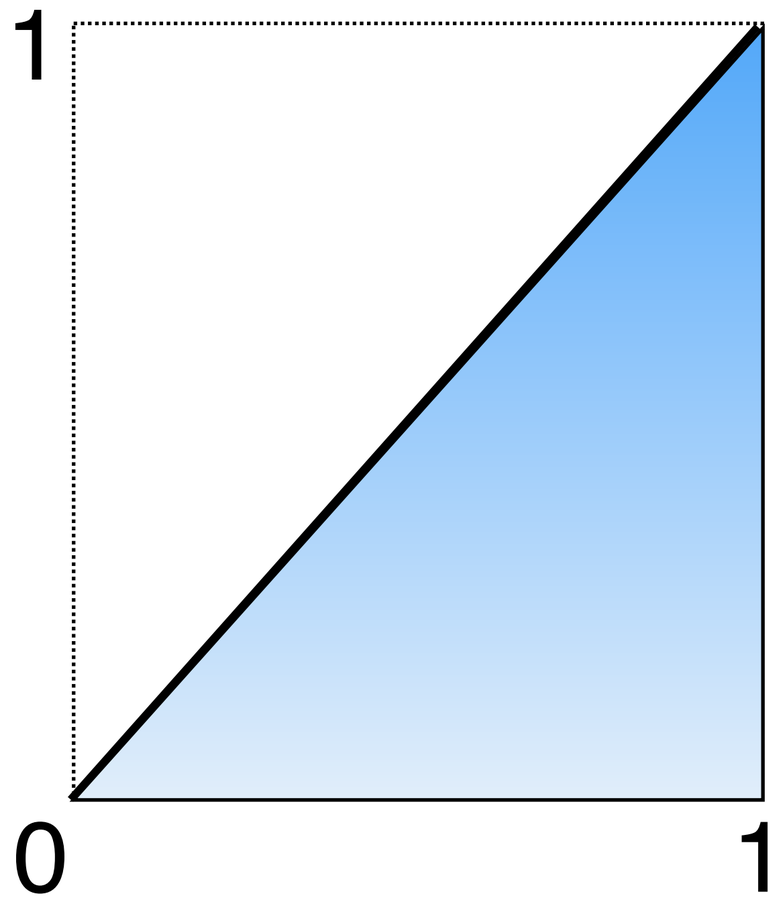
$$P(x) = \Pr(X < x)$$

$$P(1) = 1$$

$$\Pr(a \leq X \leq b) = \int_a^b p(x)\,\mathrm{d}x$$

$$= P(b) - P(a)$$



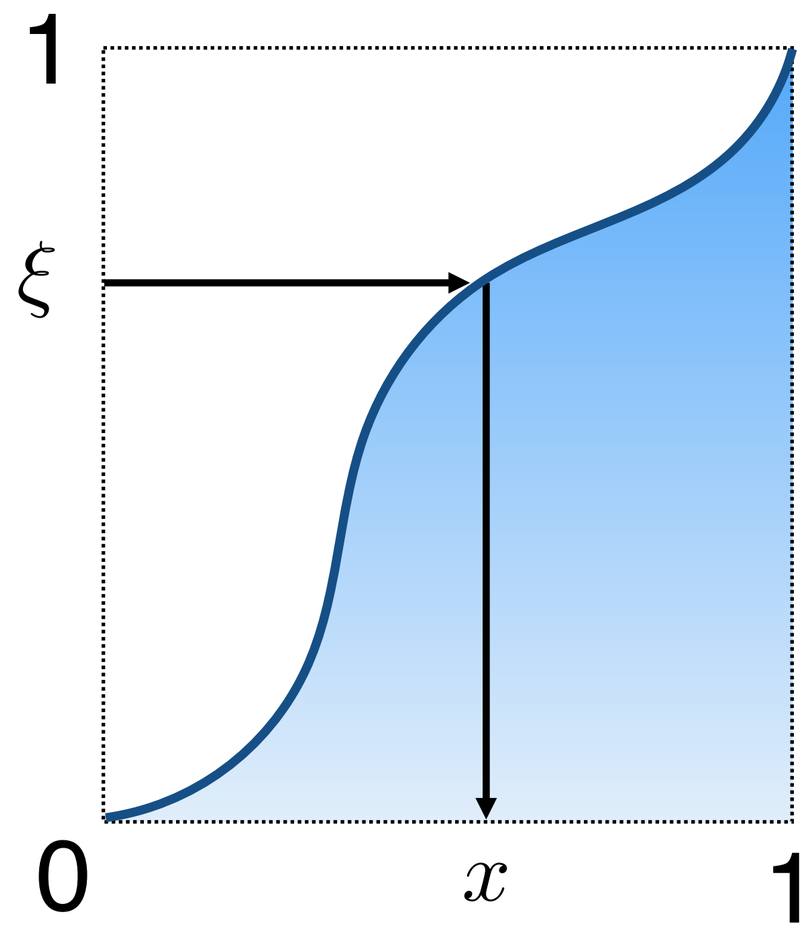


Sampling Continuous PDF Using Its CDF

Called the "inversion method"

Cumulative probability distribution function

$$P(x) = \Pr(X < x)$$



Construction of samples:

Solve for $x = P^{-1}(\xi)$

Must know the formula for:

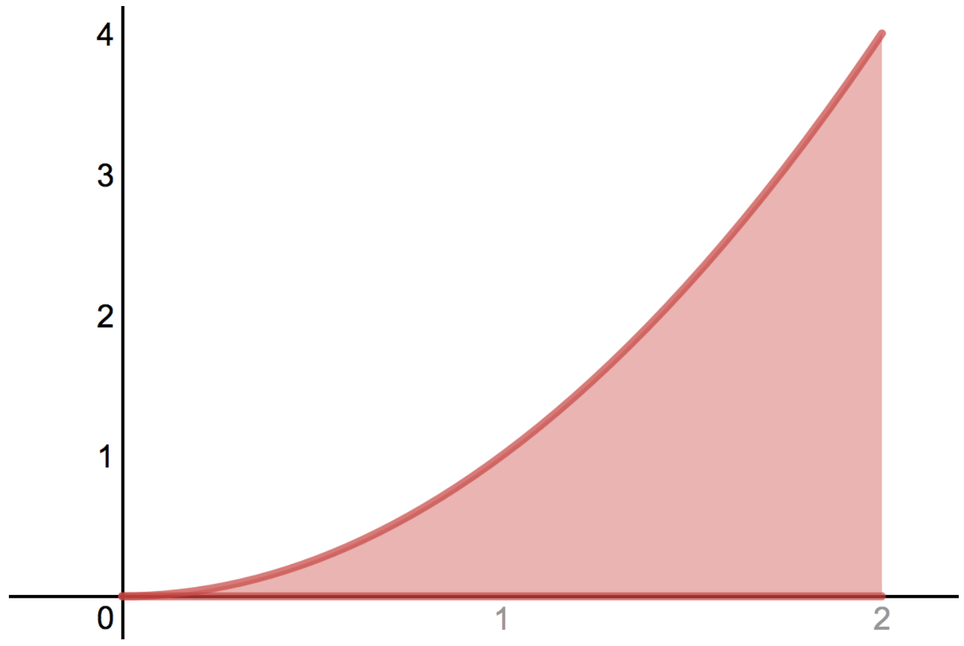
1. **The integral of** $p(x)$ **- CDF**
2. **The inverse function** $P^{-1}(x)$

Example: Sample Proportional to x^2

Given:

$$f(x) = x^2 \qquad x \in [0, 2]$$

Want to sample according to this graph:

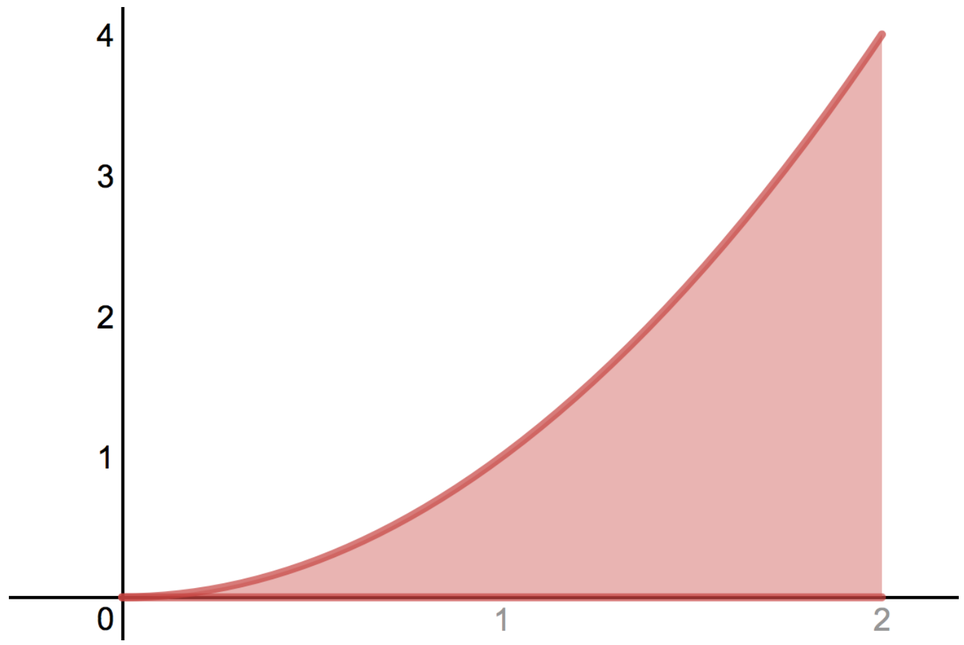


Example: Sample Proportional to x^2

Given:

$$f(x) = x^2 \qquad x \in [0, 2]$$

Want to sample according to this graph:



Step 0: compute PDF by normalizing

$$p(x) = c\, f(x) = c\, x^2$$

$$\text{Also } 1 = \int_0^2 p(x)\, dx = \int_0^2 c\, x^2\, dx = \left.\frac{cx^3}{3}\right|_0^2 = \frac{8c}{3}$$

$$\implies c = \frac{3}{8}$$

$$\implies p(x) = \frac{3x^2}{8}$$

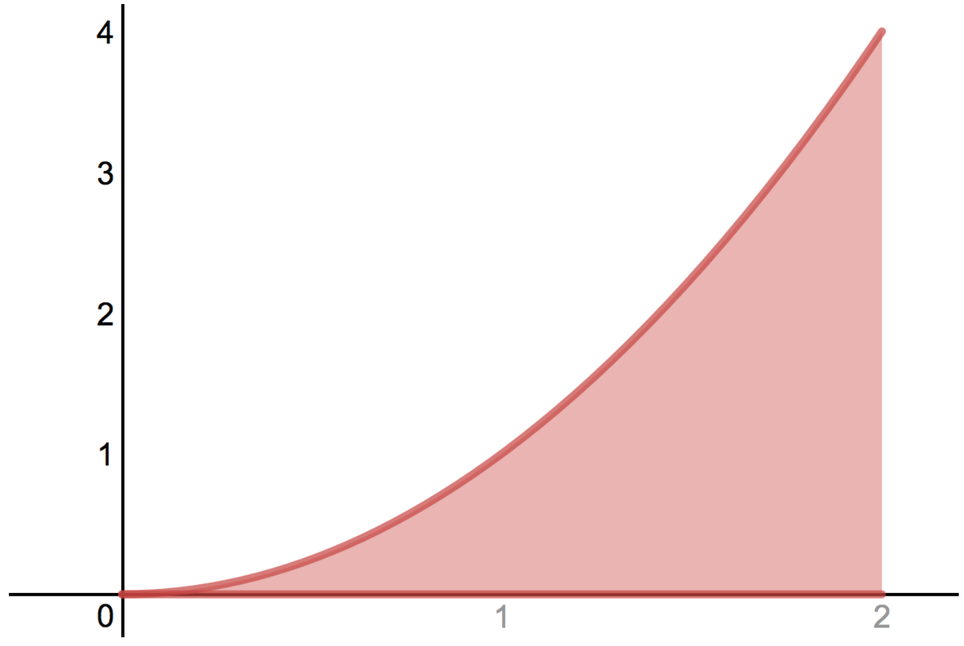
Example: Sample Proportional to x^2

Given:

$$f(x) = x^2 \quad x \in [0, 2]$$

$$\implies p(x) = \frac{3x^2}{8}$$

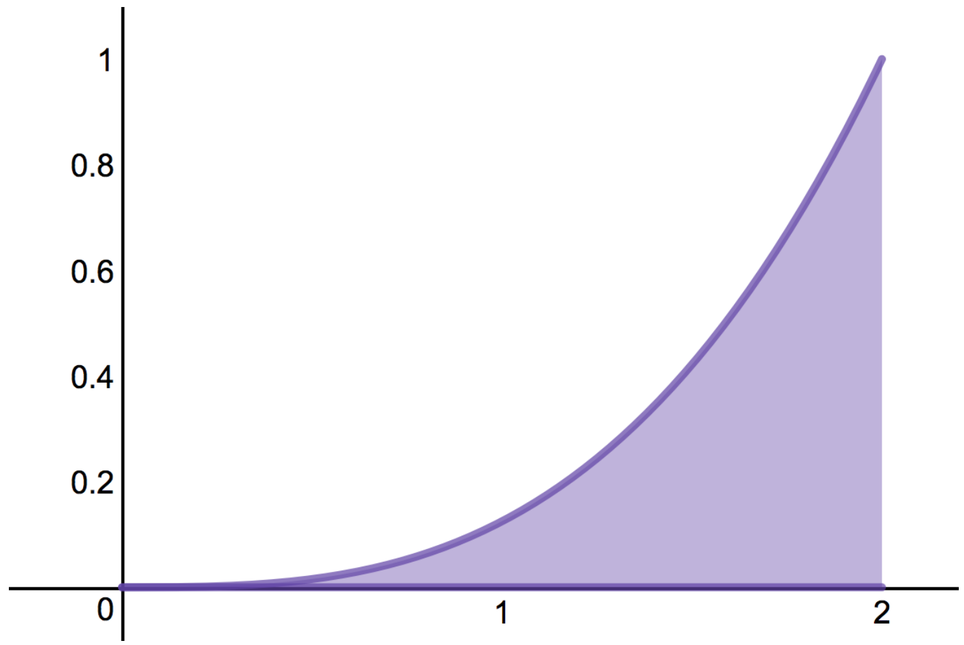
Want to sample according to this



Step 1: Compute CDF:

$$P(x) = \int_0^x p(x)\,\mathrm{d}x$$

$$= \frac{x^3}{8}$$



Example: Sample Proportional to x^2

Given:

$$f(x) = x^2 \quad x \in [0, 2]$$

$$p(x) = \frac{3}{8}x^2$$

$$P(x) = \frac{x^3}{8}$$

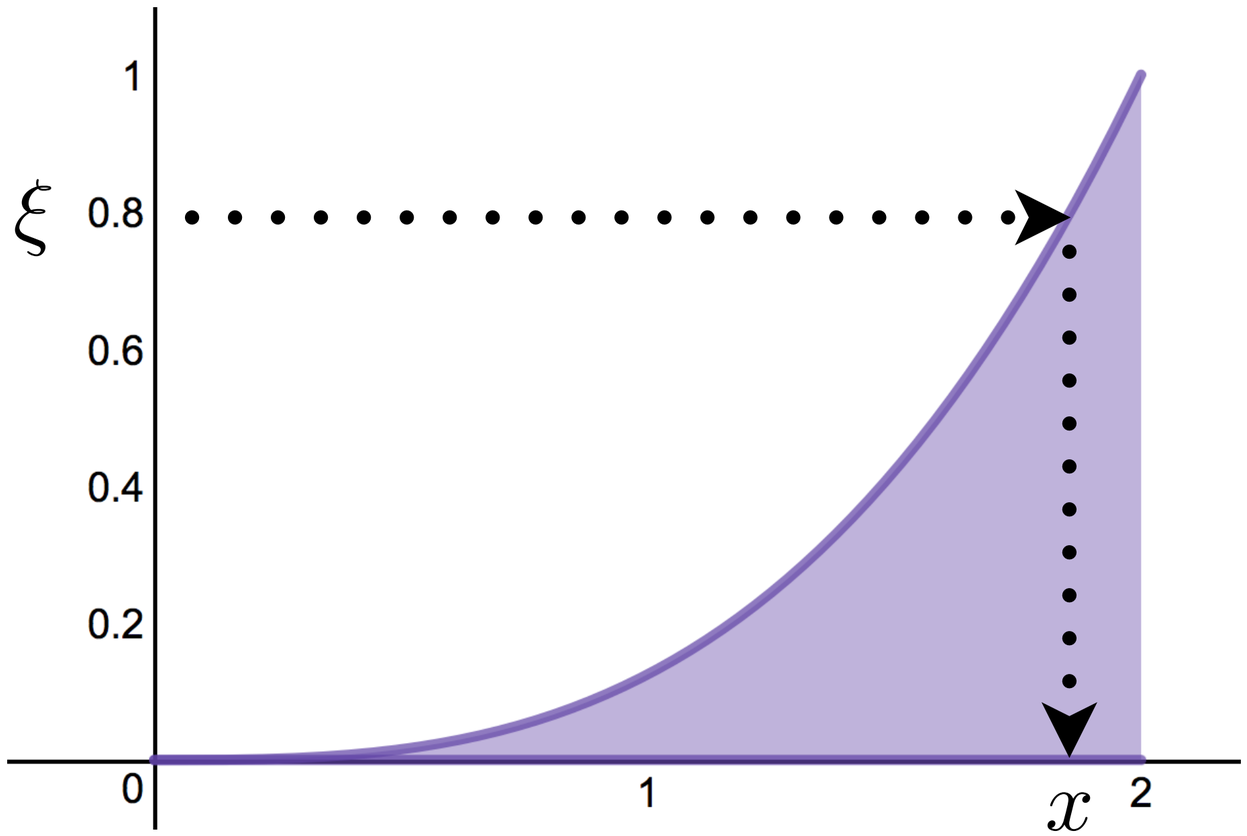
Step 2: Sample from $p(x)$

$$\xi = P(x) = \frac{x^3}{8}$$

$$x = \sqrt[3]{8\xi}$$

Applying the inversion method

Remember ξ is uniform random number in [0,1)



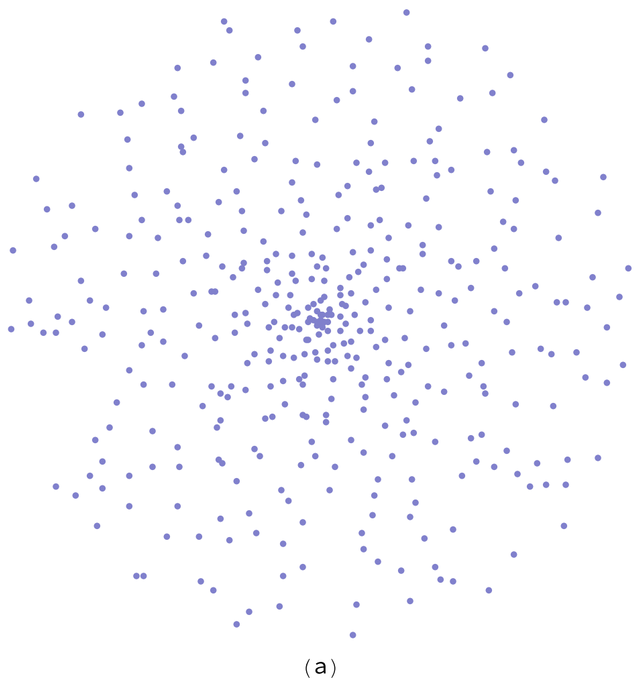
Random Sampling of Disks & Hemispheres

Sampling Unit Circle: Simple but Wrong Method

θ = uniform random angle between 0 and 2π

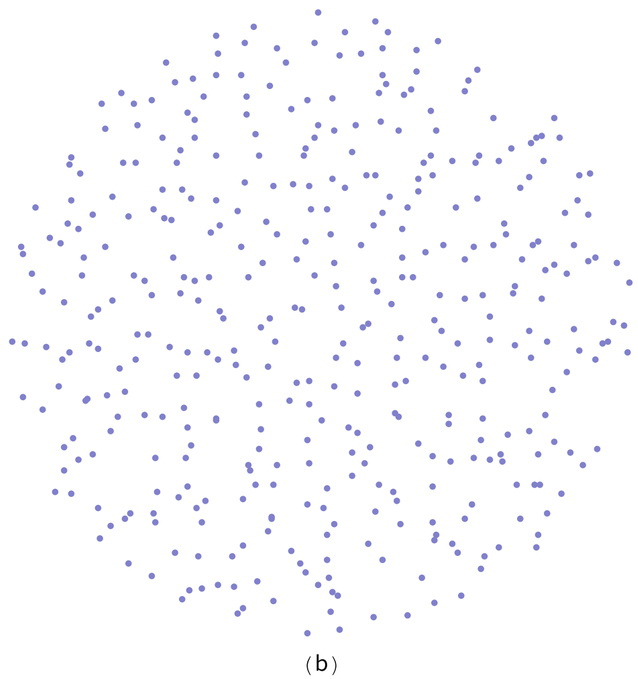
r = uniform random radius between 0 and 1

Return point: $(r \cos\ \theta, r \sin\ \theta)$



(a)

Result



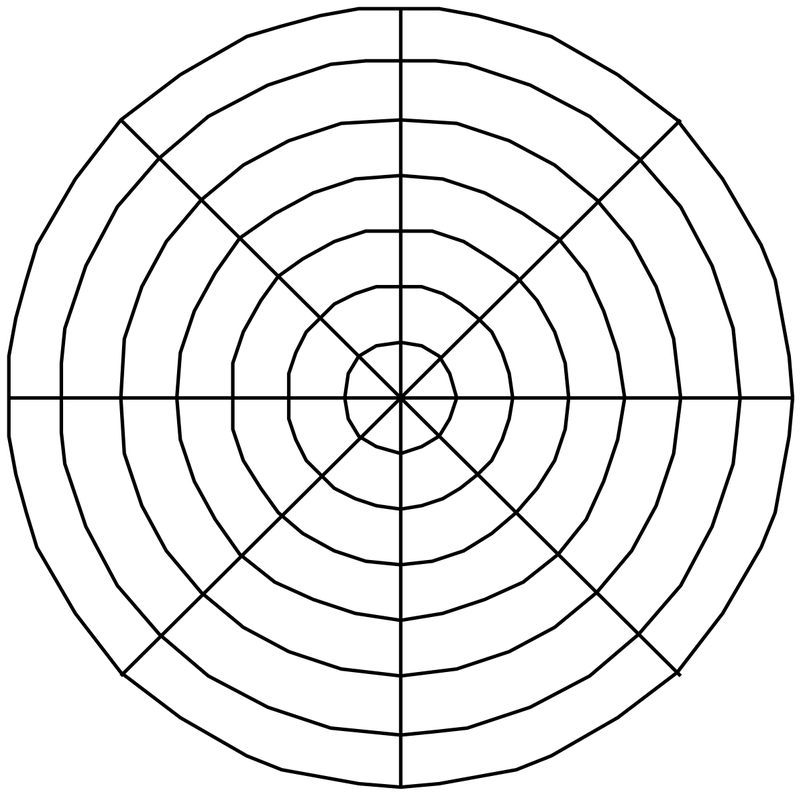
(b)

Reference

Source: [PBRT]

Need to Sample Uniformly in Area

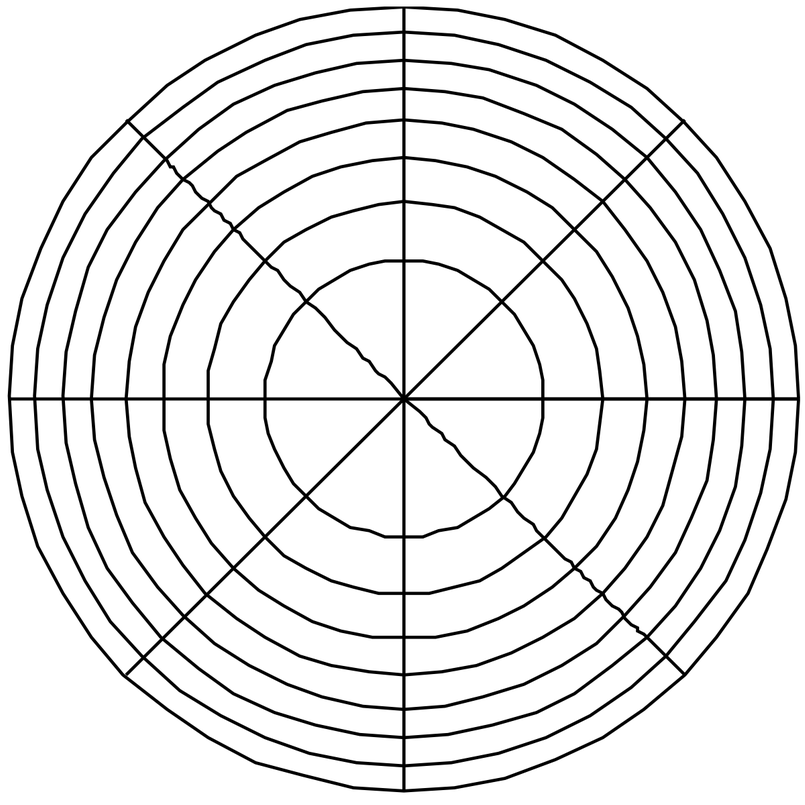
Incorrect
Not Equi-areal



$$\theta = 2\pi\xi_1$$

$$r = \xi_2$$

Correct
Equi-areal

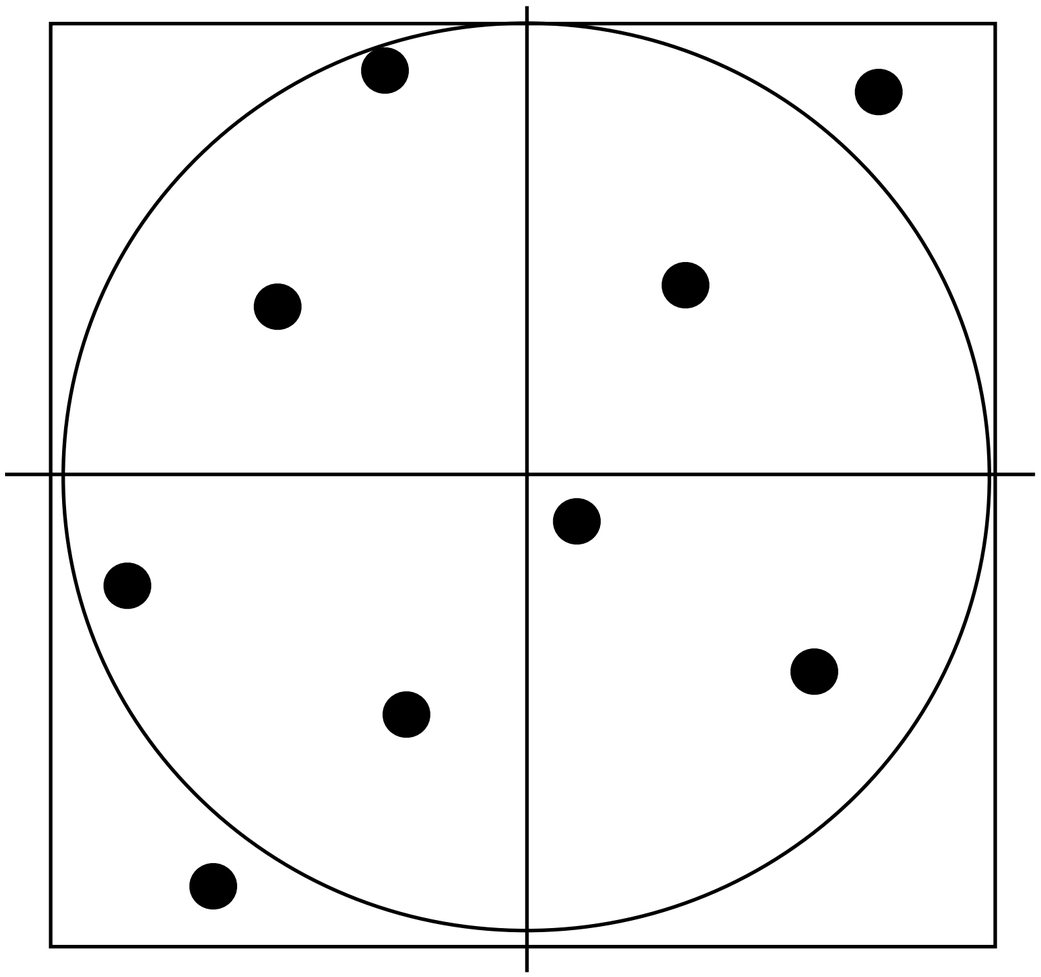


$$\theta = 2\pi\xi_1$$

$$r = \sqrt{\xi_2}$$

*** See Shirley et al. p.331 for full explanation using inversion method**

Rejection Sampling Circle's Area



```
do {
   x = 1 - 2 * rand01();
   y = 1 - 2 * rand01();
} while (x*x + y*y > 1.);
```

Efficiency of technique: area of circle / area of square

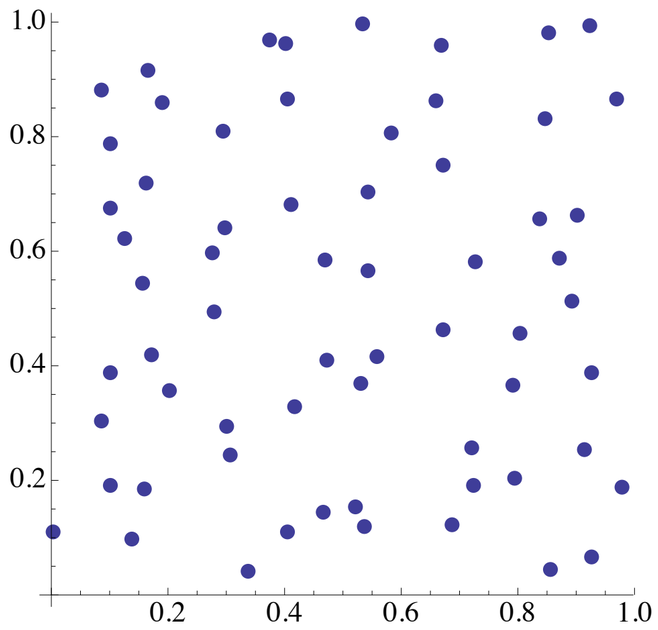
Uniform Sampling of Hemisphere

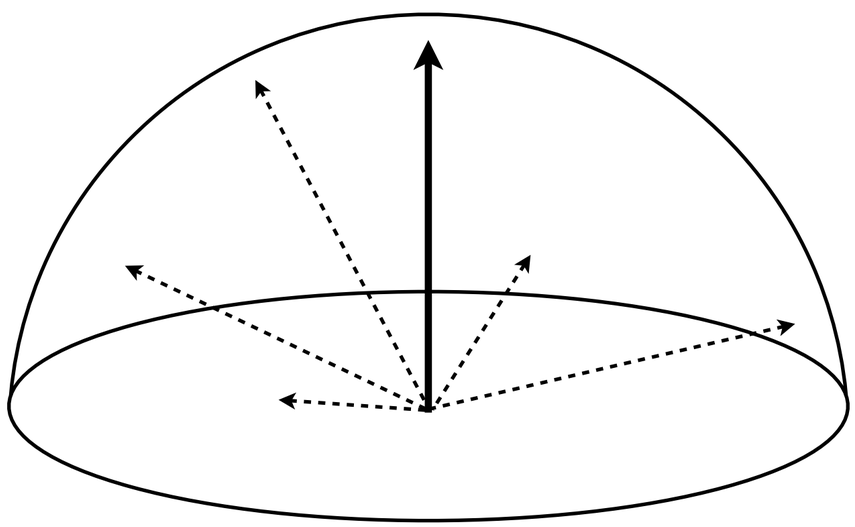
Generate random direction on hemisphere (all dirs. equally likely)

$$p(\omega) = \frac{1}{2\pi}$$

Direction computed from uniformly distributed point on 2D square:

$$(\xi_1, \xi_2) \rightarrow (\underbrace{\sqrt{1-\xi_1^2}\cos(2\pi\xi_2)}_{x}, \underbrace{\sqrt{1-\xi_1^2}\sin(2\pi\xi_2)}_{y}, \underbrace{\xi_1}_{z})$$





Full derivation: see PBRT 3rd Ed. 13.6.1

Things to Remember

Monte Carlo integration

- Unbiased estimators
- Good for high-dimensional integrals
- Estimates are visually noisy and need many samples
- Importance sampling can reduce variance (noise) if probability distribution "fits" underlying function

Sampling random variables

- Inversion method, rejection sampling
- Sampling in 1D, 2D, disks, hemispheres

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